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Corporate sustainability and firm performance

A comparative study of established and newly listed firms in the Nordic region

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Abstract

This study examines the relationship between sustainability and firm performance among listed firms in the Nordic region during the period 2010-2025, and whether this relationship differs between established and newly listed firms. Established firms are defined as firms that conducted their IPO before 2017, while newly listed firms are defined as the new cohort of firms that conducted their IPO in 2017 or later, around the period when the Nordic region experienced a clear increase in IPO activity.

The empirical data analysis is based on a quantitative research design using panel data and panel OLS regressions. Sustainability is operationalized through ESG scores from LSEG Workspace, including the combined ESG score and the individual environmental, social, and governance scores. Firm performance is measured through ROA, ROCE, and Tobin's Q, which together capture operating performance, financial performance, and market performance, respectively.

The findings show that the relationship between sustainability and firm performance is not uniform. Instead, the relationship depends on both the sustainability measure and the firm performance measure used. The combined ESG score does not show a statistically significant relationship with ROA, ROCE, or Tobin's Q. However, the individual sustainability scores provide a more nuanced result. The social score is positively and significantly related to ROCE, while the governance score is negatively and significantly related to ROA and ROCE. The environmental score does not show a statistically significant relationship with any of the firm performance measures.

The findings further show that sustainability is more positively associated with ROA and ROCE for newly listed firms than for established firms, specifically for the combined ESG score, environmental score, and social score. However, no significant difference is found for the governance score, nor for Tobin's Q. The findings suggest that the relationship between sustainability and firm performance in the Nordic region is measure-specific, pillar-specific, and partly dependent on firm listing history. The study contributes to previous research by providing updated empirical evidence from the Nordic region and by showing that the relationship differs across sustainability dimensions, performance measures, and that the relationship between sustainability and firm performance is more positive for newly listed firms than for established firms, indicating that investing in sustainability performance practices may be particularly beneficial for firms in their earlier stages of their listed life. However, these findings should be interpreted with caution due to limitations in ESG data availability, especially among newly listed firms, which means that the results should be seen as preliminary evidence, and that future research is needed.

Keywords: Firm performance, sustainability, ESG, environmental, social, governance, newly listed firms, established firms, Nordic region.

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1. Introduction

1.1. Background

Sustainability has become an increasingly important part of modern business and finance. Firms are no longer only expected to generate financial returns, but also to consider how their activities impact the environment, society, and different stakeholders connected to the firm. This increased development is closely related to the broader concept of corporate social responsibility (CSR), which can be explained as a firm's engagement with society and as part of the process through which firms develop corporate culture and social awareness (Šefránek & Mikle, 2022, p. 14). CSR has therefore become an important part of how firms communicate responsibility and respond to expectations from stakeholders, investors, and society. The development of sustainability reporting has also contributed to the growing importance of sustainability in business. Sustainability reporting started to grow during the 1990s, when environmental concerns, public pressure, and the presence of regulators and reporters increased the need for firms to disclose information about their social and environmental impact (Larrinaga & Bebbington, 2021, p. 174). And, over time, sustainability reporting became more common among large firms and is today viewed as normal part of corporate reporting. Larrinaga & Bebbington, (2021, p. 162) explain that sustainability reporting has become institutionalized, as almost all of the world's largest firms now produce some form of sustainability report. An important part of this development was the introduction of the Global Reporting Initiative (GRI) in 1999, which provided guidelines that were important for standardizing how firms communicate sustainability information publicly (Larrinaga & Bebbington, 2021, p. 162).

As sustainability became more important for firms and investors, the acronym ESG developed into a widely used framework for evaluating sustainability performance among firms. ESG refers to environmental, social, and governance sustainability. The environmental dimension concerns how firms manage issues such as emissions, recourse use, and environmental impact. The social dimension concerns how firms manage relationships with employees, customers, suppliers, and society. The governance dimension concerns how firms are controlled and managed, including issues such as transparency, accountability, board structure, and shareholder rights (Gillan et al., 2021, pp. 1-2). The term ESG was officially first used in 2004, when the United Nations Global Compact (2004) published the report Who Cares Wins, which brought these three dimensions together into one framework for investments analysis and corporate evaluation. And, since then, ESG has become a common way for investors and researchers to evaluate firms beyond their financial measures (Aydoğmuş et al., 2022, p. 122; Narula et al., 2024). In previous research, ESG scores have therefore been used as a structured and measurable proxy for sustainability performance across environmental, social, and governance dimensions (Aydoğmuş et al., 2022; Kumar & Firoz., 2022; Narula et al., 2024).

The importance of ESG was further strengthened when the Principles for Responsible Investments (PRI) were established in 2006 together with the United Nations and some of the world's largest institutional investors. The purpose of the PRI was to encourage investors to include ESG factors in investments decisions in order to manage risk and support long-term results (PRI, 2026). This development shows that sustainability has moved from being mainly an ethical or voluntary issue to becoming a financially relevant factor in investment analysis and corporate decision-making, supported by the growth of

sustainable finance, where investors increasingly use ESG information together with financial criteria's when making investment decisions (Drempetic et al., 2020).

The regulatory landscape has also increased the importance of sustainability reporting, especially in Europe. The Non-Financial Reporting Directive (NFRD) introduced in the year 2014, introduced requirements for large public-interest firms with more than 500 employees to disclose non-financial information. This included information on environmental, social and employee matters, human rights, and anti-corrupt and bribery issues. Although the directive was adopted in 2014, it applied to reporting for the financial year 2017. This directive made sustainability reporting a more formalized part of corporate reporting in Europe. More recently, the Corporate Sustainability Reporting Directive (CSRD), which entered into force in the year 2023, has increased the regulatory focus on sustainability reporting, meaning that firms are now preparing for more extensive sustainability disclosure requirements in accordance with the European Sustainability Reporting Standards (European Commission, 2025). These regulatory developments show that sustainability reporting is moving from a voluntary practice toward a more formal and mandatory part of corporate reporting. For listed firms in the Nordic region, which operate in markets with high expectations regarding transparency and responsible business conduct (Saha & Khan., 2024, pp. 794-795), these developments make sustainability an increasingly relevant subject.

Although, ESG scores should not be viewed as perfect measures of sustainability. ESG ratings are intended to assess firms' non-financial performance, including environmental impact, social responsibility, and governance processes. However, previous research shows that ESG ratings differ across rating agencies because the agencies use different definitions, data sources, criteria, and weighting methods (Abhayawansa & Tyagi, 2021, p. 51). Drempetic et al. (2020) also explain that ESG scores may be heavily influenced by firm size, data availability, and the resources firms have available for providing ESG information. This means that larger and more visible firms may receive higher ESG scores partly because they disclose more information, not necessarily because they are more sustainable in practice. Therefore, ESG scores should be understood as an operational measure of sustainability performance rather than a complete and perfect measure of sustainability. Moreover, firm performance can be understood and measure in different ways, as Narula et al. (2024, p. 1174) describe firm performance as a broad concept that includes operational, financial, and market objectives. Previous research on sustainability and firm performance therefore use several performance measures to capture different dimensions of firm performance (Aydoğmuş et al: 2022; Kumar & Firoz, 2022; Mugova et al., 2025; Narula et al., 2024). Since sustainability may affect internal profitability and external market valuation differently, it is important to study firm performance through several measures.

1.2. Problem discussion and research gap

A large number of studies have examined whether firms with stronger sustainability performance also perform better financially, but the literature does not provide one clear conclusion. Several studies suggest that sustainability may improve firm performance by strengthening reputation, stakeholder trust, investor confidence, access to capital, and financing conditions (Aydoğmuş et al., 2022, pp. 119-120; Eliwa et al., 2021, pp. 16-17; Kumar & Firoz, 2022, p. 60; Qian et al., 2024, p. 12). Other studies find negative, mixed, or insignificant relationships, suggesting that sustainability initiatives may involve costs, reporting requirements, compliance, and long-term investments that do not always

improve firm performance (King, 2025; Narula et al., 2024, pp. 1180-1181; Rahi et al., 2022, pp. 302-305; Shaikh, 2022, p. 234). One of the most influential studies in this field is Eccles et al. (2014), who argue that high sustainability firms significantly outperform low sustainability firms over the long term in both the stock market and in terms of firm performance. At the same time, this conclusion has later been challenged by King (2025), who argues that his replication and extension of the original study by Eccles et al. (2014), do not reveal evidence that corporate sustainability improves either stock market performance or firm performance. This debate further show that the relationship between sustainability and firm performance remains debated and not settled, and that the topic remains important to further knowledge, especially in new settings and with new angles.

However, previous research that examines the literature field generally show that there is a relationship between sustainability and firm performance, but that the direction and strength of this relationship varies. Friede et al. (2015, pp. 210-212) review more than 2,000 empirical studies and find that most studies report a non-negative relationship between sustainability and firm performance. They also show that the relationship differs across regions, ESG ratings, and individual ESG pillars (Friede et al., 2015, pp. 225-227). This is also reflected in more recent empirical studies on the matter, where some find positive relationships between sustainability and firm performance (Aydoğmuş et al., 2022; Kumar & Firoz, 2022; Qian et al., 2024), while others find negative, mixed, or insignificant results (King, 2025; Mugova et al., 2025; Narula et al., 2024; Rahi et al., 2022; Shaikh, 2022).

One important reason for the mixed findings is that sustainability and firm performance can be measured in different ways. The combined ESG score may hide differences between the environmental, social, and governance dimensions, since the individual pillars may have different relationships with firm performance. For example, previous studies often find that governance is more directly connected to firm performance, while environmental and social sustainability may involve higher implementation costs or longer time horizons before financial benefits appear (Rahi et al., 2022; Shaikh, 2022). In addition, the relationship may depend on the performance measure used. Accounting-based measures capture internal profitability and capital efficiency, while market-based measures capture investor expectations and future growth opportunities. Sustainability may therefore affect accounting-based and market-based performance differently.

The Nordic region provides an interesting context for further research on the relationship between sustainability and firm performance. Nordic countries are often associated with developed capital markets, high transparency, strong stakeholder expectations, and advanced sustainability practices (Saha & Khan, 2024, pp. 794-795). At the same time, existing research on this relationship in the Nordic region remains limited, with previous studies covering narrow industry scopes, and short time periods. For example, Rahi et al. (2022) focus only on listed financial firms in the Nordic region, while Saha and Khan (2024) examine a smaller Nordic sample over the period 2012-2021 using the combined ESG score as the main sustainability measure and do not separately examine the individual pillar scores specifically. This indicates that further research is needed using a broader sample of Nordic listed firms across a wide range of non-financial industries and over a more recent time period.

A further limitation in previous research is that many studies examine earlier time periods and therefore do not fully capture recent developments in sustainability reporting, ESG

data availability, and market conditions. Sustainability regulations has developed, and firms have faced changing market conditions. Recent regulatory developments, such as the CSRD, have increased the relevance of sustainability reporting, but their implications are still emerging in the literature. Another important gap concerns whether the relationship between sustainability and firm performance differs between established and newly listed firms. Established firms usually have longer public track records, more available information for investors, and more developed sustainability and governance structures. Newly listed firms, on the other hand, may face greater information asymmetry and higher uncertainty because investors have less historical information available when evaluating them. From an agency theory perspective, newly listed firms may also face higher agency costs due to less developed governance structures. In this context, ESG may be especially important for newly listed firms because it can act as a signal of firm quality, reduce uncertainty, and help attract external investors and capital (Choudhury, 2024; Economidou et al., 2023, p. 170).

This distinction is particularly relevant because the Nordic region experienced a clear increase in initial public offering (IPO) activity from around year 2017, followed by a historically strong IPO market in 2021. The post 2020 period is associated with the COVID-19 pandemic, but also with a surge in initial public offerings (IPO) under unusual market conditions (Baig & Chen, 2022, pp. 2-5). PwC (2022) presents Nordic IPO activity over the period 1998-2021 and shows that IPO activity had a peak in 2017, before reaching a record level in 2021. PwC (2022) further states that the number of IPOs across Nordic exchanges in 2021 surpassed the levels observed during the technology bubble years of 1998-2000, making the IPO activity around and post 2017 in the Nordic region especially interesting. The latest cohort of listed firms in the Nordic region can therefore be seen as newly listed firms that conducted their IPO in 2017 or later, and established firms as the old cohort of listed firms that conducted their IPO before 2017. This classification makes it interesting to compare firms with longer public histories to firms that entered public markets during a more active and recent IPO period. Newly listed firms may face greater information asymmetry, higher uncertainty, and shorter public reporting histories, which may make ESG performance disclosures a more important signal of firm quality to investors. To the best of the authors' knowledge, no previous study has directly examined whether the relationship between sustainability and firm performance differs between a group of established firms compared to a group of newly listed firms in the Nordic region, representing a gap in the existing literature.

Based on the discussion above, two distinct gaps in the existing literature can be identified. First, although previous research has examined the relationship between sustainability and firm performance, evidence from the Nordic region remains limited, especially studies using broader samples and more recent time periods. Second, limited attention has been given to whether firm listing history matters for the relationship between sustainability and firm performance. This is important because established and newly listed firms may differ in public track record, information asymmetry, governance structures, and investor expectations. Therefore, further research is needed to examine not only whether there is a relationship between sustainability and firm performance, but also whether this relationship differs across sustainability dimensions, performance measures, and firm listing history in the Nordic region.

1.3. Purpose

The purpose of this study is to investigate and analyze the relationship between sustainability and firm performance among listed firms in the Nordic region, and to examine whether this relationship differs between established and newly listed firms. The study aims to contribute to the existing body of knowledge on sustainability and firm performance in three ways. First, it tends to provide updated empirical evidence on the relationship between sustainability and firm performance in the Nordic region over an extended and more recent time period than previous Nordic studies. Second, it examines both the combined ESG score and the individual environmental, social, and governance pillars separately, contributing to a more nuanced understanding of how different sustainability dimensions relate to firm performance. Third, it introduces a novel comparison between established and newly listed firms, examining whether the relationship between sustainability and firm performance differs between these two groups in the Nordic region.

1.4. Research questions

To fulfill the purpose of this study, the following two research questions are formulated:

RQ1: Is there a relationship between sustainability and firm performance among listed firms in the Nordic region?

RQ2: Is there a difference in the relationship between sustainability and firm performance between established and newly listed firms?

1.5. Delimitations

This study focuses on publicly listed firms in the Nordic region during the period 2010-2025, including Sweden, Norway, Finland, Denmark, and Iceland. The findings are therefore limited to this period and geographic area and may not be generalized to firms in other regions with different regulatory frameworks, institutional environments, and levels of sustainability maturity, or in other time periods. The choice of year 2010 as the starting point were motivated by excluding the 2007-2009 financial crisis, as previous research show that the financial crisis strongly affected firms financing conditions and investments behavior (Campello et al., 2010), and because of low ESG data availability from LSEG Workspace prior to 2010 (De la Fuente & Velasco, 2024, p. 4).

Sustainability were operationalized through ESG scores, including the combined ESG score and the individual environmental score, social score, and governance score, sourced from LSEG Workspace. Since ESG ratings vary across rating providers due to the differences in methodology, data sources, and weighting (Abhayawansa & Tyagi, 2021, p. 51), the findings are specific for the LSEG Workspace scoring methodology. Firm performances were operationalized through three measures that together capture different dimensions of how a firm performs, return on assets (ROA) as a measure of operating performance, return on capital employed (ROCE) as a measure of financial performance, and Tobin's Q as a measure of market performance. Established firms were defined as firms that conducted their IPO prior to 2017, while newly listed firms were defined as firms that conducted their IPO in 2017 or later. This operationalization of firm performance and the classification of established and newly listed firms represent the scope of the conclusions that can be drawn from the findings of this study. The study is not intended to investigate causal claims, but rather to investigate and analyze associations between sustainability and firm performance.

2. Theoretical framework and previous findings

2.1. Sustainability and ESG

Corporate Social Responsibility (CSR) can be explained as a firm's engagement of their actions with society. It has become a commonly used term and is the process of which firms develop their corporate culture and social awareness (Šefránek & Mikle, 2022, p. 14). The authors mention that the term CSR has received a lot of attention the past decade, and that it is now considered to be the most prevalent for firms. Almost all corporate firms decide to report their CSR efforts as a way to ensure that their societal obligations are being met. Moreover, Abd Rahim (2011), cited by Šefránek and Mikle (2022, p. 14) in their article, argues that firms have now grown in society and they no longer only focus on wealth creation but also on acting sustainably towards their stakeholders. Furthermore, Šefránek and Mikle (2022, p. 24) mention a study done by Caroll (2008), which argues that there are different perspectives on CSR. One highlight from this study is that European firms are applying more CSR into their practices than American firms where they have a somewhat skeptical view of corporate responsibility.

LSEG Workspace has identified twelve themes, with 220 indicators to provide ESG scores to firms. The assessment is based on publicly available data that has been obtained from each firm through annual reports, corporate sustainability reports, and more. The twelve themes include biodiversity, emission, climate transition, etc., where each theme is related to either the environmental, social, or governance pillar. The firm is then scored across each theme and receives an aggregate score which gets integrated into each pillar (LSEG, 2026, p. 5-6).

The term ESG was first officially used in 2004, when the United Nations Global Compact (2004) published the report; Who Cares Wins. The report was developed together with twenty financial institutions with the purpose to show the financial industry how to incorporate environmental sustainability, social sustainability, and governance sustainability into firms' investment analysis and asset management (United Nations Global Compact, 2004). This was the first time the three sustainability dimensions were brought together under the ESG acronym, creating a framework that investors and researchers now widely use to evaluate firms beyond just their financial performance (Aydoğmuş et al., 2022; Narula et al., 2024).

In this study, sustainability is defined as the broad concept that refers to how firms manage environmental sustainability, social sustainability, and governance sustainability. Following previous studies (Aydoğmuş et al: 2022; Eccles et al, 2014; Kumar & Firoz, 2022; Narula et al., 2024) ESG scores provide a structured and measurable framework for capturing sustainability performance across these three dimensions, making it a suited operational proxy to examine sustainability performance among firms.

The concept of ESG ratings is to evaluate how sustainable a firm is beyond its financial performance. These ratings assess a firm's non-financial performance, including environmental impact, social responsibility, and corporate governance processes. ESG ratings provide investors with a standardized framework to evaluate the sustainability and ethical practices of potential investment opportunities. According to Abhayawansa & Tyagi (2021, p. 51), ratings vary widely across agencies. One rating agency may rank a firm as a leader in promoting ESG practices, while another may rank it as one of the worst in its industry. The authors identified two main factors contributing to the discrepancies.

First, there are theorization issues, which refer to variations in how ESG variables are perceived and interpreted, as rating agencies may prioritize different aspects of ESG, which results in varying ratings for firms' performance in these areas. Second, variations in ESG evaluation techniques can lead to consistency issues. Rating agencies may use different criteria, data sources, and weighting, resulting in varying ratings for the same firm (Abhayawansa & Tyagi, 2021, p. 51). Previous research has used a variety of rating providers. For example, Aydoğmuş et al (2022), Eliwa et al (2021) and Rahi et al (2022) used scores provided by LSEG. Other studies used Bloomberg's ratings (Kumar & Firoz, 2022; Shaikh, 2022; Demers et al., 2021) and MSCI ratings (Fu et al., 2023). There exist other rating provers as well, such as Crisil ESG that were used in the study done by Narula et al. (2024), and RGBI used by (Economidou et al., 2023). In the study by Qian et al. (2024), they created their own Green-commitment index due to assessing flaws in ESG ratings from providers. This indicates that the different previous findings associated between sustainability and firm performance can differ due to the sourced ESG scores from different rating providers.

2.2. Firm performance

Firm performance is a broad concept that refers to how well a firm achieves its operational, financial, and market objectives (Narula et al., 2024, p. 1174). Since firm performance can be measured in different ways, previous research on the relationship between sustainability and firm performance use multiple measures to capture different aspects of firm performance (Aydoğmuş et al., 2022; Kumar & Firoz, 2022; Mugova et al., 2025; Narula et al., 2024). This study follows the same approach as Narula et al. (2024, p. 1174), and distinguishes firm performance between operating performance, financial performance, and market performance.

The first aspect of firm performance, operating performance, refers to how efficiently a firm generates profit from its assets in this study. Return on assets (ROA) is a commonly used measure of operating performance in previous studies (Aydoğmuş et al., 2022; Narula et al., 2024), as it reflects how well a firm converts its assets into net income. A higher ROA indicates that the firm generates more profit relative to its total assets, which can be interpreted as stronger operational performance for a firm (Kumar & Firoz, 2022; Narula et al., 2024). Second, in this study, financial performance focuses on how efficiently a firm generates returns from the capital it employs in its operations. Return on capital employed (ROCE) measures this by relating operating profit to the capital used to generate it (Kumar & Firoz, 2022, p. 50). ROCE complements ROA by focusing on invested capital rather than just total assets, making it an additional measure of how efficiently a firm uses its financial resources (Kumar & Firoz, 2022; Narula et al., 2024). Third, market performance in this study is set out to reflect how the market values a firm relative to its assets. Tobin's Q is a measurement that is widely used in previous research on sustainability and firm performance, as it captures investor expectations of future growth and profitability rather than only reflecting accounting outcomes (Aydoğmuş et al., 2022; Shaikh, 2022; Tang et al., 2024). Unlike ROA and ROCE, which are more backward-looking accounting measures, Tobin's Q is more of a forward-looking measurement that captures how investors and external market participants value a firm.

Using all three measures together in this study is important because sustainability may affect different aspects of firm performance in different ways. Mugova et al. (2025) point this out and find that ESG is negatively related to ROA while being positively related to Tobin's Q, suggesting that sustainability investments may reduce short-term profitability

while at the same time also improving how the market values the firm. Including different measures of firm performance is set out to provide a broader picture of how sustainability relates to firm performance and reduces the risk of findings being driven by the choice of a single firm performance measure.

2.3. Theoretical perspectives on sustainability and firm performance

2.3.1. Stakeholder theory

Stakeholder theory was originally developed from the idea of strategic management by Freeman (1984). Since the theory's beginning, it has positioned itself exclusively from traditional economic theorizing. While traditional economic theorizing focuses on market competition, the stakeholder theory emphasizes cooperation. The theory holds that it is the job of managers to foster cooperation between different stakeholder and balance their interests (Bridoux & Stoelhorst, 2022, pp. 797-809). The stakeholder concept began with two principles (Friedman & Miles, 2024, p. 2). The first principle is stakeholder enabling, which states that firms should act and be managed in the best interests of their stakeholders. The second is director responsibility, which holds that directors have a duty of care to define and direct the affairs of the firm in line with the first principle. Later, a third principle was added, the principle of stakeholder resource that stated that if directors fail to perform the required duty of care, stakeholders may act against them (Friedman & Miles, 2024, p. 2).

The stakeholder theory suggests that a firm must create value for its stakeholders, including customers, suppliers, employees, and society. Together, they create something for which no one can be solely responsible (Marcon Nora et al., 2023). Stakeholder theory has proven to improve firm performance by focusing on all stakeholders, not just shareholders (Marcon Nora et al., 2023). According to Lahouel et al. (2022), incorporating stakeholder theory into firm practices enhances competitiveness by improving customer and employee relationships and operational productivity. However, stakeholder theory is not without criticism. Maharaj (2008) argues that firms should not only identify their stakeholders but also carefully evaluate their ability to influence decision-making. The same author warns that prioritizing too many stakeholder interests simultaneously can harm the firm, particularly boards of directors and management, as reducing focus on the core firm's activities may lead to lower financial returns. On the other hand, Maharaj (2008) also acknowledges that effectively managing stakeholder relationships can strengthen a firm's brand and reputation, which may offset some of these costs over time.

Stakeholder theory provides a useful lens for understanding why sustainability performance may relate to firm performance. Environmental sustainability reflects how firms manage their responsibilities toward emissions and resource usage, social sustainability captures how firms handle relationships with employees, customers, and communities, and governance sustainability concerns how firms are directed and controlled in the interest of all parties (Gillan et al., 2021, pp. 1-2). Firms that perform well across these dimensions demonstrate that they manage their operations in a way that considers a broad set of stakeholder interests, which according to stakeholder theory should support stronger long-term firm performance through enhanced stakeholder trust and long-term value creation (Marcon Nora et al., 2023; Bridoux & Stoelhorst, 2022; Lahouel et al., 2022). This makes stakeholder theory one of the central theoretical

foundations for expecting a positive relationship between sustainability and firm performance.

2.3.2. Legitimacy theory

Legitimacy is a generalized perception that the action of an entity is appropriate within social norms and beliefs (Suchman, 1995, p. 574, cited in Crossley et al., 2021, p. 3742). The theory argues that firms will employ sustainable disclosures to improve their public perception and therefore attract new business. However, Hummel & Schlik (2016, p. 457) argue that the theory also suggests that firms who are performing poor sustainably, may provide low quality information to the public, in order to maintain their legitimacy and hide their poor performance. The authors of the same article further build on the legitimacy theory and mention that low performing firms do so to avoid transparency and improve their image (Hummel & Schlik, 2016, p. 473). The theory is also closely tied to the concept of CSR, as Olateju et al. (2021, pp. 2-4) argue that for a firm to gain legitimacy, it must maintain the integrity of its value system. Historically, it has been difficult for firms to operationalize their CSR policies in practice, and legitimacy theory helps to fill this gap by emphasizing that firms should not only perform their social responsibility but also report and communicate this performance publicly.

The theory provides a relevant perspective for understanding why firms engage with sustainability and how this engagement may relate to firm performance. Firms that actively manage and disclose their sustainability practices signal to external parties that they operate within accepted social norms, which strengthens legitimacy. Empirical support of this argument is provided by studies showing that voluntary ESG disclosure improves long-run firm performance and reduces the risk of firm failure, with the effect being particularly strong for firms with limited public track records (Economidou et al., 2023, p. 170; Fu et al., 2023, p. 16). This strengthened legitimacy can in turn support firm performance by improving access to capital, attracting investors, and reducing the risk of reputational damage or regulatory intervention (Olateju et al., 2021, pp. 2-4). In this way, legitimacy theory is expected to complement stakeholder theory by suggesting that sustainability engagement is not only about satisfying stakeholder interests but also maintaining a broader societal acceptance that supports long-term firm survival and performance.

2.3.3. Shareholder theory

Different to the stakeholder theory, the shareholder theory holds that a firm's primary goal is to maximize wealth for its shareholders (Friedman, 1962). The theory is well established in finance and is regarded by many as the conventional way to operate a firm (Danielson et al., 2008). However, it has received criticism for encouraging short-term and sometimes unethical firm practices, as it prioritizes shareholder interests over those of employees, customers, and other stakeholders (Danielson et al., 2008). In practice, shareholder theory suggests that firms should maximize the present value of expected future dividends. Managers may also face formal and informal incentives to increase the firm's stock price, for example through stock option compensation, which can reinforce a short-term firm performance focus (Danielson et al., 2008). This short-term orientation has been linked to corporate scandals such as Enron and Worldcom, where the pursuit of shareholder returns led to unethical and ultimately destructive decisions. However, Danielson et al. (2008) also note that shareholder theory has long term focus, which Enron and Worldcom clearly failed to follow.

Shareholder theory offers a contrasting perspective to stakeholder and legitimacy theory on the relationship between sustainability and firm performance. From a shareholder theory perspective, allocating resources toward sustainability may be seen as costly and potentially value-reducing if these investments do not directly generate shareholder returns in the short-run. This view is further supported by what Economidou et al. (2023, p. 132) describe as the shareholder expense view of sustainability, which suggests that sustainability activities can represent a form of wasteful spending. This suggests that sustainability investments may increase costs and reduce short-term profitability (Rahi et al., 2022, pp. 302-305; Shaikh, 2022, p. 234). Shareholder theory therefore provides a theoretical foundation for expecting a negative relationship between sustainability and firm performance.

2.3.4. Trade-off theory

The trade-off theory is the longest standing theory of capital structure originally formalized by Kraus & Litzenberger (1973). The theory argues that there is an optimal debt ratio for a firm, and that the optimal amount of debt equates to the marginal benefit of a dollar of debt that arises from tax benefits. The established framework implies that changes in leverage would also change the marginal tax shield overtime (Abel, 2018, p, 95). The trade-off theory came into serious consideration after the debate on the Modigliani-Miller theorem, which also argues in favor of debt due to its tax shield benefits (Jahanzeb et al., 2013, p. 11). Abel (2018, p. 95) further mentions that the theory is conventionally interpreted as more profitable firms should have high leverage ratios. However, Jahanzeb et al. (2013, p. 13) argues this is not the case. Titman & Wessels (1988) and Rajan & Zingales (1995) find that firms that borrow less tend to be more profitable, which goes against the trade-off theory that suggests higher profitable firms should borrow more to reduce tax burden (Jahanzeb et al., 2013, p. 13). The same article published by Jahanzeb et al. (2013), mentions Marsh (1982) and Hovakimian (2004), who find results that support the theory and its predictions about leverage decisions (Jahanzeb et al., 2013, p. 13).

Trade-off theory complements shareholder theory as a contrasting perspective on the relationship between sustainability and firm performance. Sustainability investments, such as environmental initiatives, social programs, and governance improvements, require financial resources and may alter a firm's capital structure by increasing costs and reducing financial flexibility (Qian et al., 2024, p. 12; Rahi et al., 2022, p. 305). From a trade-off theory perspective, firms that allocate resources toward sustainability may face higher financial constraints, which can reduce their ability to maintain an optimal capital structure and in turn negatively affect firm performance (Jahanzeb et al., 2013). This suggests that investing in sustainability may reduce short-term profitability, particularly for firms that might already operate under tight financial conditions (Qian et al., 2024, p. 12; Rahi et al., 2022, pp. 305-306).

2.3.5. Agency Theory

Agency theory has been a major economic theory since the published formative work by Jensen & Meckling (1976). The agency theory lies in the idea that a firm is a special case, where different agents can work against each other. The agent in a firm operates under both written and unwritten contracts, and agency costs are a result of having different interests and agreements between owners or managers (Pepper & Gore, 2015, pp. 1045-1068). The same articles mention that agency theory focuses on the cost of potential conflicts of interest between principals and agents. Pepper & Gore (2015, p. 1045-1068)

further describes that Jensen and Meckling identified agency costs as the sum of monitoring, bonding, and residual costs. Tekin & Polat (2020, p. 1325) describes these costs as follows; monitoring costs refer to shareholders spending resources to supervise managerial decisions and evaluate performance. Bonding costs refer to when managers agree to rules that align with the shareholder's goals, and residual costs occur when, even after monitoring and bonding, managerial decisions still do not align with shareholder interests.

Understanding the relationship between sustainability and firm performance benefits from an agency theory perspective, particularly when considering the role of governance. Strong governance practices, such as improved board oversight, transparency, and accountability structures, can reduce agency costs by better aligning the interests of managers and shareholders (Shaikh, 2022, p. 228; Tekin & Polat., 2020). This suggests that firms with stronger governance sustainability are better positioned to reduce agency problems, improve decision-making, and generate stronger firm performance.

However, Economidou et al. (2023, p. 132) also argues that investments to improve ESG performance could themselves signal agency problems, for example when managers over-invest in sustainability activities to pursue private benefits rather than shareholder value, which may in turn negatively affect firm performance. This suggests that the relationship between sustainability and firm performance may not be straightforward from an agency theory perspective, as sustainability engagement may both reduce and reflect agency costs. At the same time, agency theory also provides insight into whether there is a difference in how established firms perform compared to newly listed firms in relation to sustainability. Firms that have recently entered public markets typically have shorter public track records and less developed governance structures and may therefore face higher agency costs than more established firms, as external investors have less information available to monitor and evaluate managerial behavior (Economidou et al., 2023, p. 132; Fu et al., 2023, p. 16). This suggests that governance sustainability may play a more important role in reducing agency costs for newly listed firms, making the relationship between sustainability and firm performance potentially stronger for this group compared to established firms.

2.3.6. Signaling theory and information asymmetry

Signaling theory originates from asymmetric information where one party has more information than another (Spence, 1973). This phenomena is common in economics and finance, where a seller usually has more information about their product or firm than a potential buyer or investor (Choudhury, 2024, p. 98). Choudhury (2024, p. 99) further argues that information asymmetry is often reduced when a seller communicates certain qualities or characteristics to the other party, an action described as sending signals, hence the name of the theory. However, signaling theory do not always benefit the receiving party, as a seller with a low-quality product may misrepresent it to earn profits, making it important for the receiving party to carefully assess the credibility of the signals they observe (Choudhury, 2024, p. 99).

Applied to firms and financial markets, signaling theory suggests that firms use various mechanisms to communicate their quality and prospects to external investors (Choudhury, 2024, p. 114). Sustainability disclosures in terms of ESG reporting, and ESG performance can serve as such signals, as they provide investors with non-financial information about how a firm manages its environmental sustainability, social

sustainability, and governance sustainability responsibilities. Economidou et al. (2023, p. 170) argue that visible ESG information can reduce information asymmetry and signal high firm quality to investors, which may in turn improve access to capital and support firm performance. This links signaling theory to the broader relationship between sustainability and firm performance, suggesting that firms with stronger sustainability performance may benefit from reduced information asymmetry and improved investor confidence.

From a signaling theory perspective, the relationship between sustainability and firm performance may differ between established and newly listed firms, as the value of sustainability signals is expected to vary with the level of asymmetry a firm faces. Firms that have recently entered public markets typically face greater information asymmetry than established firms, as they have limited public track records and less historical financial information available for investors to evaluate (Economidou et al., 2023, p. 170; Fu et al., 2023, p. 16). For these firms, ESG disclosure and strong sustainability performance may therefore serve as especially valuable signals of firm quality, helping to reduce investor uncertainty and improve access to external financing (Qian et al., 2024, p. 12). Empirical support for this argument is provided by Economidou et al. (2023, pp. 172-173), who find that firms with ESG ratings attract more sophisticated investors and demonstrate stronger initial public offering (IPO) performance, and by Fu et al. (2023, p. 16), who find that voluntary ESG disclosure reduces IPO failure risk and improves long-run performance. Together, these findings suggest that the value of ESG as a signal may be greater for newly listed firms than for established firms, as newly listed firms face higher information asymmetry and therefore have more to gain from communicating their sustainability performance to external investors. This provides a theoretical basis for expecting the relationship between sustainability and firm performance to differ between established and newly listed firms.

2.3.7. Comparison of theories

The six theories presented in this chapter approach the relationship between sustainability and firm performance from different angles, and together they form a theoretical framework that helps explain both why the relationship may exist and why it may differ across firms and contexts.

Stakeholder theory and legitimacy theory are the most closely aligned, as both emphasize the importance of managing responsibilities beyond shareholders alone. Stakeholder theory argues that firms create value by satisfying the needs of a broad set of stakeholders (Bridoux & Stoelhorst, 2022; Marcon Nora et al., 2023), while legitimacy theory focuses on how firms maintain societal acceptance through voluntary disclosure and responsible behavior (Hummel & Schlik, 2016; Olateju et al., 2021, pp. 2-4). Both theories support the expectation of a positive relationship between sustainability and firm performance, through enhanced trust, reputation, and access to capital.

Shareholder theory and trade-off theory offer a contrasting perspective. Both are rooted in financial reasoning and suggest that sustainability investments may be costly and value-reducing if they do not generate direct financial returns. Shareholder theory focuses on the primacy of shareholder wealth maximization (Danielson et al., 2008), while trade-off theory emphasizes the importance of maintaining an optimal capital structure (Jahanzeb et al., 2013). Together, these two theories provide a theoretical foundation for expecting a negative or insignificant relationship between sustainability and firm

performance, particularly for short-term accounting-based measures (Shaikh, 2022, p. 234).

Agency theory and signaling theory are most relevant for understanding why the relationship between sustainability and firm performance may differ between newly listed and established firms. Agency theory focuses on the conflicts of interest that arise from the separation of ownership and control and suggests that stronger governance sustainability can reduce these conflicts and improve firm performance (Pepper & Gore, 2015; Tekin & Polat., 2020). Signaling theory focuses on how firms communicate their quality to external investors through ESG disclosures and suggests that this signal carries more value for newly listed firms that face higher information asymmetry (Choudhury, 2024; Economidou et al., 2023, p. 170). While agency theory emphasizes internal governance mechanisms, signaling theory emphasizes external communication and together they provide complementary explanations for why newly listed firms may respond differently to sustainability engagement than established firms (Economidou et al., 2023, p. 132; Fu et al., 2023, p. 16).

2.4. Previous findings and hypothesis development

2.4.1. Previous findings

A large amount of research exists on the relationship between sustainability and firm performance, operationalized through ESG scores, and different dimensions of firm performance (Friede et al., 2015). However, previous research on the relationship between sustainability and firm performance presents extensive and broad evidence on the relationship but is often ambiguous and inconsistent. A large meta-analysis review by Friede et al. (2015, pp. 210-212), reviewing more than 2000 empirical earlier studies, find that roughly 90% of the studies report a non-negative relationship between ESG and firm performance, with the majority of these studies presenting positive findings on the relationship, while also noting that the literature has long been described as ambiguous and contradictory. Further, Friede et al. (2015, pp. 225-227) find that the strength of the relationship varies across studies, regions, and ESG score as well as the individual environmental score, social score, and governance score, and how they are reported. This suggests that ESG should not be treated as a universally identical predictor of firm performance.

One of the most influential and highly cited studies on corporate sustainability is a study by Eccles et al. (2014), and in their sample of 180 United States (U.S.) firms, they find that firms with strong sustainability practices outperform firms with weaker sustainability practices in both accounting and stock market performance over the long term (Eccles et al., 2014, p. 2035). However, these findings have been recently challenged by King (2025), that replicates the data sample and finds no evidence to justify and support the finding by Eccles et al. (2014). Even when extending the tests and robustness checking, still find no support to the findings by Eccles et al. (2014). And when King (2025, p. 109) contacted the journal and editors at Management Science, Eccles et al. (2026) published an errata corrige to explain that the analysis by King (2025) does not match the exact same as Eccles et al. (2014) but acknowledges a typo of where the word not, was missing before statistically significant. This illustrates that even highly cited studies in respectively high rated journals can reach different conclusions depending on research design and the possibility of typos and that there still is a need for further studies on the subject of sustainability and firm performance.

In a large study by Aydoğmuş et al. (2022) including 1720 listed firms from 39 countries over the period 2013-2021, using ESG scores from LSEG Workspace and in the regression model controlling for firm size and leverage, find that the overall ESG combined score is positively and significantly related with firm value, measured by Tobin's Q, and profitability measured by ROA. They specifically find that the combined ESG score, as well as the social and the governance pillars is positively related to Tobin's Q, while the environmental pillar is not significant for Tobin's Q but show positive and significant relationship with ROA (Aydoğmuş et al., 2022, pp. 119-120). These findings indicate that sustainability can both be related to accounting-based measures and market-based outcomes, but also that the relationship can differ between the different ESG pillars. Additional, in research by Tang et al. (2024), that focus on ESG and firm value as well as the effect of investor perception on a large sample of Chinese listed firms from year 2009-2022, using ESG scores from HUAZHENG ESG rating system and in the regression model controlling for firm size, profitability, and growth opportunities, find that ESG is positively related to firm value, measured by Tobin's Q, and that investors play a mediating role in this relationship. Their result further shows that all three individual ESG pillars are also positively related to firm value, although the effect is strongest for governance and weakest for the environmental pillar (Tang et al., 2024, pp. 1-2, 7). This suggests that ESG may influence market-based performance through improved investors perception of risk, reputation, and long-term prospects.

A related positive argument of a positive relationship between sustainability and firm performance appears in studies that also emphasize disclosures and visibility rather than only sustainability through ESG scores. Economidou et al. (2023, pp. 127-128) examine 1856 U.S. IPOs during 2007-2018, using ESG ratings from RGBI and in the regression model with controls such as firm size, and leverage among others, find that firms with available ESG performance information before going public experience higher underpricing because of a positive market response. They argue that visible ESG information can reduce information asymmetry and signal high firm quality to investors (Economidou et al., 2023, p. 170). They also find that firms with ESG ratings attract more sophisticated investors and demonstrate stronger post-IPO performance (Economidou et al., 2023, pp. 172-173), suggesting that ESG influences firm performance not only through internal operations but also through external market assessments. A similar empirical finding is presented by Fu et al. (2023, pp. 3-6), who study 1102 U.S. IPOs issued between 1999-2016, using ESG scores from MSCI ESG Research with a survival analysis controlling for firm size, profitability and leverage. They find that voluntary ESG disclosure reduces IPO failure risk, and also improves long-run performance, and that earlier disclosure is related with a greater likelihood of survival (Fu et al., 2023, p. 16). In addition, firms with higher ESG scores are less likely to fail, and this is mainly driven by the social and governance pillar scores rather than the environmental pillar (Fu et al., (2023, p. 16). These findings show that ESG disclosures and high ESG performance may reduce information asymmetry and improve investor confidence, particularly for young firms with higher uncertainty to survive long-term.

Further support of a positive relationship between sustainability and firm performance is in the study by Kumar & Firoz (2022, p. 60), using ESG scores from Bloomberg and in the regression model controlling for firm size, leverage, book-to-market value, and industry, who find that the ESG score is positively related to firm performance measured as ROA and ROCE, in their sample of 77 listed Indian firms during the period 2015-2019.

Their results also show that the environmental and governance pillars are positively associated with firm performance, while the social pillar is not statistically significant. This suggests that different individual ESG pillars contribute unevenly to firm performance. Kumar & Firoz (2022, p. 60) explain that these findings may be driven by improved stakeholder trust, enhanced corporate reputation, and more efficient governance structures, which together support stronger firm performance. Further research by Qian et al. (2024), constructing firm-level green commitment index, representing their own ESG rating to avoid the flaws in rating from LSEG Workspace and Bloomberg (Qian et al., 2024, p. 3) and in the regression model controlling for firm size, book-to-market value, beta among others, analyze a large sample of 5054 listed firms in China over the period 2015-2021, and find that ESG performance improves firm performance indirectly by reducing financial constraints. Their finding indicates that firms with stronger ESG profiles face fewer financing frictions, as they are perceived as more transparent and less risky by investors and financial institutions, which facilitates access to external capital and supports firm growth (Qian et al., 2024, p. 12). Complementing and supporting this, a study by Eliwa et al. (2021, p. 14) using a European plus Norway sample of 6018 firm-year observations over the period 2005-2016, using ESG scores from LSEG Workspace and regression model controlling for firm size, leverage, profitability, and firm risk, find that ESG performance is related with a lower cost of debt. Their result suggest that lenders incorporate ESG information into their risk assessments, leading to more favorable borrowing conditions for high ESG performance firms, which in turn improves firm performance through reduced financing costs rather than from just internal operational improvements (Eliwa et al., 2021, pp. 16-17).

Not all studies find a positive relationship between sustainability and firm performance, often explained by implementation costs, agency concerns, or short-term profitability pressures (Rahi et al., 2022, pp. 305-306; Shaikh, 2022, p. 232). In a study by Rahi et al. (2022) focusing on the Nordic region and only on the financial industry sector with 39 listed financial firms over the period 2015-2019, using ESG scores from LSEG Workspace and in the regression model controlling for firm size and leverage, find a negative relationship between ESG combined score and firm performance, while the individual governance pillar show a positive relationship with ROA as performance measure. Their result suggests that although certain aspects of the ESG pillar, particularly governance, may contribute positively to firm performance, the overall effect of ESG can be negative when the costs associated with sustainability initiatives outweigh the short-term financial benefits. Rahi et al. (2022, pp. 302-305) argue that sustainability investments often require long-term commitments, such as increased compliance, reporting, and operational adjustments, which may increase costs and reduce profitability in the short run. This effect may be particularly visible in financial firms, where efficiency and cost structures are closely related to performance outcomes, Furthermore their findings indicate that even in developed and sustainability oriented markets such as the Nordic region, ESG does not automatically improve financial performance in the financial industry sector, suggesting that the relationship is conditional and highly depend on the balance between long term benefits and short term costs.

Negative evidence is also found in broader international research. Shaikh (2022, p. 218) analyze over 500 firms across 17 countries over the period 2010-2018, using ESG scores from Bloomberg and employing regression model controlling for firm size, leverage, growth, and industry, and find that the combined ESG score is negatively related with firm performance measures, including ROA, and Tobin's Q. At the same time, they find

a positive relationship with the governance pillar and firm performance, while the environmental pillar and social pillar show a negative relationship. This further indicates that the overall ESG score may conceal variation across its separate individual pillars. Shaikh (2022, p. 234) suggests that governance practices, such as improved transparency and accountability, are more directly linked to firm performance, as they reduce agency problems and improve decision-making within the firm. In contrast, environmental and social initiatives may involve higher implementation costs, regulatory compliance, and long-term investments, which can reduce profitability in the short run. These findings suggest that aggregating ESG into a single measure may obscure important differences between the three separate pillars, and that the relationship between sustainability and firm performance is not constant but rather depends on which pillar is being considered and over what time horizon the effects are evaluated.

There is also previous research that finds evidence on the relationship that is mixed or insignificant rather than clearly positive or negative. Narula et al. (2024, p. 1170), using CRISIL ESG scores and regression model controlling for firm size, leverage and book-to-market value, study 220 Indian listed firms over the period 2018-2020 and find that the relationship between sustainability and firm performance differs across the individual ESG pillars. Their results show that the governance pillar is positively related to ROA, ROCE, and Tobin's Q, while the social pillar is not statistically significant for any of the three firm performance measures. In contrast, the environmental pillar shows a negative relationship with ROCE and Tobin's Q and no relationship with ROA. They argue that ESG related activities are not yet tangibly visible in overall firm performance when measured using ROA, ROCE, and Tobin's Q. Their result indicates that the relationship between ESG and firm performance may be weak or insignificant in certain contexts, particularly when sustainability practices are still developing or not fully integrated into firm strategy (Narula et al., 2024, pp. 1180-1181). Narula et al. (2024, pp. 1174-1176) argue that ESG effects may take time to materialize and depend on institutional conditions, such as regulatory frameworks and market maturity. Similarly, Saha & Khan (2024, pp. 809-810), using ESG scores from LSEG Workspace and regression model controlling for firm size, leverage, and firm age, analyze 136 firms in the Nordic region over the period 2012-2021 and find the relationship between sustainability and firm performance varies, when using measures as ESG combined score, ROE, ROA, and net profit margin, their country and industry level analysis show that the relationship varies across context. For ROA specifically, ESG is not significantly related to ROA in country-level test, including Denmark, Finland, Norway, and Sweden, and many industry-level test also show no significant relationship with ESG and ROA (Saha & Khan, 2022, pp. 805-807).

Additional mixed evidence on the relationship between sustainability and firm performance is supported by Kumar & Puthur Joseph (2025, pp. 733-735), who, using ESG scores from Bloomberg and regression model controlling for firm size, leverage and growth, analyses a global sample of 466 firms in the retail sector during 2022 and find that ESG is initially negatively related to firm performance, measured using Tobin's Q and ROA. This indicates that ESG effects are conditional rather than universal, and that firms with stronger internal capabilities may be better able to convert sustainability efforts into firm performance. Mugova et al. (2025), using ESG scores from LSEG Workspace and regression model controlling for firm size, leverage, growth, inflation among others on a sample of 74 listed firms in South Africa over the period 2010-2022, further show that ESG is negatively related to accounting-based performance measures such as ROA,

while being positively related to marked-based measures such as Tobin's Q. This further suggest that sustainability investments may reduce short-term profitability due to implementation costs, but at the same time enhance firm value through improved reputation and investor perception. In addition, Demers et al. (2021, pp. 433-434), using ESG scores from MSCI ESG Research and LSEG Workspace and employing regression model controlling for firm size, book-to-market value, and profitability among others, analyzing approximately 1600 listed firms in the U.S. during the COVID-19 pandemic in 2020, find that ESG does not significantly explain stock return resilience once firm characteristics are controlled for, indicating that ESG effects may also vary depending on macroeconomic conditions and periods of market uncertainty.

2.4.2. Hypothesis development

The theoretical framework and previous findings show that the relationship between sustainability and firm performance is ambiguous and inconsistent, and that both theoretical and previous empirical findings suggest different possible directions of the relationship. One part of the literature suggests that there is no significant relationship between sustainability and firm performance. From a theoretical perspective, shareholder theory suggests that sustainability investments may represent a misallocation of resources that do not generate firm performance in the short run (Danielson et al., 2008). This suggests that sustainability investments may increase costs without necessarily improving firm performance. This is empirically supported by studies that find no significant, weak, or non-replicable relationship between sustainability and firm performance in different contexts. For example, King (2025, p. 109) finds no support for the positive findings presented by Eccles et al. (2014), while Demers et al. (2021, pp. 433-434) find that ESG does not significantly explain stock return resilience once firm characteristics are controlled for. Previous findings also show insignificant relationships for individual ESG pillars and firm performance measures. Narula et al. (2024, pp. 1180-1181) find that the social pillar is not significantly related to ROA, ROCE, or Tobin's Q, while Kumar and Firoz (2022, p. 60) find that the social pillar is not significantly related to ROA and ROCE. In addition, Aydoğmuş et al. (2022, pp. 119-120) find that the environmental pillar is not significantly related to Tobin's Q, and Saha and Khan (2024, pp. 805-807) find that the combined ESG score is not significantly related to ROA in several country and industry level analyses. Based on this, the null hypothesis is formulated.

$H1_0$: There is no significant relationship between sustainability and firm performance.

Another part of the literature suggests that there is a positive relationship between sustainability and firm performance. Stakeholder theory predicts that firms managing and investing in their environmental, social, and governance responsibilities effectively can generate enhanced stakeholder trust, stronger reputation, and stronger long-term firm performance (Bridoux & Stoelhorst, 2022, pp. 797-809; Lahouel et al., 2022; Marcon Nora et al., 2023). In terms of legitimacy theory, firms can strengthen their societal acceptance through sustainability disclosure, and this can improve their reputation and reduce uncertainty among stakeholders, supporting firm performance through improved access to capital and reduced reputational risk (Olateju et al., 2021, pp. 2-4; Hummel & Schlick, 2016; Crossley et al, 2021, p. 3742). Signaling theory also supports a positive relationship, as ESG disclosures can reduce information asymmetry and signal firm quality to investors, improving investor confidence and access to external financing (Choudhury, 2024, pp. 98-99; Economidou et al., 2023, p. 170). This is empirically supported by studies finding a positive relationship between sustainability and firm

performance. Aydoğmuş et al. (2022, pp. 119-120) find that the combined ESG score is positively related to both ROA and Tobin's Q, and that the social and governance pillars are positively related to Tobin's Q. Tang et al. (2024, pp. 1-2, 7) also find that ESG and all three individual ESG pillars are positively related to Tobin's Q. Kumar and Firoz (2022, p. 60) find that the combined ESG score, environmental pillar, and governance pillar are positively related to ROA and ROCE. Narula et al. (2024, pp. 1180-1181) also find that the governance pillar is positively related to ROA, ROCE, and Tobin's Q. These findings suggest that sustainability may improve both accounting-based and market-based firm performance, although the positive relationship differs depending on the ESG dimension and the firm performance measure used. Based on these previous findings and theoretical support, the following hypothesis is formulated.

H1a: There is a significant positive relationship between sustainability and firm performance.

Another part of the literature suggests a negative relationship between sustainability and firm performance. Shareholder theory and trade-off theory suggest that sustainability investments involve real costs that may outweigh their short-term financial benefits, reducing accounting-based profitability in the short run (Abel, 2018, p. 95; Jahanzeb et al., 2013, pp. 11-13; Danielson et al., 2008; Rahi et al., 2022, pp. 302-305). This is empirically supported by studies finding negative relationships between sustainability and firm performance measures in different samples and contexts. Rahi et al. (2022, pp. 302-305) find a negative relationship between the combined ESG score and firm performance in Nordic financial firms, although the governance pillar is positively related to ROA. Shaikh (2022, p. 230) finds that the combined ESG score is negatively related to firm performance measures, including ROA and Tobin's Q, while the environmental and social pillars are also negatively related to firm performance. Narula et al. (2024, pp. 1180-1181) find that the environmental pillar is negatively related to ROCE and Tobin's Q. Further, Kumar and Puthur Joseph (2025, pp. 733-735) find that the combined ESG score is negatively related to ROA and Tobin's Q, and Mugova et al. (2025) find that ESG is negatively related to ROA. These findings suggest that sustainability investments may reduce firm performance when implementation costs, compliance costs, or short-term profitability pressures outweigh the financial benefits. Based on these findings from previous studies and theoretical position, the following hypothesis is formulated.

H1b: There is a significant negative relationship between sustainability and firm performance.

Several previous studies also report mixed findings, where the direction of the relationship depends on the ESG score examined, the performance measure used, or firm-specific characteristics (Kumar & Puthur Joseph, 2025, pp. 733-735; Mugova et al., 2025; Saha & Khan, 2024, pp. 809-810). A clear pattern across the previous findings is that the individual ESG pillars contribute differently to firm performance, and that the combined ESG score may conceal important variation across its components. Governance is most consistently found to be positively related to firm performance across previous studies, including ROA, ROCE, and Tobin's Q in Narula et al. (2024), ROA in Rahi et al. (2022), and Tobin's Q in Aydoğmuş et al. (2022) and Tang et al. (2024). This can be explained by the role of governance in reducing agency costs, improving transparency, and enhancing decision-making within the firm. The environmental pillar is more inconclusive, with some studies finding a positive relationship with ROA and ROCE

(Kumar & Firoz, 2022, p. 60), while others find no significant relationship with Tobin's Q (Aydoğmuş et al., 2022, pp. 119-120) or a negative relationship with ROCE and Tobin's Q (Narula et al., 2024, pp. 1180-1181). The social pillar also shows mixed evidence, with some studies reporting a positive relationship with Tobin's Q (Aydoğmuş et al., 2022, pp. 119-120; Tang et al., 2024, p. 7), while other studies find no significant relationship with ROA, ROCE, and Tobin's Q (Narula et al., 2024, pp. 1180-1181), or with ROA and ROCE (Kumar & Firoz, 2022, p. 60). These previous findings suggest that the combined ESG score masks individual pillar level effects, and that studying both the combined ESG score and the separate individual environmental (E) score, social (S) score, and governance (G) score is necessary to capture more of the relationship between sustainability and firm performance. The three hypotheses are therefore formulated to address RQ1 and tested across four dimensions of sustainability, the combined ESG score and the individual E score, S score, and G score, and firm performance across three measures, ROA, ROCE, and Tobin's Q, reflecting operating, financial, and market dimensions of firm performance, respectively.

Regarding RQ2, the investigation focuses on whether the relationship between sustainability and firm performance differs between established and newly listed firms. Through the perspective of agency theory, newly listed firms may have shorter public track records and less developed governance structures, which can increase agency costs and make sustainability, especially governance sustainability, more important for aligning managers and shareholders (Economidou et al., 2023, p. 132; Fu et al., p. 16). Through the lens of signaling theory, newly listed firms can face higher information asymmetry, making ESG disclosures and sustainability performance potentially more valuable for signals of firm quality to external investors (Choudhury, 2024, pp. 98-99; Economidou et al., 2023, p. 170).

First, it is possible that there is no significant difference between established and newly listed firms. In a developed and high-performance sustainability market as the Nordic region (Saha & Khan, 2024, p. 796), sustainability reporting and stakeholder expectations may be relatively mature among listed firms. If sustainability is already expected by investors, customers, and regulators, the relationship between sustainability and firm performance may not differ significantly between established and newly listed. This could mean that sustainability may function as a general firm characteristic rather than a signal that has clearly different potential value for newly listed firms. Based on this, the null hypothesis is formulated.

$H2_0$: There is no significant difference in the relationship between sustainability and firm performance between established and newly listed firms.

However, there are also arguments suggesting that the relationship may be more positive for newly listed firms. Through the lens of signaling theory, firms with higher information asymmetry have stronger incentives to use credible signals to communicate firm quality to external investors (Choudhury, 2024). For newly listed firms, as they usually have shorter public track record than established firms. In this context, sustainability performance may function as a valuable signal that reduces investor uncertainty and strengthens confidence in the firm. As in the findings by Economidou et al. (2023, pp. 172-173), that firms with visible ESG information before going public receive a positive market response, attract more sophisticated investors, and show stronger post IPO performance. Similarly, Fu et al. (2023, p. 16) find that voluntary ESG disclosure reduces

IPO failure risk and improves long-run performance, and that higher ESG scores reduce IPO failure risk, mainly through the social and governance pillars. In addition, Qian et al. (2024, p. 12) show that ESG engagement can reduce financial constraints, which may be particularly important for newly listed firms that may depend on external financing. Based on this, the following hypothesis is formulated.

H2a: There is a significant positive difference in the relationship between sustainability and firm performance between established and newly listed firms.

At the same time, there are also arguments suggesting that the relationship may be weaker or more negative for newly listed firms. Sustainability investments require financial resources, management attention, and long-term commitments. This may be more costly for newly listed firms, which often face higher growth pressure, more limited recourses, and greater uncertainty after becoming public. Narula et al. (2024, pp. 1174-1176) suggest that ESG effects may take time to materialize and depend on institutional conditions, which means that newly listed firms may not immediately convert sustainability engagement into firm performance. Fu et al. (2023, p. 4) also argues that ESG initiation and disclosure may impose significant costs on IPO firms and create agency problems. In addition, Teoh et al. (1998) show that IPO firms with high discretionary accruals in the IPO year experience weaker stock performance three years after going public, suggesting that strong signals around IPOs may sometimes reflect opportunistic reporting rather than long-term quality. Although Teoh et al. (1998) study earnings management and not ESG, the argument could support the broader idea that firms that may manipulate or have incentives to pump up their ESG rating could potentially experience lower returns afterwards. If ESG is used more as impression management than as a reflection of real sustainability quality, the relationship between sustainability and firm performance may be more negative for newly listed firms than for established firms. Based on this, the following hypothesis is formulated.

H2b: There is a significant negative difference in the relationship between sustainability and firm performance between established and newly listed firms.

The three hypothesis are tested across the four dimension of sustainability, the combined ESG score, and the individual E score, S score, and G score, and firm performance across ROA, ROCE, and Tobin's Q, reflecting operating, financial, and market performance, respectively.

3. Methodology

3.1. Philosophy

Beliefs and systems have existed through history, yet conceptions of reality change constantly. Because of this, new ways to conduct research are constantly emerging, and old ways are left behind. Historically, positivism was the only research paradigm, until interpretivism was discovered. Positivism focuses on occurrences that are observable, and the paradigm holds that reality is singular and can be explained through statistics. The paradigm in conducting research is called natural sciences, to be separated from social sciences (Collis & Hussey, 2014, p. 43). There are two main paradigms that exist today, positivism and interpretivism, where the latter was developed when criticism to positivism emerged. The interpretivism paradigm focuses on the opposite of positivism and focuses on idealism and holds that reality is subjective and different for every mind (Collis & Hussey, 2014, p. 43 & 44).

The core of positivism lies in the idea that knowledge should be gathered based on natural occurrences and relationships are explained by statistics. Positivism also holds that all factual knowledge is based on information that has been obtained through observations to discover new theories (Collis & Hussey, 2014, p. 44). Moreover, the positivism paradigm holds that the authors should be separate from the research, which is done to not affect the results and therefore minimize biased results. Since the paradigm holds that reality is singular, researchers who observe a phenomena cannot be a part of that reality. Furthermore, positivist research possesses an objective approach to knowledge. According to Collis & Hussey (2014, p. 45), the approach requires researchers to formulate arguments by designing experiments and gathering empirical data. The investigators can then interpret the findings through statistical analysis to minimize bias. Finally, as the outcome is sourced from verifiable facts, the results can be seen as highly reliable and objective.

As mentioned above, interpretivism can be explained as the opposite of positivism. Instead of seeing reality as singular, the interpretivism paradigm sees reality as subjective because reality is shaped by the perceptions of it. Due to this view, the interpretivism paradigm assumes that there are unlimited amounts of realities, and each of them is different in each mind. Therefore, researchers that conduct research under the interpretivism paradigm strive to discover, describe, and translate how the social world looks for the participant. Contrary to positivism, this paradigm focuses on subjective methods to gather information. Interviews are therefore most used, and they are used to gather a full understanding of how reality is to the participant (Collis & Hussey, 2014, p. 45).

Some argue that the positivist paradigm is overly harsh and overlooks important aspects. Collis and Hussey (2014, p. 45) argue that researchers using the positivist paradigm cannot maintain an objective stance because they bring their own values into the research. They furthermore mention that a complete understanding of human behavior is impossible without investigating how individuals personally perceive their own actions. This study has adopted the positivist paradigm in conducting the research. In the financial world, returns and firm performance are often separated to someone's ideas and therefore not affected by their perception. The quantitative approach will provide the researchers of this study clear and objective results which will be used to either accept or reject the hypothesis that was presented previously. Moreover, the positivist paradigm aligns better

with this research rather than the interpretivist paradigm. The ESG scores, and the performance of the firms are presented in numerical form, and the social realities of individuals cannot affect the performance of firms. The interpretivist paradigm would be more beneficial to research that aims to understand thoughts about a matter, not numerical performance that is objective and separate.

3.1.1. Ontology

Ontology describes the nature of being, which aims to describe existence and the general idea. It influences the research method, but also the meaning of research (Nasution, 2018). There are two types of ontologies, subjective and objective, and the two have very different perceptions of reality. On the one hand, the positivist paradigm allows for the exploration and discovery of only one reality. As a result, everyone shares the same perception of reality, and research is conducted outside of the researcher's control. Therefore, the researcher cannot influence reality (Collis & Hussey, 2014, p. 47). Contrary to this, the interpretivism paradigm views reality as a collection of subjective realities. These subjective realities are socially constructed, so there are multiple realities, and it also implies that researchers can shape reality (Collis & Hussey, 2014, p. 46). This study has employed the positivist paradigm, which separates the researchers from reality.

3.1.2. Epistemology

Epistemology refers to what the researchers regard as valid knowledge. Collis and Hussey (2014, p. 47) describe the term as a strive to understand how researchers acquire knowledge and provide evidence to support their beliefs about knowledge. Positivists identify knowledge as observable, and measurable phenomena that shapes their perspective on reality. They therefore try to maintain a distance from the research to be as separate and objective as possible (Collis & Hussey, 2014, p. 47). According to the positivist paradigm, knowledge is based on "positive" information, which is supported by logic or mathematics. To prevent bias and deviations in quantitative research, a large sample size is recommended (Collis & Hussey, 2014, pp. 44-50). The researchers of this study share this view of what is regarded as valid knowledge.

Interpretivism focuses on the subjective meaning of human experiences and their social context. However, positivism seeks objective truths through empirical observation and quantitative analysis and interpretivism seeks to understand the complexities of lived experience through the perspectives of those who have experienced it (Collis & Hussey, 2014, p. 47). The interpretivist paradigm emphasizes the importance of understanding participants' perspectives, behaviors, and interactions within their unique social and cultural contexts to fully comprehend social phenomena. According to Hiller (2016, p. 103), qualitative approaches, such as interviews, participant observation, and textual analysis, enable a thorough examination of individual’s meanings, experiences, and perspectives on reality.

3.1.3. Axiology

According to Saunders et al (2023, p. 135), axiology refers to the role of ethics and values in the research process. One important factor that researchers will encounter during their research is the extent to which their own values influence the study. Researcher’s values influence how information is perceived and what is considered knowledge. Positivists believe that the objects being studied are not affected by the researcher's presence and will therefore remain visible even after the study is completed. The interpretivist assumptions contradict this assumption. The assumptions that interpretivism holds

suggest that the researcher's values are linked to the research topic and therefore believe that researchers are deeply engaged in their work (Collis & Hussey, 2014, p. 48).

As previously mentioned, this research has adopted a positivist paradigm. The researchers are assumed to be independent from the research. The author's values and ethics will therefore not influence the study. Furthermore, the research findings will be objectively evaluated and analyzed for the intended audience.

3.2. Research approach

There are two approaches to research. A deductive approach is used for quantitative research, where a positivist paradigm is developed, and an inductive approach is used for qualitative research where an interpretivist paradigm is developed. The deductive approach involves developing a theory and testing it against empirical data (Collis & Hussey, 2014, p. 44-47). Because this study uses a positivist paradigm, the authors have adopted the deductive approach. Saunders et al (2023, p. 154) mention that the deductive approach reasons from when the conclusion is derived logically from a set of theory-driven premises. Furthermore, the deductive approach starts with a theory which is developed from reading academic literature and the researchers then develop a design to test the theory. The process starts with presenting an idea or hypothesis to form a theory and decide under which conditions the theory holds by using existing literature to develop a proposition. The next step is testing the propositions by collecting empirical data and if the results are aligned with the premises, then the theory is correct. If they do not align means that the theory is incorrect and must be rejected (Saunders et al 2023, p. 155-156). In contrast, the inductive approach would start by gathering empirical data, analyze the findings, and formulate a theory based on the analysis. Under the inductive approach, theory follows data rather than vice versa as in the deduction approach (Saunders et al 2023, p. 157).

In this study, the authors begin with researching existing academic literature and use it to develop hypotheses. The identified hypotheses will then be tested against empirical data which will allow the researchers to either accept or reject the hypotheses, and the findings will be connected to existing theories rather than used to develop a theory. Unlike the inductive approach, the deductive approach is the standard for quantitative research because it is typically not attempting to develop a completely new theory.

3.3. Research method

Barnham (2015, p. 838) argues about the differences between a quantitative and qualitative study. The author argues that quantitative research aims to represent or present a phenomenon, while qualitative research aims to understand the “why” behind it. It is vital for researchers to understand what differentiates the two philosophies to select the strategy that best matches their aims. In quantitative research, it is important to adopt a well-ordered set of criteria of scope, objectivity, goal, data, and method. It concentrates on measurable variables and quantifiable data to objectively analyze a phenomenon. Furthermore, it aims to test hypotheses and to generalize the findings which consist of a large sample that employs numerical data. It offers a structured and objective framework to present a phenomenon in a larger picture, while a qualitative method offers context-sensitive exploration (Lim, 2025, p. 325-348; Collis & Hussey, 2014, p. 5).

In contrast to a quantitative study, qualitative studies focus on gathering deep insights and understanding of a complex phenomenon. Empirical data here are often non-numerical

because it aims to discover human experiences and perspective that are often hard to translate into numbers, data is therefore often gathered through interviews (Lim, 2025, pp. 199-229). Lim (2025, pp. 199-229) further argues that findings are not generalized under this philosophy, rather, it wants to keep findings original which explains the participant's "why" and "what". It is therefore common to have an unstructured approach in interviews to truly grasp and understand what the participant communicates. The authors decided to employ a quantitative method due to the nature of this research. There is currently enough quantitative information about firms for this study to be conducted. Moreover, a quantitative method is more appropriate than a qualitative due to the statistical analysis that is required.

3.4. Literature search and source criticism

Conducting a critical literature search provides context for this study by analyzing previous research. This offers background information and justifications of the relevance of the chosen topic by identifying research gaps and adding new perspectives to existing knowledge (Saunders et al., 2023, p. 78). This approach guided the authors of this study to identify and select relevant theories, and reviewing previous literature provided a research gap that highlighted previous findings.

Given this study's positivist and deductive orientation, it relied entirely on secondary sources. The literature search focused on identifying well-established theories and previous empirical findings related to the relationship between sustainability and firm performance. Peer-reviewed scientific articles were retrieved primarily through the Umeå University Library's databases and from Business Source Ultimate through EBSCO. Google Scholar was used as a supplementary search tool to identify highly cited and relevant articles across a wider range of sources. In some cases, non-peer reviewed articles were included. These articles were however cited in credible peer-reviewed articles which indicated their academic relevance and credibility. Also, to ensure the credibility and academic quality of the scientific articles used in this study, the journals in which the articles were published were checked against recognized journal ranking lists. The journals were reviewed in relation to the ABS Academic Journal Guide and the ABDC Journal Quality List. These lists were used as indicators of whether the journals were established and reliable. If a journal was not included in either of the lists, the source was evaluated more carefully to ensure that it was academically credible and that it was cited in other peer reviewed articles, and that it was not a predatory journal. Keywords were included in the literature search, which were used to find and trace established theories and previous empirical findings, such as corporate sustainability, ESG, firm performance, stakeholder theory, legitimacy theory, shareholder theory, trade-off theory, agency theory, signaling theory and information asymmetry among others. Furthermore, the study includes a few non-scientific articles which were primarily used in the Introduction chapter. The Methodology chapter utilized course books provided by the International Business Program at Umeå School of Business and Economics.

3.5. Research design

Research design explains the process to answering the research questions, achieving the research aim and meeting the objectives. Saunders et al (2023, p.178) mentions four different research designs which consist of exploratory, descriptive, explanatory, and evaluative research but that there can also be a combination of these. Collis & Hussey (2014, p. 5) further propose predictive research as a fifth design.

Exploratory research is designed to clarify understanding about a specific topic, and research questions under exploratory research typically start with a “how” or “why”. The advantage of this design is that it is adaptable to change, and researchers must therefore be willing to change direction if new data appears. A descriptive study aims to gain further knowledge and a more accurate profile of events and can therefore also extend an exploratory study. Moreover, explanatory research establishes relationships between variables and aims to explore a situation to understand and explain the relationships. Lastly, an evaluative study explains how well something works. Evaluative studies that are done in business or management typically aim to explain how well a specific strategy or policy works (Saunders et al, 2023, p. 178-181). Collis & Hussey (2014, p. 5) further mention predictive research as another design, which aims to predict future events by analyzing current events. This study is classified as an explanatory research based on the descriptions above. To answer the research questions, the authors of this study will collect quantitative data. Furthermore, the researchers will examine the area to see if there is any statistical significance between relevant variables that can describe the relationship between sustainability and firm performance.

3.5.1. Data collection and sample design

According to Aguinis et al. (2021, p. 679), data collection and preparation are foundational stages in empirical research. This is because these stages take place before the production of results and analysis. Therefore, the data used in a study must be suitable for the purpose of the study and handled in a transparent way. Aguinis et al. (2021, p. 679) further emphasize that methodological transparency is important because readers need to understand the decisions made during the research process in order to assess the reproducibility and trustworthiness of the study. In general, two types of data can be used in research, primary and secondary data. Primary data refers to data collected directly by the researcher for the specific purpose of the study, while secondary data refers to data that have already been collected and made available for other purposes but that can be reused and analyzed in relations to new purposes. Saunders et al. (2023, p. 378) define secondary data as data that were originally collected for another purpose and later processed or stored. Saunders et al. (2023, p. 345) also explain that secondary data may include both quantitative and qualitative data and can be used in descriptive and explanatory research.

This study uses secondary quantitative data because the purpose of this thesis is to investigate and analyze the relationship between sustainability and firm performance among listed firms in the Nordic region, and to examine whether this relationship differs between established and newly listed firms. Secondary data was considered the most appropriate data because sustainability information in form of ESG scores and financial information are already available through established databases. Saunders et al. (2023, p. 378) argue that secondary data can save both time and resources compared with collecting primary data, especially when the data can be downloaded and made compatible with statistical analysis software. However, secondary data also possess limitations. Since the data were originally collected for another purpose, they may not fully match the needs of a study’s purpose. Saunders et al. (2023, p. 378) explains that researchers must evaluate whether secondary data are suitable for the specific purpose and research questions, including issues such as measurement validity, coverage, reliability, and potential bias. For this reason, the secondary data collected in this study were carefully evaluated before being included in the final dataset. By applying appropriate statistical methods, secondary data were processed and adopted to the purpose of this study.

In line with Aguinis et al. (2021, pp. 681-683), the sample design is described transparently. All secondary data has been collected from the database LSEG Workspace. The database LSEG Workspace was chosen because of the accessible license provided by Umeå University and because the database provides reliable and comprehensive financial market data and that it has also been used in previous studies (Aydoğmuş et al., 2022, p. 122; Saha & Khan, 2024, p. 799). The construction of the sample composition is motivated by the structure of available data and by the research questions of this study. The following parameters were used in the sample selection process: first, public listed firms. Second, country of exchange: Sweden, Denmark, Finland, Norway and Iceland. Third, country of headquarters: Sweden, Denmark, Finland, Norway and Iceland. Fourth, all available industries. The starting point is a panel of 1513 Nordic listed firms with 24496 firm-year observations between 2010-2025. The choice of year 2010 as the starting point of the observations is motivated by not including the 2007-2009 financial crisis in the sample, as previous research show that the financial crisis strongly affected firms performance conditions and also investment behavior (Campello et al., 2010), and because of the low ESG data availability from LSEG Workspace prior to year 2010 (De la Fuente & Velasco, 2024, p. 4). The observation window extends through 2025 in order to include all firm-year observations available at the time of data extraction from LSEG Workspace. Although data coverage for 2025 was incomplete, as many firms had not yet published their annual financial statements at the time of extracting the data in late April 2026, including the partial 2025 observations preserves all available observations for the firms that had reported.

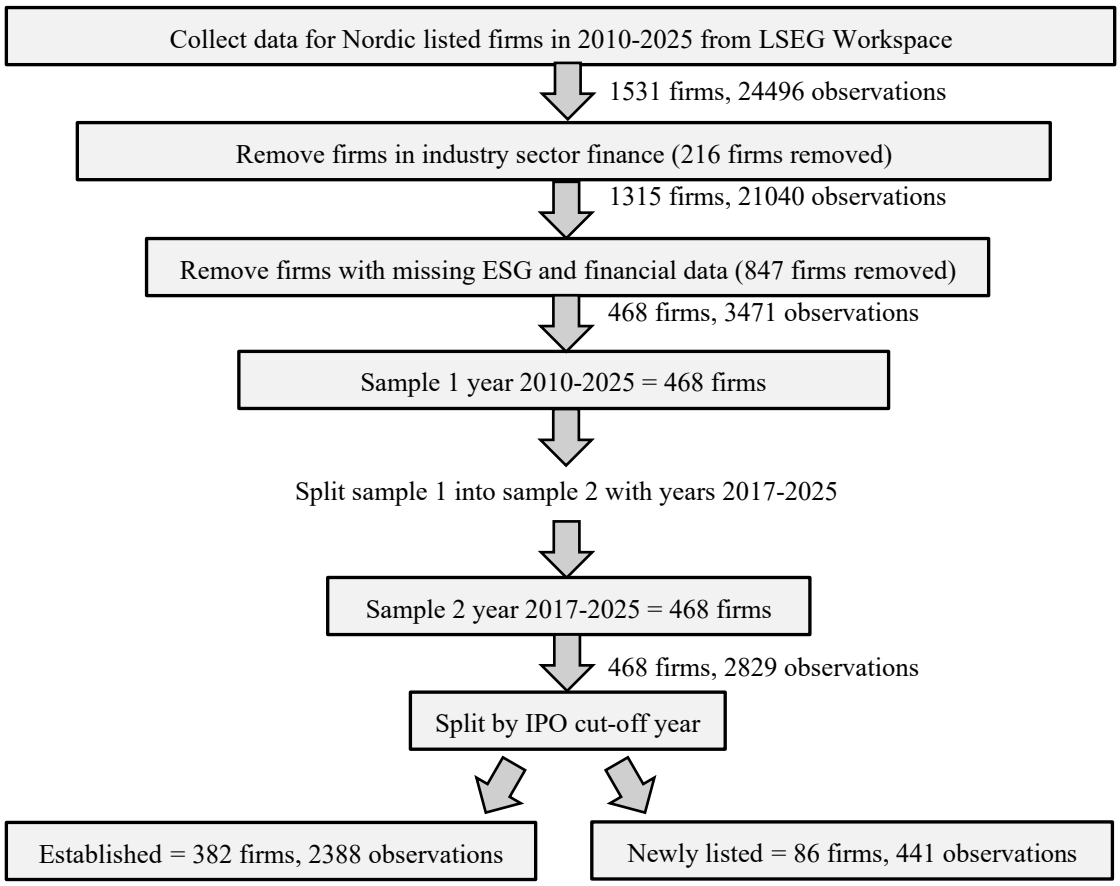


Figure 1. Data sample collection process

Following the data sample collection process in Figure 1, firms in the financial industry sector was removed, motivated by the sector-specific nature of financial firms, including differences in reporting structures, leverage, regulation, and performance measurement (Economidou et al., 2023, p. 138). This is also in line with previous studies that acknowledges the different characteristics in financial firms and either separates them or excluding them when exploring the relationship between sustainability and firm performance (Narula et al., 2024; Rahi et al., 2022). This results in a panel of 1315 firms with 21040 observations. Further, only firms with at least one non-missing ESG score during at least one year in the period 2010-2025 and with necessary financial data are included in the final sample composition, resulting in 468 firms, presented in Appendix 1, with 3471 observations. Sample 1 was used to address RQ1: Is there a relationship between sustainability and firm performance among listed firms in the Nordic region. Sample 1 was then split into sample 2, that restricts sample 1 to the period 2017-2025, containing 2829 observations, of which 86 firms had their IPO during 2017-2024. Sample 2 was used to address RQ2: Is there a difference in the relationship between sustainability and firm performance between established and newly listed firms. The reduction in sample size from 1315 firms in the panel to 468 firms in sample 1 reflects the limited coverage of ESG data among Nordic listed firms. The motivation for this restriction is the need of ESG score availability, and observations without ESG data cannot contribute to the estimation of the relationship between sustainability and firm performance. Figure 2 provides direct support for this restriction.

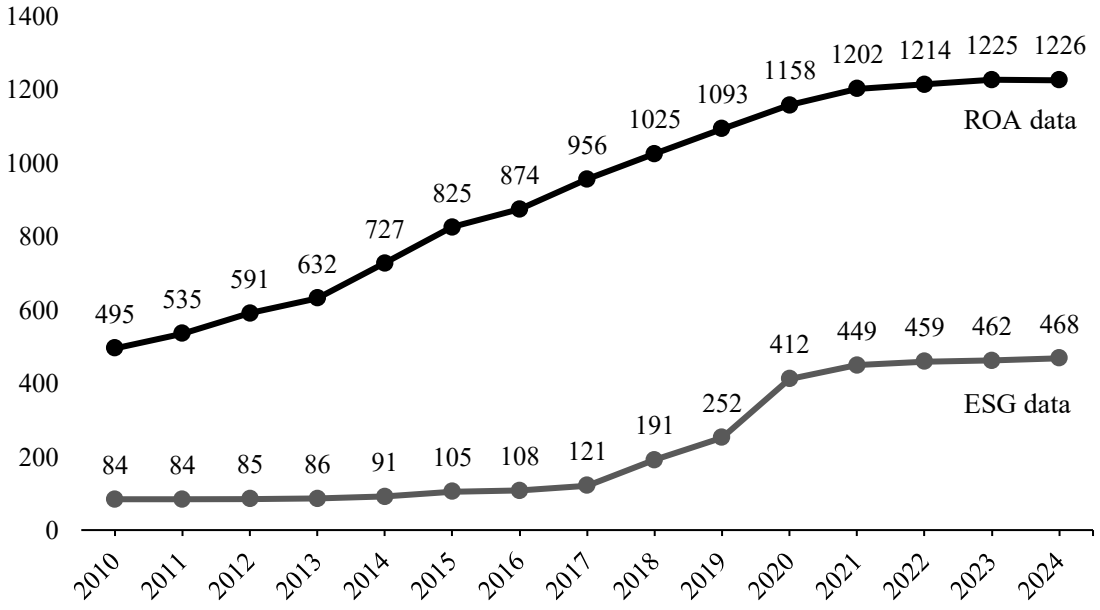


Figure 2. ESG data coverage over time

Figure 2 shows how ESG data coverage in the sample of Nordic listed firms has developed over the period 2010-2024, both in absolute terms and relative to the listed firms with available ROA data. Year 2025 is not included in the figure, as many firms had not yet published their annual financial statements at the time of extracting the data. Despite the growth in ESG reporting after 2017, a substantial gap between the two lines persist throughout the sample period, with only 84 out of 495 firms with available ROA data and ESG score in year 2010, and in year 2024 only 468 firms out of 1226 firms with available data. This gap may reflect the tendency of ESG rating providers to systematically favor larger, more established and more visible firms (Drempetic et al.,

2020, p. 335). Also, with the reporting directive NFRD, which from 2017 mandated ESG disclosure for large public firms with more than 500 employees, leaving smaller listed firms outside the reporting requirement. This could together explain why a substantial share of listed firms lack ESG data throughout the sample period.

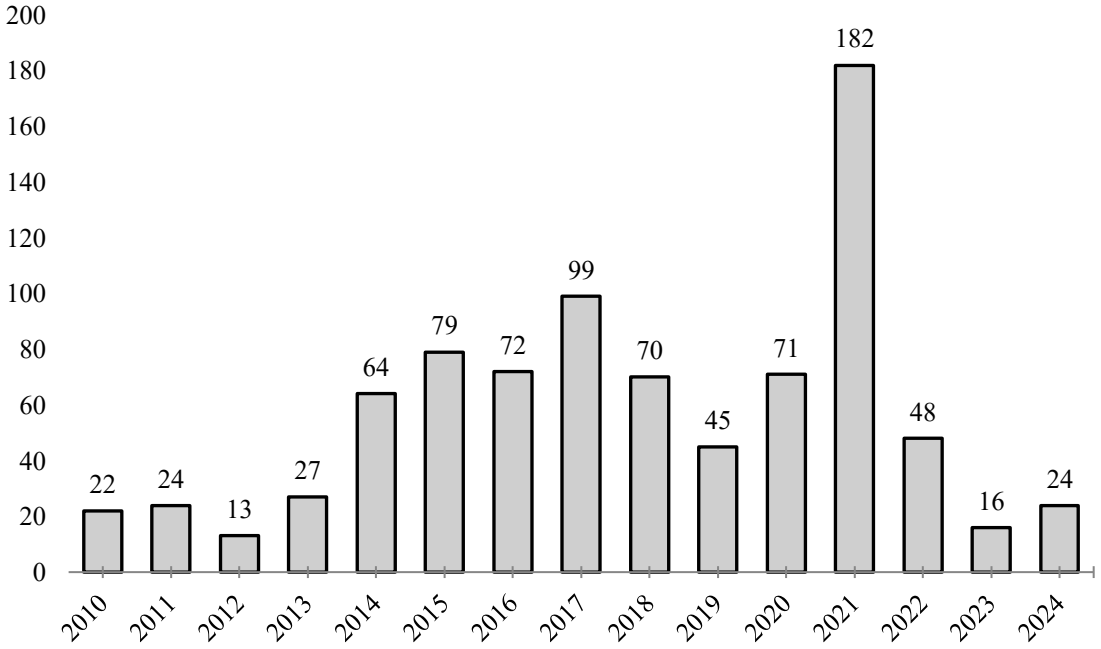


Figure 3. IPO activity in the sample

Figure 3 motivates the choice of 2017 as the starting year of sample 2. Year 2025 is not included in the figure as IPO data for that year was not yet fully available at the time of data collection. The figure shows that IPO activity in the Nordic region was low between 2010-2013, with fewer than 30 listings per year, before increasing from 2014 and onwards. A first peak of 99 IPOs is observed in 2017, followed by a temporary slowdown in 2018-2019, a recovery in 2020 with 71 IPOs, and a record 182 IPOs in 2021. After 2021, listing activity declines sharply to 48 IPOs in 2022, 16 in 2023, and 24 in 2024. The 2017-2021 IPO period coincides with a period of historically low interest rates, elevated equity valuations, and strong investor demand (PwC, 2022), and the first peak of IPO activity in 2017 motivates the comparison between established and newly listed firms in RQ2 and supports the use of 2017 as the cut-off year for sample 2. Sample 2 contains the same 468 firms as sample 1, observed during the period 2017-2025, and is divided into two subgroups based on each firm's IPO year. The first subgroup consists of 382 established firms with 2388 observations, defined as firms that conducted their IPO before 2017 and were therefore already listed at the start of the sample 2 period. The second subgroup consists of 86 newly listed firms with 441 observations, defined as firms that conducted their IPO in 2017 or later. The unbalanced split in observations between the two groups reflects the fact that established firms can in principle contribute up to nine firm-year observations (2017-2025), while a firm that listed in, for example 2021 can contribute at most five firm-year observations. This difference is acknowledged and considered in the empirical specification setup.

In the panel of 1315 firms after firms in industry sector finance was removed (shown in Figure 1), a total of 555 firms conducted their IPO in 2017 and onwards. And out of these 555 firms, 469 firms lack ESG data, resulting in only 86 firms with available ESG data.

This pattern is consistent with the discussion above, as ESG rating providers tend to favor larger, more established, and more visible firms (Drempetic et al., 2020, p. 335), and smaller firms are less likely to be covered by ESG data in the years immediately following their IPO. As a consequence, the 86 newly listed firms included in the newly listed subgroup of sample 2 are likely to be larger and more visible than the average newly listed Nordic firm during the period 2017-2025. This introduces a potential selection bias toward larger newly listed firms within the subgroup. This is acknowledged when interpreting the results of RQ2.

Table 1. Country composition in the sample

Country	Firms	Observations
Sweden	271	1886
Finland	70	575
Norway	64	490
Denmark	53	484
Iceland	10	36
Total	468	3471

The geographic distribution of the 468 firms included in both sample 1 and sample 2 is presented in Table 1. The 468 firms are distributed across the five Nordic countries, with Sweden accounting for the largest share of the sample with 271 firms and 1886 observations, followed by Finland with 70 firms and 575 observations, Norway with 64 firms and 490 observations, Denmark with 53 firms and 484 observations, and Iceland with 10 firms and 36 observations. Observations refer to the firm-year observations in sample 1 with ROA. The dominance of Swedish firms reflects the size of Nasdaq Stockholm, which is the largest Nordic exchange by both market capitalization and number of listed firms. The relatively limited representation of Icelandic firms is consistent with the size of the Icelandic equity market. However, the presence of all five Nordic countries supports the generalizability of the results to the Nordic region as a whole. By listing exchange, the 468 firms are distributed across Nasdaq Stockholm (271 firms), Nasdaq Helsinki (69), Oslo Børs (66), Nasdaq Copenhagen (52), Nasdaq Iceland (9), and Nordic Growth Market (1). The listing venue distribution differs slightly from the headquarters distribution in Table 1 because a small number of firms are headquartered in one Nordic country but listed on another Nordic exchange. The firms recorded under Nasdaq Stockholm include firms admitted to the Nasdaq First North Growth Market, which the exchange data does not separate from the main market.

Figure 4 presents the industry composition of the 468 firms included in sample 1 and 2. Industries are classified to the Thomson Reuters Business Classification (TRBC) system which is the most comprehensive, detailed, and up-to-date business sector and industry classification from LSEG in LSEG Workspace database (LSEG, n.d.). The sample is dominated by industries of industrials with 112 firms, technology with 89 firms, healthcare with 71 firms, and consumer cyclicals with 66 firms. Followed by real estate with 35 firms, basic materials with 35 firms, consumer non-cyclicals with 30 firms, energy with 20 firms, utilities with 8 firms, and academic and educational services consisting of 2 firms. These Industry sectors are included and in line with previous studies that include these industries (Narula et al., 2024, p. 47). Real estate was considered to be dropped, because of the similarities with financial industry sector characteristics, but considering many previous studies including the real estate sector in their research (Kumar & Firoz, 2022, p. 47; Narula et al., 2024, p. 1175), it was motivated to be kept.

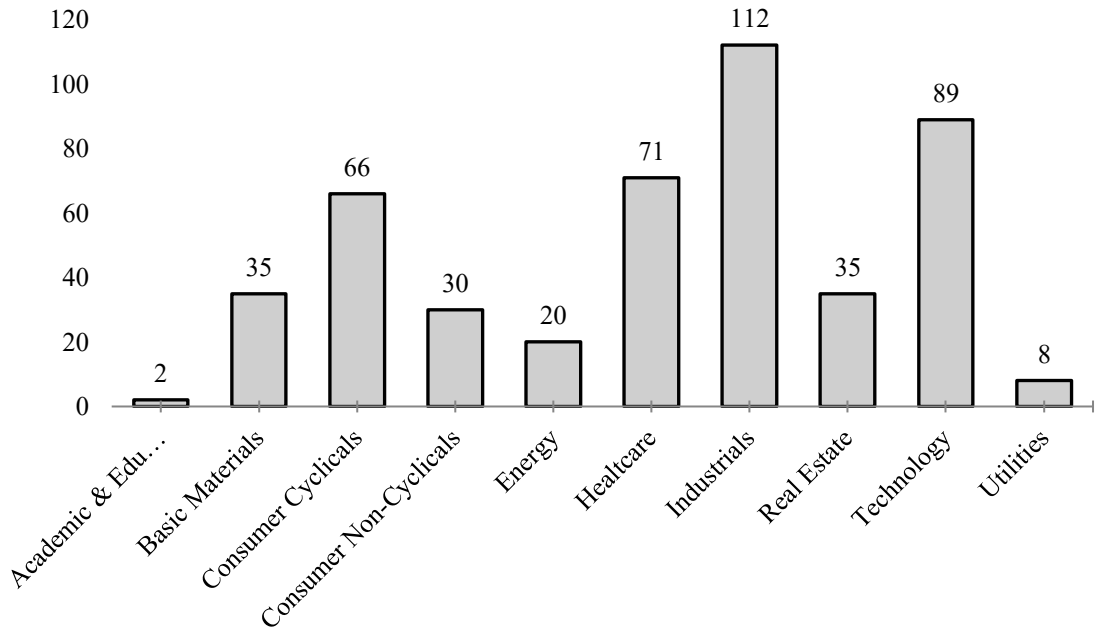


Figure 4. Industry composition in the sample

3.5.2. Data analysis method

The data used in this study consist of panel data, which refers to multiple observations of the same variables observed at different time periods, combining cross-sectional and time-series dimensions (Stock & Watson, 2015, pp. 347-348). Given that this study is based on observations of multiple Nordic listed firms over the period 2010-2025, panel data is considered an appropriate choice. A panel can be either a balanced panel, where every firm is observed in every time period, or an unbalanced panel, where the number of observations differs across firms because not all firms are observed in every year (Stock & Watson, 2015, p. 348). The panel in this study is unbalanced, meaning that the number of observations varies across firms because not all firms appear in every year of the sample. As firms enter the dataset in different years depending on when a firm conducted their IPO and on when ESG reporting became available for each firm. The unbalanced panel in this study reflects the Nordic listed firms that conducted their IPO in different years throughout the sample period, meaning that not all firms were listed and available for observation across the time period 2010-2025. Unbalanced panels are common in firm-level studies and do not consider posing a methodological concern, as standard regression techniques accounts for the variation in the number of observations per firm (Stock & Watson, 2015, pp. 348-349).

To investigate and analyze the relationship between sustainability and firm performance among listed firms in the Nordic region, and to examine whether this relationship differs between established and newly listed firms, a multiple linear regression analysis was used. This statistical method is well suited for quantifying the simultaneous influence of several independent variables on a single dependent variable (Chatterjee & Hadi, 2012, pp. 57-59) and is consistent with the methodological approach used in previous studies on the relationship between sustainability and firm performance (Aydoğmuş et al., 2022; Mugova et al., 2025; Narula et al., 2024).

3.5.3. Operationalization of variables

The financial and sustainability metrics obtained from LSEG Workspace were operationalized into representative variables that formed the basis for the empirical

analysis. Each variable was constructed in line with definitions established in previous studies on the relationship between sustainability and firm performance, ensuring comparability with previous studies and consistency with accounting and financial measures. The firm performance measures were defined as the dependent variables, while sustainability operationalized through ESG were defined as the independent variables that are used to explain or predict firm performance. Three different proxies for firm performance are employed. ROA is set to capture operating performance, and ROCE is set to capture financial performance, while Tobin's Q captures market performance. The use of multiple firm performance measures is set out to provide a more comprehensive view into the relationship between sustainability and firm performance. These operationalization of variables is in line with previous studies (Kumar & Firoz, 2022, pp. 50-51; Narula et al., 2024, p. 1177).

Control variables are an essential component of multiple linear regression analysis, as they allow the researcher to isolate the effect of the main independent variable on the dependent variable by holding other relevant factors constant (Stock & Watson, 2015, p. 196). Without appropriate control variables, the estimated relationship between the independent and dependent variables may be biased due to the influence of omitted variables that affect both the independent and dependent variables that can lead to biased coefficient estimates and misleading inference (Chatterjee & Hadi, 2012, pp. 169-172). Three control variables are included to ensure that any observed relationship between sustainability and firm performance is not driven by other firm-level characteristics. The control variables included were based on their established use in previous studies on the relationship between sustainability and firm performance (Aydoğmuş et al., 2022; Kumar & Firoz, 2022; Narula et al., 2024).

3.5.4. Dependent variables

The dependent variables used in this study represent firm performance. Since firm performance can be measured in different ways, three different variables were included to capture different dimensions of firm performance. This approach is in line with previous studies that have studied the relationship between sustainability and firm performance, where multiple firm performance measures have been used to capture both accounting-based and market-based outcomes (Aydoğmuş et al., 2022, pp. 123-125; Kumar & Firoz, 2022, p. 60; Narula et al., 2024, pp. 1175-1176).

Return on assets (ROA) was used as the first dependent variable and represents operating performance. ROA measures how efficiently a firm uses its total assets to generate profits, as it relates net income to the firm's total assets. A higher ROA indicates that the firm generated more profit from its available assets and can be interpreted as stronger operating performance. ROA was relevant to include because previous studies on the relationship between sustainability and firm performance commonly used it as a measurement of firm performance (Aydoğmuş et al., 2022, pp. 123-125; Kumar & Firoz, 2022, p. 60; Saha & Khan, 2024, p. 795). These studies show that ROA is a relevant measure because it captures the accounting-based and operational dimension of firm performance, which may be affected by ESG engagement through both the costs of implementing sustainable practices and the potential benefits generated in efficiency gains.

$$ROA = \frac{Net\ income}{Total\ assets} \quad (1)$$

Return on capital employed (ROCE) was used as the second dependent variable and represented financial performance. ROCE measures how efficiently a firm generates operating profit from the capital employed in the firm. A higher ROCE indicates that the firm was more efficient in using its invested capital to generate returns. ROCE was relevant to include in the study because it complements ROA by focusing on the return generated from capital used in operations rather than total assets alone. ROCE has also been used in previous studies (Kumar & Firoz, 2022, p. 60; Narula et al., 2024, p. 1175) which supports its use as an additional measure of firm performance. ROCE was therefore included to examine whether sustainability was related to firms' ability to generate returns from capital employed. Capital employed was calculated as total assets minus total current liabilities and ROCE was calculated as earnings before interest and taxes (EBIT) divided by capital employed, consistent with previous studies (Narula et al., 2024, p. 1177).

$$ROCE = \frac{EBIT}{Capital\ employed} \tag{2}$$

Tobin's Q (TQ) was used as the third dependent variable and represented market performance. While ROA and ROCE are accounting-based measures, TQ captures how the market valued the firm in relation to its assets and is a more forward-looking measurement of firm performance. A higher TQ indicates that investors values the firm above the accounting value of its assets, which may reflect expectations of future growth, profitability, and intangible value (Aydoğmuş et al., 2022, p. 122). TQ was relevant because sustainability may influence not only internal profitability, but also investor perceptions and market valuation. TQ is commonly used in previous studies on the relationship between sustainability and firm performance to capture market-based performance and firm value (Aydoğmuş et al., 2022, p. 125; Narula et al. 2024, pp. 1175-1176; Shaikh, 2022, p. 223; Tang et al., 2024, p. 7). TQ was therefore included to examine whether sustainability was related to firm's market valuation and was calculated as the sum of market capitalization and total assets minus shareholder's equity, divided by total assets.

$$TQ = \frac{Market\ capitalization + (Total\ assets - Shareholder's\ equity)}{Total\ assets} \tag{3}$$

3.5.5. Independent variables

The independent variables in this study were used to measure firms' sustainability performance. Since the purpose of this thesis is to investigate and analyze the relationship between sustainability and firm performance among listed firms in the Nordic region, and to examine whether this relationship differs between established and newly listed firms, sustainability was operationalized through the overall ESG score and the three individual ESG pillar scores. Following previous studies (Aydoğmuş et al., 2022; Rahi et al., 2022; Saha & Khan, 2024) the ESG score data was collected from LSEG Workspace. The score is expressed on a scale from 0-100, where a higher score indicates stronger sustainability performance (LSEG, 2026, p. 21-22).

The main independent variable was the overall ESG score. The ESG score represents the combined measure of a firms environmental, social, and governance sustainability performance. It provides an overall assessment of how well a firm manages sustainability-related issues across the three ESG dimensions (LSEG, 2026, p. 15). Following previous

studies that use the combined ESG score as the primary measure of sustainability (Aydoğmuş et al., 2022; Saha & Khan, 2024), the ESG score was included to examine whether sustainability is associated with firm performance. The ESG score is illustrated in Figure 5.

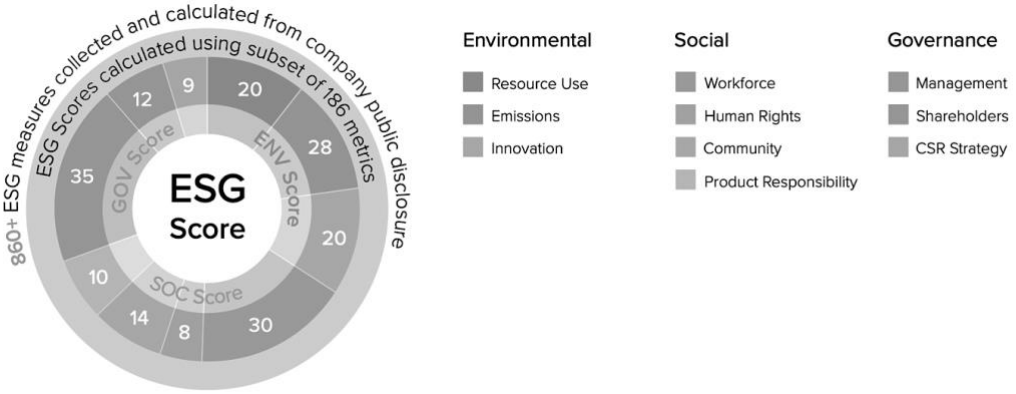


Figure 5. Illustration of ESG score. Source: LSEG. (n.d.)

Environmental sustainability was operationalized through the environmental pillar score. This score captures how well a firm manages environmental issues, such as emissions, resource use, energy efficiency, waste management and environmental innovation (LSEG, 2026, p. 15). A higher environmental score indicates stronger environmental performance and a more developed approach to managing environmental responsibility. The environmental pillar score was included to examine whether environmental sustainability is related to firm performance.

Social sustainability was operationalized through the social pillar score. It reflects how firms manage relationships with employees, customers, suppliers, and society more broadly. It includes aspects such as workforce conditions, human rights, product responsibility, diversity, and community engagement (LSEG, 2026, p. 15). A higher social score indicates that the firm performs better in managing social responsibilities and stakeholder relationships. The social pillar score was included to examine whether social sustainability is associated with firm performance.

Governance sustainability was operationalized through the governance pillar score. It captures how firms are governed and controlled, including issues such as board structure, shareholder rights, transparency, management quality, and firm ethics (LSEG, 2026, pp. 15-16). A higher governance score indicates stronger corporate governance practices and more effective control and accountability structures. The governance pillar score was included to examine whether governance related sustainability is associated with firm performance.

Firm size has been used as a control variable in previous studies (Aydoğmuş et al., 2022, p. 123; Kumar & Firoz., 2022, p. 50; Kumar & Puthur Joseph., 2025, p. 728). And it was also relevant to include in this study since larger firms tend to have both higher ESG scores and stronger firm performance, as they benefit from greater resources, economies of scale, and more developed reporting capabilities (Drempetic et al., 2020). Not controlling for firm size could therefore lead to an upward bias in the estimated relationship between sustainability and firm performance. The logarithmic transformation

was applied to reduce skewness in the distribution of total assets and is consistent with previous studies (Aydoğmuş et al., 2022, p. 123; Kumar & Firoz., 2022, p. 50).

$$SIZE = ln(Total\ assets) \tag{4}$$

Leverage is used as a control variable because firms with higher debt levels face higher financial costs that may lower firm performance and because leverage also reflects the capital structure decisions of firms, which may be related to both sustainability investments and firm performance (Jahanzeb et al., 2013; Rahi et al., 2022). Leverage may also be related to sustainability investments, as firms with higher debt levels may have less financial flexibility to invest in sustainability activities. This is especially relevant because sustainability initiatives can require long-term commitments, including compliance, reporting, and operational adjustments, which may increase costs in the short run (Rhai et al., 2021, pp. 302-305). Including leverage as a control variable is consistent with previous studies (Aydoğmuş et al., 2022, p. 123; Kumar & Firoz, 2022, p. 50; Rahi et al., 2022, p. 299).

$$LEV = \frac{Total\ liabilities}{Total\ assets} \tag{5}$$

Book-to-market value (BTMV) is included as a control variable to control for differences in growth opportunities and market valuation across firms, as high book-to-market firms tend to exhibit different performance characteristics than low book-to-market growth firms (Kumar & Firoz, 2022; Narula et al., 2024). BTMV has been included in previous studies that examine the relationship between sustainability and firm performance (Kumar & Firoz, 2022; Narula et al., 2024). Firms with low BTMV are typically high-growth firms for which the market expects strong futures, while firms with high BTMV may trade at a discount relative to their book value, reflecting weaker growth prospects or financial difficulties. BTMV is measured as the ratio of book equity to market capitalization, where book equity is defined as total assets minus total liabilities (Fama & French, 1992).

$$BTMV = \frac{Book\ equity}{Market\ capitalization} \tag{6}$$

In addition to the continuous control variables and following previous studies (Mugova et al., 2025: Tang et al., 2024) industry and year fixed effects was included in all model specifications as binary dummy variables to control for systematic differences in firm performance that are not captured by the firm-level control variables.

Industry fixed effects are included to control for systematic performance differences across sectors, as firms operating in different industries are subject to different cost structures, competitive dynamics, and regulatory environments that may independently influence both ESG scores and firm performance. Without controlling for industry affiliation, observed differences in firm performance may reflect industry-level characteristics rather than the effect on sustainability. Industry fixed effects was constructed as binary dummy variables based on the TRBC industry classification system, with one dummy variable included per industry sector. As shown in Figure 4, the sample spans 10 industries, with Industrials (23.9%), technology (19%), and healthcare (15.2%) being the most represented industry sectors.

Year fixed effects was included to control for common time trends and macroeconomic conditions that affect all firms simultaneously across a given year, such as changes in interest rates, equity market conditions, and shifts in sustainability reporting requirements. Without year fixed effects, time-varying macroeconomic shocks could bias the estimated relationship between sustainability and firm performance if they affect both ESG scores and firm performance in the same direction. Year fixed effects are constructed as binary dummy variables for each year in the sample period, with one year omitted as the reference category.

To address RQ2, an interaction specification was used, and a binary dummy variable and an interaction term was included. The newly listed dummy variable (Newly_listed) is a time-invariant binary variable that equals one if firm *i* conducted its IPO in or after 2017 and zero otherwise, as defined in equation (7).

$$Newly_listed = Firms\ with\ IPO \geq 2017 \tag{7}$$

3.5.6. Model specifications and treatment of OLS assumptions

To address the research questions of this thesis, two sets of regression models are estimated. The first one forms the baseline specification for RQ1, examining the relationship between sustainability and firm performance using sample 1. The second set introduces an interaction specification for RQ2, to examine if there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms using sample 2. Following previous studies on the relationship between sustainability and firm performance (Aydoğmuş et al., 2022; Mugova et al., 2025; Narula et al., 2024), Ordinary Least Squares (OLS) regression were used as the main statistical estimation technique. According to Stock & Watson (2015, p. 186), OLS estimates the regression coefficients (β) by minimizing the sum of squared residuals, that is, the difference between the observed values of the dependent variable and those predicted by the model, thereby identifying the line of best fit that minimizes total prediction error.

The baseline specification model includes each performance measure on the ESG score variable and three firm-level control variables. Three dependent variables are used to capture different dimensions of firm performance, ROA for operating performance, ROCE for financial performance and Tobin's Q (TQ) for market performance. Each model is estimated separately for the ESG score, and for each of the three individual pillar scores (E, S, and G), resulting in twelve regressions in total. The general form of the baseline OLS estimation regression was specified as follows in equation (8) to (10):

$$ROA_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 LEV_{i,t} + \beta_3 SIZE_{i,t} + \beta_4 BTMV_{i,t} + \gamma_s + \delta_t + \varepsilon_{i,t} \tag{8}$$

$$ROCE_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 LEV_{i,t} + \beta_3 SIZE_{i,t} + \beta_4 BTMV_{i,t} + \gamma_s + \delta_t + \varepsilon_{i,t} \tag{9}$$

$$TQ_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 LEV_{i,t} + \beta_3 SIZE_{i,t} + \beta_4 BTMV_{i,t} + \gamma_s + \delta_t + \varepsilon_{i,t} \tag{10}$$

Where *i* is the firms and *t* is the years. The dependent variables ROA, ROCE, and TQ are the operating, financial, and market performance measures. The variable ESG is the sustainability measure and takes one of four forms across the twelve regressions, ESG, E,

S, and G. The control variables are leverage (LEV), firm size (SIZE), and book-to-market value (BTMV). The terms γ_s and δ_t denote industry and year fixed effects, respectively, where s is the TRBC industry sector. The term ε is the error term, and α is the intercept. The coefficient of primary interest is β_1, which captures the association between the sustainability measure and firm performance. Hypothesis $H1_0$, H1a, and H1b are tested through the statistical significance of β_1 across the four ESG specifications.
To examine if there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms, an interaction term between the ESG score and a newly listed dummy variable is added to the baseline specification. The interaction specification is estimated for ROA, ROCE and TQ. Equation (11) presents the general form of the interaction model:

$$Y_{i,t} = \alpha + \beta_1 ESG_{i,t} + \beta_2 Newly_listed_{i,t} + \beta_3 (ESG_{i,t} \times Newly_listed_{i,t}) + \beta_4 LEV_{i,t} + \beta_5 SIZE_{i,t} + \beta_6 BTMV_{i,t} + \gamma_s + \delta_t + \varepsilon_{i,t} \quad (11)$$

Where Y is the dependent variable, taking the value of either ROA, ROCE, or TQ. The variable Newly_listed is a time-invariant dummy variable equal to one if firm i had its IPO in or after 2017 (IPO≥2017) and zero otherwise. As in the baseline model specification, ESG takes one of four forms (ESG, E, S, or G), resulting in twelve regressions in total. All other variables included are defined as in equation (8) to (10). In equation (11), the coefficient β1 captures the association between sustainability and firm performance for established firms, the reference group, while the coefficient β2 captures the baseline difference in performance between newly listed and established firms. The coefficient of primary interest is β3, the interaction term, which measures the differential interaction effect of sustainability on firm performance for newly listed firms relative to established firms. The total interaction effect of sustainability on firm performance for newly listed firms is therefore given by the sum of β1 plus β3. Hypothesis $H2_0$, H2a, and H2b are tested through the statistical significance of β3 across the twelve model specifications. The expectations are as follows. If the coefficient β3 is not statistically significant, it provides support for $H2_0$, meaning that there is no significant difference between the two groups. If β3 is positive and statistically significant, this provides support for H2a, meaning that the relationship is more positive for newly listed firms than for established firms. If β3 is negative and statistically significant, it provides support for H2b, meaning that the relationship is more negative for newly listed firms than for established firms.

The validity of OLS as an estimator depends on a set of underlying assumptions. Violations of these assumptions can lead to biased coefficients, inefficient estimates, and unreliable statistical inference. To ensure the robustness of the empirical results obtained from equation (8) to (11), this study examines and addresses the relevant OLS assumptions through a series of pre-diagnostic tests with the corresponding procedures discussed below.

The first assumption underlying OLS is that the relationship between the independent variables and the dependent variable is linear in parameters (Stock & Watson, 2015, p. 188). Linearity is examined through residual versus fitted plots for each dependent variable. Where the assumption is satisfied in levels, the variable was retained in its original form, and where it was violated, a logarithmic transformation was applied to obtain linearity in the transformed variable. This treatment is consistent with previous studies that apply log-transformations to address non-linear distributions of variables

(Aydoğmuş et al., 2022; Narula et al., 2024). A further OLS assumption requires the expected value of the error term to be zero conditional on the independent variables (Stock & Watson, 2015, p. 189), which is verified by computing the mean of the regression residuals for each model, where values close to zero confirm that the estimator is unbiased.

OLS further assumes that the variance of the error term is constant across observations, known as homoscedasticity (Stock & Watson, 2015, p. 167). Violation of this assumption, heteroscedasticity, does not bias the coefficient estimates but may produce inefficient standard errors and distort statistical inference (Stock & Watson, 2015, p. 167). The presence of heteroscedasticity was examined using the Modified Wald test for groupwise heteroscedasticity. A related assumption is that the error terms are uncorrelated across observations, and in panel data, error terms within the same firm are likely to be correlated over time, which can bias standard errors if not addressed (Wooldridge, 2010, p. 282). Autocorrelation was examined using the Wooldridge test for serial correlation in panel data. All regression models are estimated using robust standard errors clustered at the firm level, which simultaneously correct for both heteroscedasticity and within firm correlation across time, thereby providing a solution to both issues (Petersen, 2009).

Multicollinearity is not a formal OLS assumption, but it still remains an important diagnostic consideration in regression analysis. According to Gujarati & Porter (2009, p. 363), severe multicollinearity can arise when two or more independent variables are highly linearly correlated with one another and therefore can distort the estimation of the individual contribution of each independent variable and inflate standard errors, undermining the statistical reliability of the regression results (Gujarati & Porter, 2009, p. 363). To address this, multicollinearity was examined using the Variance Inflation Factor (VIF), with a conventional threshold of 10 above which multicollinearity is considered problematic (O´Brien, 2007, pp. 674-675).

A further consideration is the normality of residuals. While normality of the error term is an OLS assumption that supports exact small sample inference, it is not strictly required for valid hypothesis testing in large samples, as the Central Limit Theorem ensures that the sampling distribution of the coefficients converges to normality asymptotically (Stock & Watson, 2015, pp. 48-50). Normality was nevertheless examined as a diagnostic through the Skewness-Kurtosis test on the regression residuals, complemented by visual presentation and inspection of residual histograms.

Another assumption underlying OLS estimation is the absence of endogeneity, which arises when one or more independent variables are correlated with the error term, leading to biased and inconsistent coefficient estimates (Wooldridge, 2010, p. 50). Endogeneity may stem from omitted variable bias or reverse causality, both of which are plausible in the relationship between sustainability and firm performance, as firms with stronger financial performance may have greater capacity to invest in sustainability initiatives, just as sustainability may improve performance (Qian et al., 2024, p. 10). To examine the presence of endogeneity, the Hausman test was applied, which compares fixed effects and random effects estimators to determine whether unobserved firm specific effects are correlated with the independent variables. The fixed effects estimator controls for all time-invariant unobserved heterogeneity at the firm or industry level, thereby eliminating omitted variable bias potentially arising from stable firm level characteristics (Wooldridge, 2010, p. 285).

The random effects estimator, by contrast, assumes that the unobserved firm specific effect is uncorrelated with the independent variables and is more efficient when this assumption holds (Wooldridge, 2010, p. 286). Where the Hausman test indicates that fixed effects was the appropriate estimator, the model was evaluated to be considered. In addition, given the heterogeneity of the Nordic listed firm sample across industries and the time varying nature of the macroeconomic environment, industry and year fixed effects are applied across all model specifications. This approach is consistent with previous studies on sustainability and firm performance (Mugova et al., 2025: Tang et al., 2024), which apply industry and year fixed effects to ensure consistency in the empirical results and to absorb variation common to firms in the same industry or year. To further mitigate concerns about reverse causality, a robustness check using lagged ESG score was performed.

3.6. Validity and reliability

Reliability refers to the accuracy and precision of the measurement and the ability to get comparable results if the research were repeated (Collis & Hussey, 2014, p. 52). The authors further describe that it is common for positivist research paradigms to have high reliability, and it is therefore not necessary for the research to have qualitative measures that meet reliability standards of positivism. Because of this, focus shifts to whether observations and interpretations that are made at separate times can be consistently explained and understood (Collis & Hussey, 2014, p. 53).

The data used in this study were collected from LSEG Workspace, and it is therefore possible to reproduce and mirror this study's results. As mentioned before, the issue originates when looking at what ESG factors go into the ESG score. Also, there may be a difference in the sample and sample sizes if another database was used. Finally, the authors of this research used statistical models and will therefore not interfere with the results with their own bias, which strengthens the validity of the study.

Validity refers to which degree the test measures for what the researchers want to capture. Research errors or inaccurate samples are examples of instances that can affect a study's validity (Collis & Hussey, 2014, p. 53). Collis and Hussey (2014, p. 53) further argue that there are different types of validity; mainly face validity and constructing validity. Face validity refers to the idea that variables and data measure what they are supposed to measure, and constructing validity can be explained as something that is not directly observable. An example of something that cannot be measured is anxiety and motivation. These variables are described as "hypothetical constructs". This study will not contain any hypothetical constructs, as all data and variables that are being observed are directly observable from publicly available data. Furthermore, this study would not benefit from having any hypothetical constructs. The authors of this study have accounted for the validity by analyzing previous research designs which are suitable for this type of study. By not including any hypothetical constructs, this study will increase the validity. Moreover, this study will achieve higher validity by analyzing previous research. The authors of this research have used previous studies as guidance.

Sources that have been used in this study have only been chosen after the authors deemed them to be of high quality. Most sources are academic journals and articles that focus on related areas to the topic of this research. Furthermore, analyzing previous research on similar topics has helped the authors identify the appropriate theoretical models that will

give context to this study. Some of the theories that can be found in previous research have also been used in this study since the authors have deemed them to be of great relevance in explaining the results of this study's findings. Lastly, theories which best suited this research were later implemented as they are accepted and frequently used in other research within finance.

3.7. Ethical considerations

All data used in this study is publicly available, consisting of financial data and sustainability data from publicly listed firms sourced from LSEG Workspace. No confidential or private information has been accessed or utilized, and since the data belongs exclusively to publicly listed firms, no individuals or firms are considered to have been exposed to harm, privacy violations, or ethical risk as result of this study, consistent with the principle that risk should be assessed in terms of its likelihood to cause harm (Saunders et al., 2023, p.255). The study is based entirely on secondary data, meaning that there are no active participants and no issues of informed consent relating to data collection. All gathered data and literature have been transparently described in terms of how it was collected, in order to comply with the reproducibility and trustworthiness of the study. Collis & Hussey (2014, p. 32) argue that showing openness and transparency during the research process builds trust, and the authors have followed this approach throughout the study. Previous research has been represented accurately to avoid misleading interpretations, in line with the principles that research ethics refers to how research is conducted and presented (Collis & Hussey, 2014, p. 30). The use of LSEG Workspace as the data collection source reduces the risk of data quality issues, as it is a recognized and widely used database for gathering data in previous research (Eliwa et al., 2021, Rahi et al., 2022, Saha & Khan, 2022). However, an important ethical aspect is that ESG scores may not be a complete measure of a firm's actual sustainability, as they are based on rating methodologies and available disclosures. This means that firms with more resources and more extensive reporting may receive higher scores, and therefore ESG scores are treated only as an operational proxy for sustainability performance, and the findings are interpreted with caution. The study has been conducted with objectivity and transparency throughout, and steps have been taken to minimize bias in both the data collection and the reporting of findings, in accordance with the ethical standards for academic work outlined in Umeå University's ethical guidelines.

3.8. Use of artificial intelligence

Artificial Intelligence (AI) tools were used during the writing of this thesis in strict accordance with Umeå University's rules and guidelines on academic integrity, cheating, plagiarism, and AI. Specifically, ChatGPT (GPT-5, developed by OpenAI) and Google Translate were used for language correction, translation support, and feedback on wording, clarity and grammar. The justification of the use of AI lies in the idea of improving the study's quality. This limited use was consistent with suggestions by Aguinis (2026, pp. 8-9), that AI can be used responsibly, as long as the researcher remains in control, checks the outputs, uses clear limits, and takes full responsibility for the final work. The AI tools were not used to generate thesis content, write text drafts, develop arguments, conduct analysis or interpret results. All ideas, arguments, analysis, conclusions, and final written content are the authors own. Any AI feedback was checked critically, and the authors independently decided whether and how to revise the text. This ensured that academic integrity and the authors independence were maintained throughout the thesis process.

4. Empirical findings

4.1. Pre-diagnostics tests

Prior to estimating the main regression models, a series of diagnostic tests were conducted to verify that the data satisfy the key assumptions of OLS regression outlined in section 3.5.6. These tests are important to ensure that the estimated coefficients are unbiased, consistent, and that statistical inference is valid (Stock & Watson, 2015; Wooldridge, 2010).

The first assumption of linearity was assessed through actual versus fitted value plots for each dependent variable, presented in Appendix 2. Return on assets (ROA) and return on capital employed (ROCE) displayed no systematic non-linear pattern, satisfying the linearity assumption in winsorized levels at the 1st and 99th percentiles. Tobin's Q in levels, by contrast, show a clear violation with a right skewed distribution, which is common for market-to-book ratios in firm-level data (Stock & Watson, 2015, p. 188). This was resolved by applying a natural logarithm transformation, after which the fitted line tracked the data more consistently with a less remaining systematic pattern. Consequently, a logarithmic Tobin's Q is used in all specifications which is in line with previous studies (Aydoğmuş et al., 2022; Narula et al., 2024, p. 1175). The expected mean of residuals is confirmed to equal zero across all three models, yielding values of 5.556e-11, –7.12e-11, and 5.06e-11 for ROA, ROCE, Tobin's Q respectively, as it is guaranteed by the inclusion of a constant term in OLS estimation (Stock & Watson, 2015, p. 189).

The Wooldridge test for first-order serial correlation in panel data rejected the null hypothesis of no autocorrelation for all three models, with F-statistics of 34.109 ($p<0.001$), 5.922 ($p=0.015$), and 86.110 ($p<0.001$) for ROA, ROCE, and Tobin's Q respectively. The modified Wald test similarly confirmed the presence of groupwise heteroskedasticity across all models (all $p<0.001$). Both violations are well-documented in firm-level panel data and are explicitly addressed through heteroskedasticity-robust standard errors clustered at the firm level, which corrects for both issues simultaneously (Petersen, 2009).

To address endogeneity, two measures was employed. First, industry and year fixed effects are included in all model specifications via indicator variables, controlling for time-invariant unobserved heterogeneity across industries and common time trends, thereby mitigating omitted variable bias. A firm fixed effects estimator is not used, as it would absorb the majority of cross-sectional variation in the ESG scores, which change slowly within firms over time, and eliminating the between firm variation central to the purpose of this study. OLS with industry and year fixed effects was furthermore the chosen specification and motivated in line with previous studies (Mugova et al., 2025: Tang et al., 2024). The chosen specifications is retained consistently across all models despite a performed Hausman test rejecting the null hypothesis for the Tobin's Q model, but additionally, a robustness check using one-year lagged ESG score is performed to address reverse causality presented in section 4.2.4.

Multicollinearity was assessed for all independent variables, as the concern for this relates solely to linear dependence among the regressors and has no relevance on the dependent variable (Stock & Watson, 2015, pp. 199-202). VIF values for all variables fell well below the conventional threshold of 10 (O'Brien, 2007, pp. 674-675), without the inclusion of industry and year fixed effects, VIF values are very low with ESG range from

1.08 to 1.22, leverage (LEV) at 1.08, firm size (SIZE) from 1.02 to 1.05, and book-to-market value (BTMV) from 1.22 to 2.11. When industry and year fixed effects are added as indicator variables, their mutual collinearity inflates the overall mean VIF of the models. However, the other variables remain low, all between a VIF value of 1-4. Since the fixed effects dummies are binary control variables, not the variables of interest, their contribution to a high mean VIF is not considered a concern (O'Brien, 2007). This is further supported by the correlation matrix presented in section 4.2.2, where no pairwise correlation between the independent variables exceeds the conventional threshold of 0.8, above which multicollinearity is considered a meaningful concern (Stock & Watson, 2015, p. 202), with the exception of observed correlations between the ESG, E, S, and G scores, but that never appears in the same regression model and therefore do not pose a multicollinearity concern for the interpretation of the results.

The performed Skewness-Kurtosis test rejected normality of residuals for all three models at the 1% level, a pattern visible in the residual histograms in Appendix 3. However, this does not invalidate inference. With the sample sizes ranging from 2951 to 3471 observations, the Central Limit Theorem ensures that the sampling distribution of the coefficients converges to normality asymptotically, making hypothesis testing and confidence intervals valid even when the residuals themselves are not normally distributed (Stock & Watson, 2015, p. 50).

4.2. Empirical result of sample 1

4.2.1. Descriptive statistics sample 1

Table 2 presents the descriptive statistics for sample 1, including and covering 468 Nordic listed firms over the period 2010-2025. The number of observations varies across the dependent variables due to the unbalanced panel structure, with 3471 observations for return on assets (ROA), 3220 for return on capital employed (ROCE), and 2951 for Tobin's Q (TQ).

Table 2. Descriptive statistics sample 1

Variables	Observations	Mean	Std. Dev.	Min	Max
ROA	3471	0.010	0.200	−1.971	0.307
ROCE	3220	0.089	0.276	−2.399	0.913
TQ	2951	0.600	0.706	−0.625	3.140
ESG	3471	50.343	20.258	1.319	92.855
E	3471	45.300	26.389	0.000	98.370
S	3471	53.023	23.236	0.649	96.389
G	3471	49.836	22.954	1.242	98.207
LEV	3471	0.534	0.195	0.032	1.467
SIZE	3471	23.011	1.926	16.106	26.130
BTMV	3471	0.654	0.777	−0.298	6.282

Note. ROA=Return on assets. ROCE=Return on capital employed. TQ=ln(Tobin's Q). ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value. All variables except ESG, E, S, G, winsorized at the 1st and 99th percentiles.

The mean ROA value of 0.010 indicates that on average, Nordic listed firms generate a small positive return of 1% of their total assets. The standard deviation of 0.200 shows

that there is high variation in profitability across firms. The minimum value of –1.971 indicates that some firms experienced severe losses during the sample period, while the maximum value of 0.307 shows that the most profitable firms achieved returns of approximately 30.7% on their total assets. ROCE shows a higher mean of 0.089, indicating that firms on average generate a return of 8.9% on their capital employed. The standard deviation 0.276 reflects that there is high differences in how efficiently firms use their capital across industries, with values ranging from –2.399 to 0.913. For TQ, the mean value of 0.600 suggest that on average firms are valued above their book value by the market, indicating expectations of future growth. The standard deviation of 0.706 shows that there is high variation in market valuations across firms, and some firms trade above book value, and some trade below.

ESG score has a mean of 50.343 out of 100 and a standard deviation of 20.258, indicating that ESG performance varies across the firms in the sample and that firms differ substantially in how far along they have come in their sustainability work. The minimum value of 1.319 and maximum value of 92.855 show that the sample contains both firms with very low and limited sustainability reporting and firms with very high ESG scores. Among the individual pillars, the social score (S) reports highest score on average with a mean value of 53.023 and a standard deviation of 23.236, followed by the governance score (G) with the mean value of 49.836 and a standard deviation of 22.954. and the environmental score (E) with the mean value of 45.300 and a standard deviation of 26.389. The higher standard deviation of E suggest that environmental performance is the most unevenly distributed pillar across the firms, meaning that some firms score very high on the environmental practices while others score much lower. The lower E scores likely reflects that environmental reporting was less developed among some firms in the sample period, and a minimum E score of 0.000 reflects firms with zero E reported.

The mean leverage (LEV) of 0.534 and a standard deviation of 0.195 indicate that on average, the firms' total liabilities represent 53.4% of their total assets, with values ranging from minimum 0.032 and maximum 1.467, showing that there is a large difference in how firms finance themselves, where some firms have total liabilities exceeding their total assets, may reflect financial distress, while others carry very little debt. The mean firm size (SIZE), the natural logarithm of total assets, is 23.011 with a standard deviation of 1.926, reflecting a sample with variation in sizes of firms. The minimum of 16.106 and maximum of 26.130 indicating that the sample covers a broad range of firm sizes, from smaller listed firms to very large, long-established firms. The mean book-to-market value (BTMV) of 0.654 and a standard deviation of 0.777 indicate that on average the firms book value represents 65.4% of their market value. The minimum value –0.298 indicates that some firms have negative book value, while the maximum value of 6.282 shows that some firms trade at a significant discount to their book value, suggesting that the sample includes both high-growth firms and firms facing financial difficulties.

4.2.2. Correlation matrix

Table 3 presents the pairwise correlation matrix, and the purpose is to provide insight into the relationship between all the included variables through examining the correlations between the variables and detect any multicollinearity issues, using all observations within sample 1 covering the period 2010-2025. The number of observations varies across the dependent variables due to the unbalanced panel structure. The correlation coefficient measures the strength and direction of the linear relationship between two variables,

ranging from –1 to 1, where values close to 1 indicate a strong positive relationship, values close to –1 indicate a strong negative relationship, and values close to 0 indicate no linear relationship (Chatterjee & Hadi, 2012, pp. 27-28). In this study, correlations are considered statistically significant at the 5% level, meaning that a significant relationship can be confirmed with 95% confidence.

Table 3. Correlation matrix

	ROA	ROCE	TQ	ESG	E	S	G	LEV	SIZE	BTMV
ROA	1.000									
ROCE	0.699***	1.000								
TQ	0.116***	0.203***	1.000							
ESG	0.259***	0.229***	−0.137***	1.000						
E	0.286***	0.244***	−0.193***	0.859***	1.000					
S	0.285***	0.271***	−0.090***	0.908***	0.752***	1.000				
G	0.088***	0.058***	−0.079***	0.723***	0.416***	0.472***	1.000			
LEV	−0.057***	0.090***	−0.261***	0.161***	0.176***	0.164***	0.090***	1.000		
SIZE	0.391***	0.313***	−0.321***	0.720***	0.717***	0.685***	0.412***	0.252***	1.000	
BTMV	−0.066***	−0.159***	−0.610***	−0.039**	0.023	−0.050***	−0.072***	−0.061***	0.101***	1.000

*Note. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. ROA=Return on assets. ROCE=Return on capital employed. TQ=ln(Tobin's Q). ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

The dependent variables return on assets (ROA) and return on capital employed (ROCE) show a strong positive correlation of 0.699, which is expected since both measure accounting-based profitability. The high correlation between ROA and ROCE does not pose a multicollinearity threat, since the dependent variables are never included in the same regression model. Tobin's Q (TQ) shows weaker positive correlations with both ROA of 0.116 and ROCE of 0.203, reflecting that market-based performance measures capture different dimensions of firm performance and do not necessarily move together.

Moving to the sustainability variables, the ESG score is positively correlated with both ROA, of 0.259, and ROCE of 0.229, indicating that firms with higher ESG scores tend to also have higher operating and financial performance. Further, ESG shows a negative correlation with TQ of –0.137, indicating that firms with higher ESG scores does not necessarily trade at higher market valuations than firms with lower ESG. Among the individual pillars, environmental score (E) and the social score (S) show strongest positive correlations with ROA at 0.286 and 0.285 respectively, while the governance score (G) shows a weaker positive correlation of 0.088. The observed high correlation between ESG and the three individual pillars, E of 0.859, S of 0.908, and G of 0.723, confirm that the pillars are closely related to the main ESG score. As discussed in section 4.1. this is why the four sustainability variables were never included in the same regression model.

Regarding the control variables, firm size (SIZE) shows a strong positive correlation with ESG of 0.720, suggesting that larger firms tend to have higher ESG scores, likely reflecting greater access to resources and reporting activity. Leverage (LEV) is negatively correlated with ROA at –0.057, suggesting that higher financial obligations reduce

profitability, and the positive correlation with ESG at 0.161, could reflect that larger firms tend to carry more debt while simultaneously investing more in sustainability. Book-to-market value (BTMV) shows a strong negative correlation with TQ at –0.610, which is expected, as firms with high book value relative to their market value, meaning that the market places a lower value on these firms compared to what their book value suggest. However, this could potentially bias our results, where both TQ and BTMV are included in the same regression model, therefore BTMV was tested to be excluded in the models with TQ, and it had no significant impact on the results presented. All pairwise correlations between the independent variables remain below the conventional threshold of 0.8.

4.2.3. Regression analysis sample 1

Table 4 presents the regression results with return on assets (ROA) as the dependent variable across four model specifications, each including one sustainability variable at a time, alongside the control variables, industry and year fixed effects. The multiple linear regression include ROA with 3471 observations within sample 1 covering the period 2010-2025.

Table 4. Regression analysis ROA

Variables	(1) ROA	(2) ROA	(3) ROA	(4) ROA
ESG	–0.006			
	(0.004)			
E		–0.001		
		(0.003)		
S			0.002	
			(0.003)	
G				–0.008**
				(0.003)
LEV	–0.245***	–0.245***	–0.246***	–0.246***
	(0.058)	(0.058)	(0.058)	(0.058)
SIZE	0.049***	0.045***	0.042***	0.049***
	(0.008)	(0.007)	(0.007)	(0.007)
BTMV	–0.035***	–0.033***	–0.033***	–0.035***
	(0.006)	(0.005)	(0.005)	(0.006)
Constant	–0.887***	–0.804***	–0.764***	–0.891***
	(0.159)	(0.142)	(0.138)	(0.146)
Observations	3471	3471	3471	3471
R-squared	0.249	0.247	0.248	0.254
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG, E, S, and G scores are scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. ROA=Return on assets. ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

The model explains between 24.7% and 25.4% of the variation in ROA across the four specifications, as indicated by the R-squared values. In column (1), the coefficient on ESG score is negative of –0.006 with a standard error of 0.004 meaning that a 10-point increase in ESG is associated with a decrease in ROA of 0.006, holding all other variables

constant. However, the coefficient on ESG is not statistically significant at any of the conventional significance levels, and no significant relationship between ESG and ROA can therefore be confirmed. This suggests that the overall ESG score, as the main sustainability measure, does not have a meaningful association on the operating performance of Nordic listed firms. In column (2), the coefficient on environmental score (E) is negative of –0.001 with a standard error of 0.003 and is similarly not statistically significant, suggesting that environmental practices alone are not meaningfully related to ROA in terms of operating performance. In column (3), the coefficient on the social score (S) is positive of 0.002 with a standard error of 0.003, however not statistically significant, suggesting also that social practices alone are not meaningfully related to the operating performance of Nordic listed firms.

Column (4) reports a negative and statistically significant coefficient on governance score (G) of –0.008 with a standard error of 0.003, significant at the 5% level. This means that a 10-point increase in G is associated with a decrease in ROA of 0.008, holding all other variables constant. This is the only statistically significant finding among the sustainability variables in the ROA model, suggesting that firms with stronger governance structures may face higher governance-related costs, such as increased compliance and reporting requirements, that reduce short-term operating performance.

Turning to the control variables, which show consistent signs and significance across all four model specifications. The coefficient on leverage (LEV) ranges negative of –0.245 to –0.246 with a standard error of 0.058 and is significant at the 1% level, meaning that one-unit increase in LEV is associated with a decrease in ROA of 0.245 to 0.246. This suggests that more leveraged firms have lower operating performance, indicating that higher debt levels may increase financial costs and reduce returns. The coefficient on firm size (SIZE) is positive and ranges from 0.042 to 0.049 with standard errors between 0.007 and 0.008 and is significant at the 1% level, meaning that a one-unit increase in SIZE is associated with an increase in ROA of 0.042 to 0.049. This suggest that larger firms generate higher ROA, likely because they benefit from greater efficiency and scale. The coefficient on book-to-market value (BTMV) ranges from negative –0.033 to –0.035 with standard errors between 0.005 and 0.006 and is significant at the 1% level, meaning that a one-unit increase in BTMV is associated with a decrease in ROA of 0.033 to 0.035. This suggests that firms with higher BTMV have lower operating performance. The control variables is supported by findings from Narula et al. (2024), who also find LEV and BTMV is significantly negative to ROA, and ROCE, also Aydoğmuş et al. (2022) find SIZE positive in direction related to ROA but not significant. This supports the relevance of including these control variables in the regression models.

Table 5 presents the regression results with return on capital employed (ROCE) as the dependent variable across four model specifications, each including one sustainability variable at a time alongside the control variables, industry and year fixed effects. The regression includes 3299 observations within sample 1 covering the period 2010-2025. The model explains between 17% and 17.5% of the variation in ROCE across the four specifications, as indicated by the R-squared values. A bit lower explanatory power in this model specification with ROCE as the dependent variable, in contrast with the model with ROA in Table 4. In column (1), the coefficient on ESG is negative of –0.002 with a standard error of 0.005, meaning that a 10-point increase in ESG is associated with a decrease in ROCE of 0.002, holding all other variables constant. The coefficient on ESG is not statistically significant at any conventional significance level, and no significant

relationship between ESG and ROCE can therefore be confirmed. This result is consistent with the result for ROA in Table 4, suggesting that the overall ESG score does not have a meaningful association on the operating and the financial performance of Nordic listed firms. In column (2), the coefficient on E is positive of 0.002 with a standard error of 0.004 and is also not statistically significant, suggesting that environmental practices alone are not meaningfully related to ROCE in terms of financial performance. This result is also in line with the insignificant finding for E in the ROA model in Table 4.

Table 5. Regression analysis ROCE

Variables	(1) ROCE	(2) ROCE	(3) ROCE	(4) ROCE
ESG	–0.002			
	(0.005)			
E		0.002		
		(0.004)		
S			0.009**	
			(0.004)	
G				–0.010**
				(0.004)
LEV	–0.071	–0.072	–0.074	–0.071
	(0.062)	(0.062)	(0.062)	(0.063)
SIZE	0.049***	0.045***	0.039***	0.053***
	(0.011)	(0.010)	(0.010)	(0.010)
BTMV	–0.073***	–0.072***	–0.070***	–0.075***
	(0.012)	(0.012)	(0.012)	(0.012)
Constant	–0.867***	–0.795***	–0.698***	–0.950***
	(0.242)	(0.221)	(0.217)	(0.227)
Observations	3299	3299	3299	3299
R-squared	0.170	0.170	0.172	0.175
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG, E, S, and G scores are scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. ROCE=Return on capital employed. ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

In column (3), the coefficient on S is positive of 0.009 with a standard error of 0.004 and is statistically significant at the 5% level, meaning that a 10-point increase in S is associated with an increase in ROCE of 0.009, holding all other variables constant. This is a contrasting finding compared to the ROA model in Table 4, where S was not statistically significant, suggesting that social practices may be more closely related to how efficiently firms use their capital than to their overall asset returns. In column (4), the coefficient on G is negative of –0.010 with a standard error of 0.004 and is statistically significant at the 5% level, meaning that a 10-point increase in G is associated with a decrease in ROCE of 0.010, holding all other variables constant. This negative finding is consistent with the result for G in the ROA model in Table 4, further suggesting that stronger governance structures may increase governance-related costs that reduce both operating and financial performance of Nordic listed firms.

Regarding the control variables, which show consistent signs across all four model specifications, the coefficient on LEV ranges from negative –0.071 to –0.081 with standard errors between 0.062 and 0.064 and is not statistically significant in any specification. This suggest that LEV does not have a significant impact on ROCE. This contrast with the ROA model in Table 4 where LEV was significant at the 1% level in all model specifications, indicating that the effect of LEV on firm performance may differ depending on the performance measure used. The coefficient on SIZE ranges from 0.042 to 0.055 with standard errors between 0.010 and 0.013 and is significant at the 1% level, consistent with the positive and significant finding in the ROA model in Table 4, suggesting that larger firms generate higher ROCE and ROA. The coefficient on BTMV ranges from negative –0.068 to –0.073 with standard errors between 0.013 and 0.014 and is significant at the 1% level, also consistent with the ROA model in Table 4, suggesting that firms with higher BTMV have both lower operating and financial performance.

Table 6 presents the regression results with Tobin's Q (TQ) as the dependent variable across four model specifications, each including one sustainability variable at a time alongside the control variables, industry and year fixed effects. The regression includes 2951 observations within sample 1 covering the period 2010-2025. The R-squared values range from 53% and 53.2%, indicating the explanatory power of the TQ model is notably higher than the R-squared values observed in the ROA and ROCE models.

Table 6. Regression analysis TQ

Variables	(1) TQ	(2) TQ	(3) TQ	(4) TQ
ESG	0.015			
	(0.014)			
E		0.014		
		(0.010)		
S			0.022*	
			(0.012)	
G				–0.007
				(0.009)
LEV	–0.760***	–0.759***	–0.764***	–0.760***
	(0.129)	(0.129)	(0.129)	(0.130)
SIZE	–0.069***	–0.069***	–0.075***	–0.052***
	(0.019)	(0.016)	(0.017)	(0.016)
BTMV	–0.565***	–0.566***	–0.562***	–0.570***
	(0.057)	(0.057)	(0.056)	(0.057)
Constant	2.787***	2.810***	2.879***	2.492***
	(0.419)	(0.373)	(0.371)	(0.380)
Observations	2951	2951	2951	2951
R-squared	0.531	0.531	0.532	0.530
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG, E, S, and G scores are scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. TQ=ln(Tobin's Q). ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

In column (1), the coefficient on ESG is positive of 0.015 with a standard error of 0.014, however not statistically significant at any conventional significance level, and therefore no significant relationship between ESG and TQ can be confirmed. This is consistent with the insignificant findings for ESG in both the ROA and ROCE models in Table 4 and Table 5, respectively. This further suggest that the overall ESG score does not have a meaningful association on firm performance among Nordic listed firms, neither in operating performance, financial performance or market performance. In column (2), the coefficient on E is positive of 0.014 with a standard error of 0.010 and is similarly not statistically significant, which is also in line with the insignificant findings for E in the ROA and ROCE models.

The coefficient on S in column (3) is positive of 0.022 with standard error of 0.012 and is statistically significant at the 10% level, meaning that a 10-point increase in S is associated with an increase in TQ of 0.022, holding all other variables constant, however it falls outside the 5% significance level established in this study. In column (4), the coefficient on G is negative of –0.007 with a standard error of 0.009 and is not statistically significant, contrasting with the significant and negative findings for G in both ROA and ROCE models in Table 4 and Table 5, respectively. This suggest that while governance practices appear to reduce short-term accounting profitability, they do not have a significant impact on how the market values Nordic listed firms.

Moving to the control variables, which show consistent signs across all four model specifications. The coefficient on LEV ranges from negative –0.0759 to –0.764 with standard errors between 0.129 and 0.130 and is significant at the 1 % level, meaning that one-unit increase in LEV is associated with a decrease in TQ of approximately 0.760. This is consistent with the negative findings for LEV in the ROA model in Table 4, suggesting that more leveraged firms are valued lower by the market, likely because higher debt levels increase financial risk. The coefficient on SIZE ranges from –0.052 to –0.075 with a standard error between 0.016 and 0.019 and is significant at the 1% level, meaning that a one-unit increase in SIZE is associated with a decrease in TQ of 0.052 to 0.075. This contrasts the positive and significant findings for SIZE in both the ROA and ROCE models in Table 4 and Table 5, respectively, suggesting that while larger firms generates higher accounting returns, they tend to trade at lower market valuations, possibly because markets expect lower future growth from larger and more mature firms. The coefficient on BTMV ranges from negative –0.562 to –0.570 with standard errors between 0.056 and 0.057 and is significant at the 1% level, consistent with the negative findings in the ROA and ROCE models, suggesting that firms with higher BTMV have lower market valuations. Further, to assess the robustness of these findings in all three models of ROA, ROCE, and TQ, two additional checks are performed and presented in the following section 4.2.4.

4.2.4. Robustness check sample 1

Table 7 presents the first robustness check, where the main ESG score is replaced with its one-year lagged value (ESG_LAG) to address the potential of reverse causality, the possibility that it is firm performance that drives ESG scores, and not ESG that drives firm performance. By using the prior year's ESG score to explain the following year's firm performance, ensuring that ESG is determined before the firm performance outcome is observed. The sample size is decreased slightly compared to the main models, as some firms in their first observed year have no ESG value available.

Table 7. Robustness check - Lagged ESG score

Variables	(1) ROA	(2) ROCE	(3) TQ
ESG_LAG	–0.011**	–0.003	0.017
	(0.004)	(0.005)	(0.014)
LEV	–0.253***	–0.106	–0.782***
	(0.052)	(0.070)	(0.130)
SIZE	0.055***	0.052***	–0.059***
	(0.008)	(0.012)	(0.019)
BTMV	–0.036***	–0.071***	–0.544***
	(0.006)	(0.013)	(0.057)
Constant	–0.941***	–0.908***	2.546***
	(0.172)	(0.253)	(0.425)
Observations	3082	2883	2620
R-squared	0.261	0.175	0.522
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG_LAG score is scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. ROA=Return on assets. ROCE=Return on capital employed. TQ=ln(Tobin's Q). ESG_LAG=One-year lagged ESG score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

In column (1), the coefficient on ESG_LAG is negative –0.011 with a standard error of 0.004 and is statistically significant at the 5% level, meaning that a 10-point increase in the lagged ESG score is associated with a decrease in ROA of 0.011 in the following year. While this result differs slightly from the insignificant finding in the main ROA model in Table 4, the direction is the same and the size of the coefficient is similar. The negative and significant lagged ESG coefficient for ROA may reflect the negative and significant G score found in the main model, as G was the strongest driver of the negative direction in the main ROA model and is incorporated in the ESG score. This suggest that the lagged significance may be partly governance driven rather than representing a different result overall. In column (2), the coefficient on ESG_LAG is negative of –0.003 with a standard error of 0.005 and is not statistically significant, which is fully consistent with the findings for ESG in the main ROCE model in Tabel 5. In column (3), the coefficient on ESG_LAG is positive of 0.017 with a standard error of 0.014 and is not statistically significant, which is also consistent with the insignificant finding for ESG in the main TQ model in Table 6. The control variables remain their sign and significance levels across all three model specifications, consistent with the main models, and the R-squared values are closely in line with the main models, respectively. When the individual E, S, and G pillars are lagged separately, the sign and significance levels are consistent with the main findings, but the results are untabulated for brevity. This result further support the robustness of the main results. The lagged ESG robustness check largely supports the main findings and suggest that reverse causality is not to be considered a concern of the main findings.

Table 8 presents the second robustness check, where sample 1 is split by country of headquarters into two subsamples. The splitting by headquarters is used as it captures the institutional and regulatory environment to which firms are subject to. This check is motivated by the fact that Swedish firms in the sample constitute approximately 54% of the sample size, and it is therefore considered important to examine whether the main

findings are driven by Swedish firms or hold across the Nordic region as a hole. Columns (1) to (3) reports results for Swedish firms, and columns (4) to (6) reports results for non-Swedish firms including Denmark, Finland, Norway, and Iceland.

Table 8. Robustness check - Sweden vs. non-Sweden

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	ROA	ROCE	TQ	ROA	ROCE	TQ
ESG	–0.005	0.001	0.013	–0.006	–0.003	0.017
	(0.005)	(0.007)	(0.019)	(0.005)	(0.007)	(0.017)
LEV	–0.301***	–0.190**	–0.860***	–0.196***	0.033	–0.610***
	(0.082)	(0.085)	(0.167)	(0.054)	(0.052)	(0.201)
SIZE	0.052***	0.055***	–0.091***	0.034***	0.020	–0.044
	(0.010)	(0.013)	(0.022)	(0.012)	(0.015)	(0.028)
BTMV	–0.031***	–0.064***	–0.597***	–0.035***	–0.080***	–0.500***
	(0.008)	(0.013)	(0.074)	(0.007)	(0.017)	(0.082)
Constant	–0.934***	–0.926***	3.351***	–0.647**	–0.340	1.996***
	(0.199)	(0.273)	(0.463)	(0.263)	(0.354)	(0.634)
Observations	1918	1772	1617	1634	1527	1334
R-squared	0.312	0.227	0.541	0.155	0.154	0.557
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Sweden	Sweden	Sweden	Non-Sweden	Non-Sweden	Non-Sweden

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG score is scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. Columns 1-3 reports results for Swedish firms only. Columns 4-6 reports result for Denmark, Finland, Norway, and Iceland. ROA=Return on assets. ROCE=Return on capital employed. TQ=ln(Tobin's Q). LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

In column (1) to (3), the coefficient on ESG for Swedish firms is negative –0.005 with a standard error of 0.005 for ROA, positive of 0.001 with a standard error of 0.007 for ROCE, and positive of 0.013 with a standard error of 0.019 for TQ. None of these coefficients are statistically significant at any conventional significance level, and no significant relationship between ESG and firm performance can therefore be confirmed for Swedish firms. In columns (4) to (6), the coefficient on ESG for non-Swedish firms is negative of –0.006 with a standard error of 0.005 for ROA, negative of –0.003 with a standard error of 0.007 for ROCE, and positive of 0.017 with a standard error of 0.017 for TQ. Similarly, none of these coefficients are statistically significant, and no significant relationship between ESG and firm performance can therefore be confirmed for non-Swedish firms either.

The ESG coefficients are of similar size and move in the same direction across both subsamples of Swedish firms and non-Swedish firms, with the exception of ROCE where the coefficient shifts from positive for Swedish firms of 0.001 to negative –0.003 for non-Swedish firms. However, since none of the coefficient are statistically significant in either subsample, this different in direction does not affect the conclusion of the findings. This robustness check indicates that the main finding of no significant relationship between ESG and firm performance is consistent across both Swedish firms and non-Swedish firms, suggesting that the results are not driven by Swedish firms dominating sample 1 and that the absence of significance of the relationship between ESG and firm performance holds across the Nordic region. The control variables show consistent signs

across both subsamples and are largely in line with the main models. The robustness check give support that the main findings are not country driven and that they are robust across the Nordic region.

To summarize the findings from the regression analysis of sample 1, the overall ESG score does not show a statistically significant relationship with any of the three firm performance measures, ROA, ROCE, or TQ. This is an indication that sustainability performance as a combined measure as the ESG score does not have a meaningful association with firm performance among Nordic listed firms. From the result of the individual pillar scores, S shows a positive and significant relationship with ROCE, indicating that social practices may contribute positively to financial performance. G shows a negative and significant relationship with both ROA and ROCE, suggesting that stronger governance structures may increase short-term costs that reduce operating and financial performance. The E score does not show a significant relationship with any performance measure. The control variables show expected signs and are most consistent across all model specifications and performance measures, with the exception of LEV which is only significant in the ROA model. Both robustness checks give support to the findings. The lagged ESG score robustness check confirms that reverse causality is unlikely to drive the results, and the Sweden vs. non-Sweden check confirms that the findings are not driven by just the Swedish firms that dominates sample 1. As a last initial sensitivity check of the regression models of ROA, ROCE, and TQ in Table 4, Table 5, and Table 6, standard errors were clustered at the industry level rather than the firm level. However, this did not provide stronger significance for the sustainability variables, and firm-level clustering is retained throughout all models, as it is also considered a more conservative approach. The results from industry-level clustering are kept untabulated for brevity.

4.3. Empirical result of sample 2

4.3.1. Descriptive statistics sample 2

Table 9 presents the descriptive statistics for sample 2 including 468 Nordic listed firms, which covers the period 2017-2025 and is split into two subgroups based on the IPO year of the Nordic listed firms. Established firms are defined as firms with an IPO year prior to 2017, and newly listed firms as firms with an IPO year of 2017 or later. The number of observations varies across the dependent variables due to the unbalanced panel structure. For return on assets (ROA), the established firm group consists of 2388 observations compared to 441 observations for newly listed firms. For return on capital employed (ROCE), the established firm group consists of 2185 observations compared to 412 for newly listed firms. For Tobin's Q (TQ), the established firm group consists of 2118 observations compared to 380 for newly listed firms.

The mean ROA value of 0.011 for established firms indicate that on average, the established firms generate a small positive return of 1.1% of their total assets, with values ranging from –1.971 to 0.307 and a standard deviation of 0.196. In contrast, newly listed firms show a mean ROA value of –0.067, meaning that on average, newly listed firms generate a negative return of 6.7% on their total assets, with a standard deviation of 0.296 and the same range of min and max values of –0.971 to 0.307. The difference in the mean ROA value between the two groups suggest that newly listed firms are considerably less profitable than established firm, likely because they are in earlier stages of development with higher costs and lower revenues in the years following their IPO. ROCE shows a

similar pattern, established firms with a mean ROCE value of 0.090 and a standard deviation of 0.259, compared to a mean of negative –0.028 and a standard deviation of 0.426 for newly listed firms, further suggesting that newly listed firms generate lower and more variable on their returns on capital employed. The TQ mean value of 0.590 for established firms and 0.700 for newly listed firms suggest that both groups are valued above their book value by the market, but newly listed firms tend to be valued higher. The higher standard deviation of 0.835 for newly listed firms compared to 0.698 for established firms reflect higher variation in market valuations among newly listed firms. This suggest that despite lower operating and financial performance, newly listed firms are valued higher by the market, possibly reflecting higher growth expectations from investors for firms that have recently gone public.

Table 9. Descriptive statistics sample 2

Variables	Observations	Mean	Std. Dev.	Min	Max
Established firms (IPO<2017)					
ROA	2388	0.011	0.196	−1.971	0.307
ROCE	2185	0.090	0.259	−2.399	0.913
TQ	2118	0.590	0.698	−0.625	3.140
ESG	2388	50.347	20.427	1.319	92.855
E	2388	44.609	26.217	0.000	97.836
S	2388	52.907	23.186	0.649	95.859
G	2388	50.955	23.111	1.242	98.207
LEV	2388	0.531	0.198	0.032	1.467
SIZE	2388	22.917	1.938	16.106	26.130
BTMV	2388	0.652	0.783	−0.298	6.282
Newly listed firms (IPO≥2017)					
ROA	441	−0.067	0.296	−1.971	0.307
ROCE	412	−0.028	0.426	−2.399	0.913
TQ	380	0.700	0.835	−0.625	3.140
ESG	441	40.564	19.513	2.642	86.669
E	441	30.154	21.995	0.000	98.370
S	441	42.568	23.102	1.020	93.654
G	441	45.055	23.231	2.498	89.258
LEV	441	0.523	0.244	0.032	1.467
SIZE	441	21.895	1.928	17.130	26.130
BTMV	441	0.641	0.787	−0.298	6.282

Note. The number of observations varies across dependent variables due to the unbalanced panel structure. ROA=Return on assets. ROCE=Return on capital employed. TQ=ln(Tobin's Q). ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value. All variables except ESG, E, S, G, winsorized at the 1st and 99th percentiles.

For the sustainability variables, the ESG score has a mean of 50.347 and a standard deviation of 20.427 for established firms, while newly listed firms show a notably lower mean of 40.564 and a standard deviation of 19.513. This difference of approximately 10

points indicates that newly listed firms have considerably lower ESG score on average, suggesting and may indicate that ESG reporting practices take time to develop after a firm goes public. For the individual pillars, the environmental score (E) shows the largest difference between the two groups, with a mean of 44.609 for established firms compared to 30.154 for newly listed firms, a difference of approximately 14 points, suggesting that environmental reporting is particularly less developed among newly listed firms. The social score (S) averages 52.907 for established firms compared to 42.568 for newly listed firms, and the governance score (G) averages 50.955 for established firms compared to 45.055 for newly listed firms. All the three pillars show lower scores for newly listed firms, with E showing the biggest gap between the two groups. The minimum value of 0.000 of E is observed in both groups, indicating that some firms reported no environmental data at all during some years.

Regarding the control variables, the mean leverage (LEV) value of 0.531 for established firms and 0.523 for newly listed firms are similar, suggesting that the two groups have comparable LEV on average. However, the standard deviation of LEV is higher for newly listed firms at 0.244 compared to 0.198 for established firms, indicating bigger variation in how newly listed firms finance themselves, which is expected given the different capital structures that firms may adopt around the time of their IPO. The mean firm size (SIZE) value of 22.917 for established firms is higher than 21.895 for newly listed firms, with similar standard deviation of 1.938 and 1.928 respectively, the higher mean is expected for established firms as they have had more time to grow their asset base. The mean book-to-market value (BTMV) of 0.652 for established firms and 0.641 for newly listed firms are similar, with standard deviations of 0.783 and 0.787 respectively, suggesting that the two groups have comparable BTMV on average.

Overall, the descriptive statistics reveal clear and meaningful differences between established and newly listed firms, particularly with the firm performance variables and the sustainability variables, where newly listed firms show lower values across all measures, except for TQ. These differences provide early initial motivation for examining if there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms in the Nordic region. The pairwise correlations between all variables in sample 2 are broadly consistent with those in sample 1 and reported in Table 3, which is expected given that sample 2 is a subsample of sample 1, and the observations is broadly similar in size. Therefore, no additional multicollinearity concerns is raised.

4.3.2. Regression analysis sample 2

Table 10 presents the regression results comparing the established and newly listed firms, with return on assets (ROA) as the dependent variable in columns (1) to (4) and return on capital employed (ROCE) in columns (5) to (8). Each model specification includes one sustainability variable at a time alongside the control variables, industry and year fixed effects. The interaction term between the sustainability variables and the newly listed dummy variable, captures whether the relationship differs between the two groups. The regression was also additionally estimated with Tobin's Q (TQ), however no significant results was presented, neither for the sustainability coefficients and the interaction terms in any model specification. These results are therefore kept untabulated for brevity.

Table 10. Regression analysis comparing established and newly listed firms

	ROA				ROCE			
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Established firms (IPO<2017)								
ESG	–0.012***				–0.008			
	(0.004)				(0.007)			
E		–0.004				–0.002		
		(0.003)				(0.005)		
S			–0.003				0.003	
			(0.003)				(0.005)	
G				–0.012***				–0.013***
				(0.003)				(0.005)
Newly listed firms (IPO≥2017)								
Newly_listed	–0.154**	–0.122**	–0.121**	–0.121**	–0.229***	–0.198**	–0.185**	–0.178**
	(0.062)	(0.060)	(0.058)	(0.055)	(0.084)	(0.090)	(0.078)	(0.072)
ESG×Newly_listed	0.030**				0.039***			
	(0.012)				(0.014)			
E×Newly_listed		0.029**				0.042**		
		(0.013)				(0.019)		
S×Newly_listed			0.021**				0.028**	
			(0.010)				(0.012)	
G×Newly_listed				0.020*				0.024
				(0.011)				(0.015)
LEV	–0.264***	–0.264***	–0.266***	–0.265***	–0.081	–0.081	–0.086	–0.080
	(0.060)	(0.059)	(0.060)	(0.061)	(0.064)	(0.063)	(0.064)	(0.066)
SIZE	0.052***	0.046***	0.046***	0.052***	0.051***	0.045***	0.042***	0.055***
	(0.008)	(0.007)	(0.007)	(0.008)	(0.011)	(0.010)	(0.010)	(0.010)
BTMV	–0.032***	–0.031***	–0.031***	–0.032***	–0.070***	–0.069***	–0.068***	–0.070***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.013)	(0.013)	(0.013)	(0.013)
Constant	–0.908***	–0.786***	–0.780***	–0.893***	–0.870***	–0.774***	–0.732***	–0.968***
	(0.153)	(0.145)	(0.142)	(0.163)	(0.240)	(0.218)	(0.215)	(0.224)
Observations	2829	2829	2829	2829	2597	2597	2597	2597
–Established	2388	2388	2388	2388	2185	2185	2185	2185
–Newly listed	441	441	441	441	412	412	412	412
R-squared	0.280	0.280	0.275	0.281	0.191	0.196	0.190	0.190
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG, E, S, and G scores are scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. Established firms are defined as firms with an IPO year prior to 2017, and newly listed firms as firms with an IPO year of 2017 or later. The newly listed dummy equals one for firms with an IPO year of 2017 or later and zero otherwise. The interaction term captures the differential ESG-firm performance relationship for newly listed firms relative to established firms. ROA=Return on assets. ROCE=Return on capital employed. ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

The regression model explains approximately 28% of the variation in ROA across the four specification models, and approximately 19% of the variation in ROCE, indicated by the R-squared values. For established firms, the coefficient on ESG score in column

(1) is negative of –0.012 with a standard error of 0.004 and is statistically significant at the 1% level, meaning that a 10-point increase in ESG is associated with a decrease in ROA of 0.012 for established firms, holding all other variables constant. This negative and significant finding is consistent with the negative direction finding for ESG in the ROA model in sample 1 (Table 4), but in contrast is statistically significant in sample 2 with established firms. In column (2), the coefficient on environmental score (E) is negative of –0.004 with a standard error of 0.003 and is not significant. The coefficient in column (3) on the social score (S) of negative –0.003 with a standard error of 0.003 is also not significant. However, in column (4), the coefficient on governance score (G) is negative of –0.012 with a standard error of 0.003 and is statistically significant at the 1% level, meaning that a 10-point increase in G is associated with a decrease in ROA of 0.012 for established firms, consistent with the negative and significant finding in sample 1 (Table 4).

For established firms in the ROCE columns (5) to (8), the coefficient on ESG in column (5) is negative of –0.008 with a standard error of 0.007 and is not significant, in contrast with the ROA specification. The coefficient on E in column (6) is negative of –0.002 with a standard error of 0.005 and is not significant. The coefficient on S in column (7) is positive of 0.003 with a standard error of 0.005 and is similarly as E not significant. However, consistent with the ROA specification, the coefficient on G in column (8) is negative of –0.013 with a standard error of 0.005 and is statistically significant at the 1% level, meaning that a 10-point increase in G is associated with a decrease in ROCE of 0.013 for established firms, again consistent with the negative and significant finding for G in sample 1 (Table 5).

Moving on to newly listed firms, the interaction dummy variable is negative and statistically significant across all eight model specification, giving support of evidence that newly listed firms have a significantly lower baseline profitability, both in operating and financial performance, in contrast with established firms. For ROA, the coefficient in columns (1) to (4) ranges from negative –0.121 to –0.154 with standard errors between 0.055 and 0.062 and is statistically significant at the 5% level. For ROCE, the coefficient ranges from negative –0.178 to –0.229 with standard errors between 0.072 and 0.084 and is also significant at the 5% level across column (5) to (8). This large and consistent difference in baseline profitability between the two groups is consistent with the descriptive statistics in Table 9, suggesting that newly listed firms are typically in earlier stages of development with higher costs and lower revenues in the years following their IPO.

The interaction terms are of key interest, as they capture whether there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms. For ROA, the interaction term ESG×Newly_listed in column (1) is positive of 0.030 with a standard error of 0.012 and is statistically significant at the 5% level. This means that the negative relationship between ESG and ROA for established firms is reduced by 0.030 for newly listed firms, resulting in a total ESG effect of (–0.012+0.030) plus 0.018 for newly listed firms, turning form negative to positive. While newly listed firms still have a lower baseline profitability than established firms, this positive total effect suggest that higher ESG helps improve their operating performance.

Among the separate sustainability pillars, the interaction term E×Newly_listed in column (2) is 0.029 with a standard error of 0.013 and is significant at the 5% level, resulting in a total E effect of plus 0.025 nor newly listed firms. The interaction term S×Newly_listed in column (3) is 0.021 with a standard error of 0.010 and is also significant at the 5% level, resulting in a total S effect of plus 0.018 for newly listed firms. Similarly, the interaction term G×Newly_listed in column (4) is 0.020 with a standard error of 0.011 and is significant at the 10% level, resulting in a total G effect of plus 0.008 for newly listed firms. Three out of four interaction terms for ROA are positive and statistically significant at the 5% level, while the interaction term for G×Newly_listed is significant at the 10% level. Overall, all four interaction terms are positive, which is a consistent result suggesting that sustainability engagement is more positively associated with operating performance for newly listed firms than for established firms.

For ROCE in the columns (5) to (8), the interaction terms show a similar pattern as they did with ROA. The interaction term ESG×Newly_listed in column (5) is 0.039 with a standard error of 0.014 and is statistically significant at the 1% level, resulting in a total ESG effect of (–0.008+0.039) plus 0.031 for newly listed firms. The interaction term E×Newly_listed in column (6) is 0.042 with a standard error 0.019 and is significant at the 5% level, resulting in a total E effect of plus 0.040 for newly listed firms. The interaction term S×Newly_listed in column (7) is 0.028 with a standard error of 0.012 and is significant at the 5% level, resulting in a total S effect of plus 0.031 for newly listed firms. The only exception to the similar pattern with ROA is the interaction term G×Newly_listed in column (8), which is 0.024 with a standard error of 0.015 and is not significant at any conventional level, meaning that no significant difference in governance sustainability and ROCE between the two groups can be confirmed.

Regarding the control variables, which show consistent signs across all eight model specifications, Leverage (LEV) is negative and significant at the 1% level for ROA across columns (1) to (4), with coefficient ranging from negative –0.264 to –0.266, consistent with the findings in sample 1. For ROCE, LEV is negative but not significant in any specification, also consistent with the ROCE results in sample 1. Firm size (SIZE) is positive and significant at the 1% level across all eight specifications, the coefficient ranging from 0.042 to 0.055, and book-to-market value (BTMV) is negative and significant at the 1% level across all eight model specifications, with coefficient ranging from –0.031 to –0.070, both consistent with the findings in sample 1.

Taken together, the interaction terms are positive and statistically significant across six out of eight model specifications, providing partially evidence that there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms. Newly listed firms start from considerably lower baseline in operating (ROA) and financial (ROCE) performance than for established firms, suggesting that sustainability practices may be particularly beneficial for firms in their earlier stages of their listed life. To further assess the robustness of these findings, a robustness check is performed, where the IPO cutoff year is changed from 2017 to 2018 to examine whether the results may be sensitive to the choice of IPO cutoff year.

4.3.3. Robustness check sample 2

Table 11 presents the robustness check for sample 2, where the IPO cutoff year is changed from 2017 to 2018 to examine whether the results are sensitive to the cutoff year.

Table 11. Robustness check - IPO cutoff 2018

	ROA				ROCE			
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Established firms (IPO<2018)								
ESG	–0.007*				–0.003			
	(0.004)				(0.006)			
E		0.000				0.003		
		(0.003)				(0.005)		
S			0.000				0.007	
			(0.003)				(0.005)	
G				–0.010**				–0.011**
				(0.004)				(0.005)
Newly listed firms (IPO≥2018)								
Newly_listed	–0.020	0.014	–0.007	–0.050	–0.096	–0.031	–0.066	–0.135
	(0.062)	(0.047)	(0.058)	(0.048)	(0.104)	(0.073)	(0.093)	(0.086)
ESG×Newly_listed	0.005				0.018			
	(0.011)				(0.018)			
E×Newly_listed		–0.003				0.005		
		(0.009)				(0.014)		
S×Newly_listed			0.003				0.011	
			(0.009)				(0.015)	
G×Newly_listed				0.012*				0.024*
				(0.007)				(0.013)
LEV	–0.266***	–0.267***	–0.267***	–0.269***	–0.086	–0.086	–0.088	–0.086
	(0.063)	(0.063)	(0.063)	(0.064)	(0.069)	(0.069)	(0.069)	(0.070)
SIZE	0.053***	0.047***	0.047***	0.053***	0.052***	0.048***	0.044***	0.057***
	(0.008)	(0.007)	(0.007)	(0.008)	(0.012)	(0.010)	(0.011)	(0.011)
BTMV	–0.032***	–0.029***	–0.029***	–0.032***	–0.069***	–0.067***	–0.066***	–0.070***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.013)	(0.013)	(0.013)	(0.013)
Constant	–0.930***	–0.830***	–0.820***	–0.930***	–0.923***	–0.849***	–0.794***	–1.008***
	(0.165)	(0.147)	(0.143)	(0.154)	(0.247)	(0.224)	(0.222)	(0.230)
Observations	2829	2829	2829	2829	2597	2597	2597	2597
–Established	2578	2578	2578	2578	2357	2357	2357	2357
–Newly listed	251	251	251	251	240	240	240	240
R-squared	0.268	0.266	0.266	0.274	0.178	0.177	0.179	0.183
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*Note. Robust standard errors clustered at the firm level in parentheses. *** p<0.01, ** p<0.05, * p<0.1 indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All regressions include industry and year fixed effects. ESG, E, S, and G scores are scaled by dividing by 10 (range 0-10) to ease interpretation of coefficients. Established firms are defined as firms with an IPO year prior to 2018, and newly listed firms as firms with an IPO year of 2018 or later. The newly listed dummy equals one for firms with an IPO year of 2018 or later and zero otherwise. The interaction term captures the differential ESG-firm performance relationship for newly listed firms relative to established firms. ROA=Return on assets. ROCE=Return on capital employed. ESG=ESG score. E=Environmental score. S=Social score. G=Governance score. LEV=Leverage. SIZE=ln(Total assets). BTMV=Book-to-market value.*

In Table 11, firms that conducted their IPO in 2018 or later are defined as newly listed firms, while all other firms that conducted their IPO prior to 2018 are defined as established firms. This new definition and cutoff year reduces the newly listed group from

441 to 251 observations for ROA and from 412 to 240 observations for ROCE, as the firm-year observations from firms that conducted their IPO in 2017 are reclassified as established firms. The coefficient for established firms in column (1) to (8) are broadly consistent with the main model in Table 10, with ESG score (ESG), environmental score (E), and social score (S) remaining insignificant across both ROA and ROCE, and governance score (G) remaining negative and significant for both ROA and ROCE.

The newly listed dummy variable is not statistically significant in any of the eight specifications, in contrast to the finding of consistent negative and significant coefficient observed in the main model in Table 10. The loss of significance may be because the considerably smaller subgroup of newly listed firms, which reduces the statistical power to detect a significant difference in the baseline of operating and financial performance between established and newly listed firms. For the interaction terms, most lose statistical significance compared to the main model. For ROA, the interaction terms ESG×Newly_listed in column (1), E×Newly_listed in column (2), and S×Newly_listed in column (3) show no statistical significance, with the only significant interaction term being G×Newly_listed in column (4) at 0.012 with a standard error of 0.007, but only significant at the 10% level. For ROCE, similarly, only G×Newly_listed in column (8) is significant at the 10% level, with the positive coefficient of 0.024 with a standard error of 0.013, while the interaction term for ESG, E, and S in column (5) to (6) respectively, is insignificant.

The loss of significance across most of the interaction terms is considered being attributed to the reduced size of the newly listed group rather than a reversal of the underlying relationship. With the exception of the interaction term E×Newly_listed for ROA in column (2), which shows a small negative coefficient, all other interaction terms remain consistently positive across all sustainability variables in all specification models, consistent with the main model in Table 10. The 2017 cutoff is therefore retained as the primary specification as it has more observations in the newly listed subsample. The result from Table 10 holds and is kept, and the consistent positive direction of the interaction terms across both cutoff years provides further supporting evidence that sustainability engagement is more positively associated with firm performance for newly listed firms than for established firms.

4.4. Relationship analysis and hypothesis testing

4.4.1. Analysis discussion of the relationship between sustainability and firm performance

The empirical results from the regression analysis of sample 1 show mixed results of the relationship between sustainability and firm performance among Nordic listed firms during the period 2010-2025. The direction and significance of the relationship seems to highly depend on which sustainability measure that is used in the regressions and which firm performance measure that is examined. This finding is in line with Friede et al. (2015, pp. 210-213) who report from reviewing more than 2000 empirical studies about the relationship, that the findings are often ambiguous and inconsistent, but at the same time also find that about 90% of the studies report a non-negative relationship, which is in contrast with the findings in this study for the operating performance ROA, and the financial performance ROCE which both show a negative direction, but not with the market performance Tobin's Q (TQ), where the direction is positive, however not statistically significant in neither of the performance measures.

The finding that the overall sustainability score ESG do not show a statistically significant relationship with any of the three performance measures, ROA, ROCE, or TQ, is an interesting finding. The coefficient on ESG score show a negative value of –0.006 for ROA, negative –0.002 for ROCE, and in opposite direction positive 0.015 for TQ, none of which are statistically significant at any conventional significance level.

The stakeholder theory and legitimacy theory both suggest a positive relationship, as firms that manage their sustainability responsibilities effectively should build stronger stakeholder trust, better reputation, and improve access to capital, which should in turn improve firm performance (Olateju et al., 2021, pp. 2-4; Bridoux & Stoelhorst, 2022; Hummel & Schlick, 2016; Marcon Nora et al., 2023). However, the insignificant finding suggest that this positive effect is not visible for Nordic listed firms when the combined ESG score is used. Shareholder theory and trade-off theory in contrast predict a negative relationship, suggesting that sustainability investments do involve real costs that may reduce short-term profitability (Danielson et al., 2008; Jahanzeb et al., 2013). While the coefficients for ROA and ROCE are negative in direction, they do not reach statistical significance, meaning that the findings are directionally consistent with, but do not fully confirm, the predictions of shareholder theory and the trade-off theory.

The insignificant finding for the ESG score contrasts with the positive and significant findings reported by Aydoğmuş et al., (2022, p. 125), using LSEG Workspace ESG scores, who find that the overall ESG score is positively and significantly related to both TQ and ROA in a large international sample of 1720 listed firms across 39 countries. Similarly, Tang et al. (2024, pp. 6-7), using the HUAZHENG ESG rating system, find a positive relationship between the ESG score and TQ among Chinese listed firms, and Kumar & Firoz (2022, p. 61), using Bloomberg ESG scores, find a positive relationship between ESG core and both ROA and ROCE among Indian listed firms. In contrast, Kumar & Puthur Joseph (2025, pp. 733-735) find that ESG is negatively related to both ROA and TQ, while Mugova et al. (2025) find that ESG is negatively related to ROA but positively related to TQ. The different findings in these studies may reflect both the use of different rating providers, as ESG ratings vary across providers (Abhayawansa & Tyagi, 2021, p. 51), and in different institutional regions, where ESG variation could be more pronounced and the marginal value of ESG engagement could be potentially higher than in the already mature sustainability market of Nordic firms (Saha & Khan, 2024, p. 796). The insignificant result for the ESG score is however consistent with a strand of the previous studies that find no significant, weak or non-replicable relationship between sustainability and firm performance in certain contexts (Demers et al., 2021; King, 2025), suggesting that ESG effects may depend on institutional conditions, market maturity, and also to the degree to which sustainability practices are the most integrated into firm strategy.

Further, the environmental score (E) did not show a significant relationship with any of the three firm performance measures. The coefficient on E is negative of –0.001 for ROA, positive of 0.002 for ROCE, and also positive of 0.013 for TQ, but not statistically significant at any conventional level. This insignificant finding is partially consistent with Aydoğmuş et al. (2022, p. 125), who similarly find no significant relationship between the environmental pillar and TQ, which aligns with the insignificant finding in this study. However, Aydoğmuş et al. (2022, p. 125) also find a positive and significant relationship between the environmental pillar and ROA, which contrasts the insignificant finding for

ROA in this study, suggesting that environmental sustainability may translate into operational performance in more diverse international samples but not among only a sample of Nordic listed firms. In contrast, Kumar & Firoz (2022, p. 61) find a positive and significant relationship between the environmental pillar and both ROA and ROCE among Indian listed firms, which also contrasts with the insignificant finding in this study. This difference may reflect the use of different rating providers, as well as the different institutional context, as some may be at an earlier stage of environmental reporting where the marginal signal value of environmental disclosure is higher than in the already mature sustainability market in the Nordic region.

From a trade-off theory perspective, the insignificant finding is not surprising. Environmental initiatives typically require substantial capital commitments, such as investments in cleaner production processes, investing in renewable energy, and emission reduction, with potentially long payback periods that may extend beyond the sample period of this study (Jahanzeb et al., 2013; Rahi et al., 2022; Shaikh, 2022). These costs may reduce short-term firm performance without generating returns that are immediately visible in the performance measures used in this study. From a shareholder theory perspective, such investments may be seen as a poor use of resources if they do not generate direct financial returns for shareholders in the short run (Danielson et al., 2008) The environmental sustainability score also shows the largest variation across firms in the sample compared to the other sustainability pillars, with a standard deviation of 26.389 and a minimum value of 0.000, indicating that environmental reporting practices are uneven across Nordic listed firms during the sample period. This may explain the insignificant finding, as the environmental scores may not fully reflect the actual environmental practices across all firms, making it more difficult to identify a significant relationship between environmental sustainability and firm performance.

Moving on to the social score (S), the positive and significant relationship between S and ROCE is one of the more statistically significant findings of this study. The coefficient on S is 0.009 for ROCE, significant at the 5% level, meaning that a 10-point increase in the social score is associated with an increase in ROCE of 0.009. The coefficient on S for TQ is 0.022, which is significant at the 10% level, however, since this study applies a 5% significance threshold, this finding is considered marginally significant and interpreted with caution regarding this. The coefficient on S for ROA is positive of 0.002 but not statistically significant, suggesting that the positive association between social sustainability and firm performance is more visible in financial performance as measured by ROCE than in operating performance among Nordic listed firms. The marginally positive finding for TQ, although interpreted with caution, is consistent with Tang et al. (2024, p. 7), who find that the social pillar is positively related to TQ among Chinese listed firms. The finding is also consistent with Fu et al. (2023, p. 16), using MSCI ESG scores, who find that social performance is among the main drivers of reduced IPO failure risk and improved long-run performance, suggesting that social sustainability contributes positively to firm performance over time. In contrast, Narula et al. (2024), using CRISIL ESG scores, find no significant relationship between the social pillar and any firm performance measure of ROA, ROCE or TQ, and Shaikh (2022, p. 229), using Bloomberg ESG scores, find a negative relationship between the social pillar and firm performance, which represent contrasting findings to this study. The difference between these studies and the findings of this study may reflect differences in institutional context, sample composition, rating providers, and time periods. As argued by Saha & Khan (2024, p. 796), the Nordic region is characterized by strong social norms and high stakeholder

expectations, which may suggest that the value of social sustainability may be more visible in Nordic listed firm performance than in other institutional contexts.

In terms of a stakeholder theory perspective, this finding is consistent with the argument that firms with stronger social practices, including employee relations, community engagement, and customer satisfaction, generate enhanced stakeholder trust, which can translate in more efficient use of capital (Bridoux & Stoelhorst, 2022; Lahouel et al., 2022; Marcon Nora et al., 2023). Firms that invest in their employees and communities may benefit from higher productivity, lower staff turnover, and stronger customer relationships, all of which can improve capital efficiency as measured by ROCE. From a signaling theory perspective, strong social sustainability performance may serve as a visible signal of firm quality to investors, reducing information asymmetry and improving investor confidence in the firm's long term prospects (Choudhury, 2024, p. 114; Economidou et al., 2023, p. 170), which is consistent with the marginally positive finding for the market performance measure TQ. The fact that S is significant for ROCE but not for ROA could reflect that social practices are more directly linked to how efficiently firms use their capital employed, for example through better employee productivity and lower turnover, than to how efficiently the firms convert total assets into net income.

Turning to governance sustainability (G), the negative and significant relationship between G and both ROA and ROCE is perhaps the most surprising and theoretically interesting finding of this study. The coefficient on G is negative of –0.008 for ROA, significant at the 5% level, and negative of –0.010 for ROCE, also significant at the 5% level, meaning that a 10-point increase in the governance score is associated with a decrease in ROA of 0.008 and a decrease in ROCE of 0.010. The coefficient on G for TQ is negative of –0.007 but not statistically significant.

This negative finding directly contrasts with previous studies, which consistently find governance to be positively related to firm performance. Aydoğmuş et al. (2022, p. 120) find that the governance pillar is positively and significantly related to TQ and Tang et al. (2024, p. 7) find that the governance pillar has the strongest positive effect among the three pillars (E, S, and G) on firm value measured by TQ. Kumar & Firoz (2022, p. 61) find a positive and significant relationship between the governance pillar and both ROA and ROCE. Shaikh (2022, p. 229) similarly finds a positive relationship between the governance pillar and firm performance, arguing that governance practices such as improved transparency and accountability reduce agency problems and improve decision-making within the firm. The use of different rating providers in Kumar & Firoz (2022) and Shaikh (2022) may partly contribute to the different finding, as well as the different context and geographic settings. Rahi et al. (2022), using the same ESG rating provider as this study, find a positive relationship between governance score and ROA among Nordic financial firms. However, since this study excludes the financial industry sector, the samples are not directly comparable, and the difference in findings may reflect the different industry composition rather than a contradiction.

The negative finding for governance sustainability in this study may therefore be best explained from a trade-off theory and shareholder theory perspective. In the Nordic region, where baseline governance standards are generally considered high, the marginal cost of further governance improvements may outweigh the marginal benefit. Governance improvements involve real costs, including increased compliance requirements, more extensive reporting obligations, larger and more independent and

diversified boards, and stronger internal controls, that may reduce short-term accounting profitability even if they improve the long-term quality of decision-making (Danielson et al., 2008; Jahanzeb et al., 2013). This is consistent with the argument by Rahi et al (2022, pp. 302-305) that sustainability investments often require long-term commitments that increase costs and reduce profitability in the short run, particularly in markets where baseline sustainability standards are already high.

Agency theory offers an additional perspective. While agency theory predicts stronger governance reduces agency costs and improves firm performance (Pepper & Gore, 2015, pp. 1045-1068; Tekin & Polat, 2020, p. 1325), it also acknowledges that investments in governance can themselves reflect agency problems if managers over-invest in governance activities beyond what is financially optimal (Economidou et al., 2023, p. 132). In a market where governance standards are already high, the marginal benefit of additional governance investments may be diminishing, while the associated costs remain real. The fact that G is not significant for TQ, with a coefficient of negative –0.007, further supports this potential cost-based interpretation. While governance improvements appear to reduce short-term accounting profitability as measured by ROA and ROCE, the market does not appear to penalize firms with stronger governance in terms of valuation and market performance. This suggests that investor's view governance costs as temporary or acceptable rather than as a structural concern, and that the market takes longer term view of the value of governance improvements than is captured in the short-term operating and financial performance of ROA and ROCE. This is also consistent with the argument by Tang et al. (2024, p. 7), who find that governance has the strongest positive effect on market performance TQ, suggesting that the market rewards governance over a longer time horizon than what is captured in short-term accounting measures.

4.4.2. Hypothesis testing of the relationship between sustainability and firm performance

Based on the empirical findings for sample 1. The three hypothesis formulated to address RQ1 are evaluated and tested below.

$H1_0$: There is no significant relationship between sustainability and firm performance.

$H1_0$ fails to be rejected for the combined ESG score and the environmental score across all three performance measures, ROA, ROCE, and Tobin's Q (TQ), and for the social score in relation to ROA and TQ, and for the governance score in relation to TQ. No statistically significant relationship is found at the 5% level for these sustainability measures and performance measure combinations. This is consistent with the view that ESG investments may not generate measurable firm performance in the short run (Danielson et al., 2008), and with previous studies finding no significant, weak or non-replicable relationships between sustainability and firm performance in certain contexts (Demers et al., 2021, pp. 433-434; King, 2025, p. 109). The insignificant findings for the social score and ROA and TQ are also in line with Narula et al. (2024, pp. 1180-1181), who find no significant relationship between the social score and these performance measures. The insignificant finding for the environmental score and TQ is in line with Aydoğmuş et al. (2022, pp. 119-120), who also find no significant relationship between the environmental score and TQ. The two robustness checks largely support this. However, the lagged ESG score shows a negative and significant relationship with ROA at the 5% significance level, but this may partly reflect the negative and significant governance score found in the main model, as governance score was the only individual

pillar score with a negative and significant relationship with ROA, and governance score is incorporated in the combined ESG score. This interpretation is supported by the untabulated robustness checks, where the individual E, S, and G scores were lagged separately and show signs and significance levels consistent with the main findings. Given that the lagged ESG finding is limited to one firm performance measure and that the combined ESG score in the main models shows no statistically significant relationship with ROA, ROCE, or TQ, the overall evidence does not consistently reject $H1_0$ for the combined ESG score.

H1a: There is a significant positive relationship between sustainability and firm performance

H1a is partially supported. The social score shows a positive and significant coefficient of 0.009 for ROCE at the 5% level, indicating a significant positive relationship between social sustainability and financial performance. This finding suggests that firms with stronger social practices, such as better employee relations and community engagement, may benefit from higher productivity and more efficient use of capital, which is consistent with stakeholder theory, that managing stakeholder relationships effectively may improve financial performance (Bridoux & Stoelhorst, 2022; Lahouel et al., 2022). This finding contrasts with Kumar & Firoz (2022, p. 61), who find no significant relationship between the social score and ROCE among Indian firms, and Narula et al. (2024, pp. 1180-1181), who also find no significant relationship between the social score and ROCE. These contrasting findings suggest that a positive relationship between social score and financial performance may be more visible in the Nordic context where social norms and stakeholder expectations are particularly strong. No other sustainability measure shows a significant positive relationship with any performance measure at the 5% level. H1a is therefore partially supported for the social score and ROCE only.

H1b: There is a significant negative relationship between sustainability and firm performance.

H1b is partially supported, as the governance score shows a negative and statistically significant coefficient of –0.008 for ROA and –0.010 for ROCE, both at the 5% significance level, indicating a significant negative relationship between governance sustainability and operating and financial performance. This could be seen as consistent with the perspective of the trade-off theory, which suggest that governance improvements involve real costs that may reduce short-term firm performance (Danielson et al., 2008; Rahi et al., 2022, pp. 302-305). However, the negative governance finding contrast with Kumar & Firoz (2022, p. 61), who find the governance score to be positively related to both ROA and ROCE, and also Narual et al. (2024, pp. 1180-1181), who find governance to be positively related to ROA and ROCE. H1b is therefore partially supported for the governance score and ROA and ROCE.

Taken everything together, $H1_0$ fails to be rejected for the combined ESG score and the environmental score across all performance measures, for the social score in relation to ROA and TQ, and for the governance score in relation to TQ. However, H1a is partially supported for the social score and ROCE, while H1b is partially supported for the governance score and ROA and ROCE. These findings show that the relationship between sustainability and firm performance among Nordic listed firms differs depending on which sustainability measure is examined and which performance measure is used. The

combined ESG score alone does not capture these differences, as it masks the variation across the individual separated sustainability scores.

4.4.3. Analysis discussion of the comparison between established and newly listed firms

The empirical findings from the regression analysis of sample 2 show that there is a significant difference in the relationship between sustainability and firm performance between established and newly listed firms. The interaction terms between the sustainability variables and the newly listed dummy are positive and statistically significant across six out of eight model specifications for ROA and ROCE, suggesting that the relationship between sustainability and firm performance is more positive for newly listed firms than for established firms when performance is measured through operating and financial performance.

Before discussing the interaction results, it is important to discuss the large and consistent baseline difference in profitability between the established and newly listed firms. The newly listed dummy variable is negative and statistically significant across all eight ROA and ROCE specifications, with coefficients ranging from –0.121 to –0.154 for ROA and from –0.178 to –0.229 for ROCE, all significant at the 5% level. This indicates that newly listed firms have significantly lower baseline profitability than established firms even after controlling for the sustainability scores and other firm characteristics, which is consistent with the descriptive statistics for sample 2, where newly listed firms show a mean ROA of –0.067 compared to positive 0.011 for established firms, and a mean ROCE of –0.028 compared to a positive 0.090 for established firms. This baseline difference could be seen as not surprising form a theoretical perspective, as from a trade-off theory view, newly listed firm often operate with constrained cash flows and compete on growth rather than profitability, making their short-term accounting returns lower than those of established firms with more mature operations (Danielson et al., 2008; Jahanzeb et al., 2013; Qian et al., 2024, p. 12). From a signaling theory perspective, newly listed firms face greater information asymmetry than established firms, because of their limited or lesser public track record, which may also affect their ability to attract resources and generate returns in the short run (Economidou et al., 2023, p. 170).

For established firms, the overall ESG score show a negative and significant coefficient of –0.012 for ROA at the 1% significance level. This is stronger than the insignificant coefficient of –0.006 in the main ROA model in sample 1, suggesting that the negative relationship between ESG and ROA is more pronounced for established firms specifically when the sample is restricted to the period 2017-2025 and is compared against newly listed firms directly. However, the ESG coefficient for ROCE is –0.008 and not statistically significant. This finding is in line with the trade-off theory perspective that sustainability investments involve real costs that may outweigh the short term financial benefits, reducing operating performance in the short run (Danielson et al., 2008; Jahanzeb et al., 2013) This suggest that for established firms, which already operate in a sustainability-mature Nordic market with high baseline sustainability standards, the marginal benefit of further sustainability investments may be limited while the costs associated with the investments remain real.

The central finding of RQ2 is captured by the interaction terms, which measures if there is a difference in the relationship between sustainability and firm performance between established and newly listed firms. For newly listed firms, the positive and significant

interaction terms show that the relationship between sustainability turns positive for both ROA and ROCE. This is consistent with signaling theory, which predicts that ESG disclosures may carry more value for firms facing greater information asymmetry (Economidou et al., 2023, p. 170). Newly listed firms, with their limited public track record, potentially face higher information asymmetry than established firms, making ESG performance a potentially more credible and valuable signal to external investors and stakeholders. The positive interaction effect is in line with Economidou et al. (2023, p. 170), who find that ESG information availability around the time of IPO reduces information asymmetry and generates a positive market response, and with Fu et al. (2023, p. 16) who find that voluntary ESG disclosure reduces IPO failure risk and improves the long run performance. The alignment with these previous findings and the positive interaction terms in this study suggest that the mechanism through which ESG engagement benefits newly listed firms may operate through reduced information asymmetry and improved investor confidence. This rather than potentially direct operational improvements, which would explain why the association is stronger for newly listed firms that face higher uncertainty from investors than for established firms with longer and more transparent track records.

This can also be interpreted through the lens of signaling theory, as newly listed firms typically have shorter public track records and less developed governance structures, meaning that outside investors have less information available to evaluate how well the firm is managed (Choudhury, 2024). Higher ESG engagement, particularly through stronger governance and transparency, may help reduce this investor uncertainty and align managerial behavior with stakeholder interests, which in turn supports firm performance. In addition, Qian et al. (2024, p. 12) find that ESG engagement reduces financial constraints, suggesting that firms with stronger ESG profiles gain better access to external financing, an effect that is likely more meaningful for newly listed firms, which potentially face tighter financing conditions than established firms with longer credit histories and more established investor relationships. From a stakeholder and legitimacy theory perspective, the positive interaction effect may also reflect that newly listed firms that actively manage stakeholder's relationships and demonstrate sustainability commitment gain legitimacy more quickly from investors and stakeholders, which may explain the positive interaction terms (Olateju et al., 2021, pp. 2-4; Hummel & Schlik, 2016). An opposing argument, however, is that sustainability benefits such as reputational capital and stakeholder trust may take time to build (Narula et al., 2024, pp. 1174-1176), which should place newly listed firms at a disadvantage in converting ESG engagement into short-term performance, but the consistently positive and significant interaction terms across six out of eight model specifications suggest that in this sample of Nordic firms, the signaling, agency and legitimacy benefits of ESG engagement are more visible than this concern, at least for operational and financial performance.

Further, the interaction effect holds consistently across most individual sustainability pillars, suggesting that signaling value of ESG engagement for newly listed firm is broad rather than limited to a specific sustainability dimension. The main exceptions are G×Newly_listed for ROA and ROCE and the absence of significant interaction terms for Tobin's Q (TQ) in any specification, in which suggest that the market do not differentiate between established and newly listed firms in terms of how ESG affects firm valuation. One explanation is that market valuations are driven more by broader growth expectations than by ESG performance specifically, or that ESG effects on market valuation may take longer time to materialize (Narula et al., 2024, pp. 1174-1176).

An important limitation of these findings is that the newly listed subgroup consist of only 86 firms out of 555 firms that conducted their IPO in 2017 or later, as the remaining firms lack ESG data, introducing a potential bias toward larger and more visible newly listed firms than the potentially average newly listed firm during this period, since ESG rating providers systematically tend to favor larger and more established firms (Drempetic et al., 2020). These findings should therefore only be interpreted as preliminary evidence, and that more data covering a broader set of newly listed firms with ESG ratings is needed to draw firmer conclusions about the relationship between sustainability and firm performance between established and newly listed firms.

4.4.4. Hypothesis testing of the comparison between established and newly listed firms

Based on the empirical findings from sample 2. The three hypothesis formulated to address RQ2 is evaluated and tested.

$H2_0$: There is no significant difference in the relationship between sustainability and firm performance between established and newly listed firms.

$H2_0$ fails to be rejected for market performance, measured by Tobin's Q (TQ), as no interaction terms are statistically significant in any of the TQ specifications. This suggest that in this specification setup for sample 2, there is no evidence that the relationship between sustainability and market performance differs between established and newly listed firms. A possible interpretation is that sustainability may function more as a general firm characteristic in the Nordic market, rather than a signal that creates a clearly different market valuation effect for newly listed firms. This is in line with the argument that sustainability expectations may already be relatively mature in the Nordic context (Saha & Khan, 2024, p. 796). However, $H2_0$ is rejected for ROA and ROCE in relation to the combined ESG score, the environmental score and social score, since the corresponding interaction terms are positive and statistically significant at the 5% level for both performance measures. $H2_0$ also fails to be rejected for the governance score at the 5% significant level, as the G×Newly_listed interaction term is only significant at the 10% level for ROA and insignificant for ROCE.

H2a: There is a significant positive difference in the relationship between sustainability and firm performance between established and newly listed firms.

H2a is partially supported, as the interaction terms are positive and statistically significant for the combined ESG score, environmental score, and social score in relation to ROA and ROCE. However, H2a is not supported for the governance score at the 5% significance level, neither for TQ, as discussed under $H2_0$. The interaction term ESG×Newly_listed show a positive and statistically significant coefficient of 0.030 for ROA at the 5% significance level and 0.039 for ROCE at the 1% significance level. The individual sustainability pillar interaction terms are also consistently positive, with E×Newly_listed show a coefficient of 0.029, significant at the 5% level for ROA and coefficient of 0.042 also significant at the 5% level for ROCE. S×Newly_listed show coefficient of 0.021 for ROA, significant at the 5% level, and 0.028 for ROCE, also significant at the 5% level. These findings suggest that ESG, environmental, and social performance has a more positive relationship with operating and financial performance for newly listed firms than for established firms. This can be understood through signaling

theory, as the positive interaction terms indicate that sustainability performance may have greater value when firms have shorter public track records and higher information asymmetry (Choudhury, 2024, p. 114; Economidou et al., 2023, p. 170). This suggest that sustainability engagement may help newly listed firms reduce investor uncertainty and strengthen confidence in their operating and financial performance. This can be seen in line with Economidou et al. (2023, pp. 172-173) and Fu et al. (2023, p. 16), who find positive IPO-related outcomes from ESG visibility and disclosure. The findings can also be seen as in line with Qian et al. (2024, p. 12), as ESG engagement may reduce financial constraints, which may be especially beneficial for newly listed firm that depend on external financing.

H2b: There is a significant negative difference in the relationship between sustainability and firm performance between established and newly listed firms.

H2b is not supported, as none of the interaction terms are negative and statistically significant. Instead, the significant interaction terms for ROA and ROCE are positive, indicating that the relationship between sustainability and ROA and ROCE is more positive for newly listed firms than for established firms. This means that the findings do not support the argument that sustainability engagement is more costly or less beneficial for newly listed firms in the short run. The results are therefore not in line with the expectation that ESG effects may take time to materialize, as suggested by Narula et al. (2024, pp. 1174-1176), at least not for ROA and ROCE in this sample 2. The findings also do not support the broader concern that strong signals around IPOs may reflect opportunistic reporting rather than long-term firm quality, a contrast with the logic derived from Teoh et al. (1998), although their study focuses on earnings management rather than ESG performance.

The robustness check presented in Table 11, where the IPO cutoff year is changed from 2017 to 2018, show that most interaction terms lose statistical significance. This is likely associated with the reduction in the newly listed firm subsample from 441 to 251 observations for ROA and from 412 to 240 for ROCE, which possibly reduces the statistical power to detect significant differences between the two groups. However, the interaction terms remain consistently positive in direction across most specifications, only exception is the E×Newly_listed for ROA, but it overall suggest that the underlying relationship is not reversed by the choice of cutoff year.

Taken together, H2a is partially supported, as positive and statistically significant interaction terms are found for the combined ESG score, environmental score, and social score in relation to ROA and ROCE. $H2_0$ fails to be rejected for TQ, where no significant interaction terms are found, and also for the governance score at the 5% significance level. H2b is not supported, as no negative and statistically significant interaction terms are found. As stated before, given the data limitations, including the small newly listed sample and the potential selection bias toward larger and more visible newly listed firms, these findings should be interpreted as preliminary early evidence, and that more data is needed to do draw firmer conclusions about whether the relationship between sustainability and firm performance differs between established and newly listed firms.

5. Conclusion

5.1. General conclusions

The purpose of this thesis was to investigate and analyze the relationship between sustainability and firm performance among listed firms in the Nordic region, and to examine whether this relationship differs between established and newly listed firms. Further, to answer the two research questions formulated:

RQ1: Is there a relationship between sustainability and firm performance among listed firms in the Nordic region?

RQ2: Is there a difference in the relationship between sustainability and firm performance between established and newly listed firms?

Regarding RQ1, the findings from sample 1, covering 468 Nordic listed firms over the period 2010-2025, are based on panel OLS regressions examining the relationship between sustainability and firm performance. The findings show that this relationship is not uniform but depends on both the sustainability measure and the firm performance measure used. The combined ESG score does not show a statistically significant relationship with ROA, ROCE, or Tobin's Q. This indicates that sustainability, when measured as one combined ESG score, does not have a clear direct relationship with firm performance among Nordic listed firms. This finding does not support the positive relationship predicted by stakeholder theory and legitimacy theory, nor does it confirm the negative relationship suggested by shareholder theory and trade-off theory.

However, the individual sustainability scores provide a more nuanced result. The social score shows a positive and significant relationship with ROCE, suggesting that social sustainability may be particularly important for financial performance among Nordic listed firms. This can be explained by the fact that social sustainability concerns relationships with employees, customers, suppliers, and society, which may strengthen stakeholder trust and support more efficient use of capital. This finding is therefore more in line with stakeholder perspective, where firms that manage stakeholder relationships well can benefit through stronger financial performance. The governance score shows a negative and significant relationship with both ROA and ROCE, which is found to be one of the most notable finding of this study as it challenges the agency theory prediction that stronger governance should reduce agency costs and thus improve firm performance. This suggest that in the Nordic context, where baseline governance standards are already high, further governance improvements may increase costs, reporting demands, or administrative complexity without generating operating or financial performance returns.

This makes the finding more consistent with the trade-off theory, since additional governance practices may involve short-term costs that reduce operating and financial performance. Further, the findings show that environmental sustainability does not show a significant relationship with any performance measure, suggesting that environmental initiatives and investments may not be directly reflected in operating, financial, or market performance among Nordic listed firms. Suggesting that environmental investments require longer time horizons before firm performance benefits become visible, and a possibility that the market already expects Nordic firms to maintain relatively high environmental standards. Overall, the findings suggest that the combined ESG score masks important differences across its individual pillars, and that the relationship between

sustainability and firm performance is specific for the individual sustainability scores and also firm performance measure specific.

Regarding RQ2, the findings from sample 2, covering 468 firms over the period 2017-2025 divided into 382 established firms and 86 newly listed firms, are based on panel OLS regression with interaction terms examining whether the relationship between sustainability and firm performance differs between the two groups. The findings show that the relationship differs between the two groups in both operating and financial performance, but not in market performance. The interaction terms for the combined ESG score, environmental score, and social score are positive and statistically significant in relation to both ROA and ROCE. This indicates that these sustainability dimensions are more positively associated to operating and financial performance for newly listed firms than for established firms. Newly listed firms start from a considerably lower baseline in both operating performance and financial performance, compared to established firms. Since newly listed firms have less historical information available to investors, ESG engagement may help reduce information asymmetry and improve confidence in the firm. The findings are also partly in line with agency theory, as sustainability practices may help newly listed firms reduce uncertainty related to governance structures, transparency, and managerial behavior. However, this relationship is not found for market performance (Tobin's Q), suggesting that the market does not value sustainability differently between established and newly listed firms, indicating that market valuation is influenced more by broader growth expectations and investor sentiment than by sustainability performance and that sustainability effects may require longer time to become visible.

These findings answer RQ2 by showing that firm listing history matters for the sustainability and firm performance relationship, but only for certain sustainability dimensions and performance measures. However, this indicates that sustainability practices may be particularly relevant for firms in their earlier stages of their listed life, where firms often have shorter public track records and higher uncertainty from investors. In this context, sustainability may function as signal of firm quality and help newly listed firms strengthen confidence among investors and stakeholders which in turn increases operating and financial performance.

The main theoretical contribution of this study is that the findings show the relationship between sustainability and firm performance is more complex than a simple positive or negative relationship, as the direction and significance of the relationship varies across sustainability measures and firm performance measures and when grouping firms into a cohort of newly listed firms that went public in or after 2017 and comparing them to an older cohort of established firms that went public before 2017. This findings suggest that stakeholder theory and legitimacy theory do not fully explain the relationship in the Nordic context, while signaling theory and agency theory better explain why ESG engagement is more positively associated with firm performance for newly listed firms. The negative governance finding further suggest that the conventional positive relationship between governance and firm performance from previous research may not hold in markets where baseline governance standards are already high. However, these findings should be interpreted with caution due to limitations in ESG data availability, especially among newly listed firms, which means that the results should be seen as preliminary evidence, and that future research on this is needed.

5.2. Practical and managerial implications

The findings of this study have several practical implications for several players connected to Nordic listed firms, especially managers, investors, creditors and other stakeholders. The insignificant relationship between the combined ESG score and firm performance suggest that in the Nordic market, where sustainability practices are already well developed, further ESG investments may not generate a clear financial return in the short run. This may partly reflect that the Nordic market has reached a level of sustainability maturity where strong ESG performance is expected rather than rewarded (Saha & Khan, 2024, p. 796), meaning that ESG no longer serves as a clear competitive advantage in terms of short-term financial metrics. Managers should therefore not view sustainability as a promise for quick financial returns, but rather as a long-term strategy for managing risk, maintaining legitimacy, and meeting stakeholder expectations, and in line with the argument by Qian et al. (2024, p. 12) that ESG engagement however may instead generate benefits over a longer time horizon through improved access to financing and reduced financial constraints for firm that is highly engaged with sustainability practices.

The individual sustainability pillars provide more specific implications for managers. The positive relationship between the social score and financial performance measured by ROCE, suggest that practices connected to employees, customers, suppliers, and society may be an especially important part of sustainability strategy in generating financial performance. This however does not mean that managers should only focus on social practices, but it indicates that it has a clear implication for financial performance. The negative relationship between the governance score and both ROA and ROCE suggest that additional governance practices may involve increased costs without generating operating and financial performance returns. Managers should therefore evaluate whether additional governance initiatives create real value or mainly increase costs in the short run. The insignificant environmental finding, suggest that environmental initiatives may not generate visible short-term financial returns, however this does not necessarily mean that environmental sustainability is without value, but rather that managers and investors should evaluate carefully the environmental investments over a longer time horizon, as the cost of these investments such as emissions reduction and renewable energy investments may occure before their financial benefits become visible.

For managers of firms that have recently gone public, the findings suggest that ESG engagement may be particularly valuable in the early years of being publicly listed. Although newly listed firms is found to have lower baseline ROA and ROCE than established firms, sustainability engagement may be important for newly listed firms to reduce uncertainty and strengthen confidence among investors and stakeholder. This implication is in line with Economidou et al. (2023, pp. 172-173) and Fu et al. (2023, p. 16), who argue that ESG visibility and disclosure can be especially valuable for new IPO firms. Firms that intend to conduct an IPO, or have recently gone public, should therefor consider sustainability disclosure and ESG engagement as part of their strategy for building trust in the public market, which may help them increase operating and financial performance, as this study's finding suggest.

For investors and creditors evaluating firms that have recently gone public in the Nordic region, the findings also suggest that ESG engagement may be a relevant signal of firm quality and future operating and financial performance potential, as sustainability engagement appears to be more positively associated with operating and financial

performance for newly listed firms than for firms with longer public track records. However, the absence of significant ESG effects on market performance, measured by Tobin's Q, for both groups, suggest that ESG engagement alone is unlikely to generate a market premium in the short run, and that investors should consider ESG as one of several factors when evaluating firm quality in terms of sustainability performance rather than as a standalone predictor of market performance.

5.3. Ethical and societal implications

From a broader ethical and societal perspective, the finding that the combined ESG score does not show a significant relationship with firm performance raises an important ethical question about the motivation behind sustainability engagement among Nordic listed firms. If ESG engagement does not clearly translate into improved firm performance in the short run, firms may be tempted to reduce their sustainability efforts or treat sustainability reporting as a compliance exercise rather than a genuine commitment to responsible business conduct. This would be concerning from a societal perspective, as the environmental, social, and governance issues that ESG is designed to measure and evaluate, such as climate change, employee welfare, corporate transparency, and accountability among others, are important societal issues that extend beyond the financial interests of individual firms. The findings therefore suggest that sustainability should not only be evaluated on its relationship with firm performance or the financial returns from the investments required. Instead, sustainability should also be understood in relation to its broader contribution to society, stakeholders, and its potential to support more responsible long-term development into making the world a better place.

Social sustainability is especially relevant from an ethical and societal perspective because it concerns how firms treat employees, customers, suppliers, communities, and society. The finding of a positive relationship between the social score and ROCE suggest that responsible stakeholder management does not necessarily stand in conflict with financial performance. This indicates that firms that take social responsibility seriously may create value not only for shareholders, but also for broader stakeholder groups. For employees, this may involve better working conditions, stronger attention to employee welfare, workplace safety, increased salaries, and opportunities for development. For customers and suppliers, it may involve more responsible relationships and higher trust. For local communities and society, it may mean that firms take greater responsibility for their social impact. Through the lens of stakeholder theory, this is important because firms are expected to consider the interest of several groups connected to the firm, not only the interest of shareholders (Marcon Nora et al., 2023). Governance also has an important ethical role as it concerns transparency, accountability, and responsible control of firms. Although the governance score is found to be negatively associated with ROA and ROCE, this should not be understood as evidence that governance is less important for society. Strong governance practices remain important for reducing agency problems, improving corporate transparency, and protecting the interest of shareholders and other stakeholders (Pepper & Gore, 2015; Tekin & Polat, 2020). From a societal perspective, weak governance practices can reduce trust in firms and financial markets, even if stronger governance does not generate immediate operating and financial performance benefits. Therefore, the ethical implication is that governance should be evaluated not only by its relationship with firm performance, but also by its role in maintaining transparency, accountability, and trust.

Firms that has recently become publicly listed also face ethical responsibilities, since they often have shorter public track records and higher uncertainty from investors and other stakeholders (Economidou et al., 2023, p. 170). In this context, sustainability disclosure may help reduce information asymmetry and strengthen trust between firms and external stakeholders. However, this also means that sustainability reporting must be credible and reflect real practices. If ESG disclosure is mainly used as a signal to attract investors without being supported by genuine sustainability engagement, stakeholders may be misled. This is especially important for newly listed firms, where external stakeholders have less historical information available when evaluating firm quality in terms of sustainability engagement. For investors and creditors, credible sustainability information may reduce uncertainty, and for employees', customers, and suppliers, it may also provide important information about whether the firm is committed to responsible long-term behavior. Furthermore, the finding that only 86 out of 555 firms that conducted their IPO in 2017 or later have available ESG data raises an important societal concern about the unequal coverage of ESG rating, specifically from LSEG Workspace, the provider of ESG data used in this study. Since ESG rating providers tend to cover larger and more visible firms (Drempetic et al., 2020, p. 335), smaller and less visible newly listed firms are systematically excluded from ESG research and investor analysis. This creates an uneven playing field where smaller publicly listed firms may face greater difficulty in signaling their sustainability performance to investors, regardless of their actual sustainability practices. From a societal perspective, this highlights the need for more inclusive and accessible ESG rating systems that cover more publicly listed firms.

5.4. Limitations and future research direction

While this study contributes to the understanding of the relationship between sustainability and firm performance in the Nordic region, several limitations should be acknowledged, and the conclusions should be interpreted with caution. These limitations may have influenced the findings and should be considered when interpreting the results. Each limitation also points to directions for future research that could build on and extend the findings of this study.

A main limitation is that this study relies entirely on ESG scores from LSEG Workspace, since ESG ratings differ between providers (Abhayawansa & Tyagi, 2021, p. 51). Previous studies have used different ESG data sources, including Bloomberg ESG scores used by Kumar & Firoz (2022) and Shaikh (2022), and MSCI ESG scores used by Fu et al. (2023), and differences in rating providers may partly explain the different findings across studies. Future research could therefore examine whether the separate sustainability pillar findings of this study, particularly the negative governance finding and the positive social finding, remain consistent when using ESG scores from other providers, or construct an independent sustainability measure, similar to the approach by Qian et al. (2024, p. 3), to reduce dependence on existing ESG rating system flaws that may be present.

Another limitation concerns the sample composition of this study. The sample of 468 firms with available ESG data represents only a subset of all publicly listed firms in the Nordic region, as a large number of listed firms do not have available ESG data from LSEG Workspace. This means that the findings cannot be generalizable to all Nordic listed firms and may therefore be more representative of larger and more visible firms that are more likely to be covered by ESG rating providers (Drempetic et al., 2020, p. 335). The limitation is particularly found in sample 2, where only 86 out of 555 firms that

conducted their IPO in 2017 or later had available ESG data, creating a risk of selection bias toward larger and more visible listed firms. The findings for newly listed firms should therefore be interpreted as preliminary evidence rather than as conclusions representative of all newly listed Nordic firms. This is perhaps the most important limitation of this study, given that the finding that ESG engagement is more positively associated with performance for newly listed firms. However, future research should revisit this investigation of the relationship, as ESG data coverage for newly listed firms improves over time, and could examine whether the positive interaction effect is stronger for smaller and less visible newly listed firms that potentially face even greater information asymmetry than the potentially larger newly listed firms included in this study.

Furthermore, a limitation also concerns the listing venue of the firms included in the sample. As mentioned in section 3.5.1, the sample includes firms that are admitted to growth markets such as Nasdaq First North, but these firms could not be separated from the main market firms, as the exchange data identifies the operating exchange and not the specific listing segment. This is considered relevant because the Nordic IPO activity during the period 2017-2025 was largely concentrated on growth markets, which suggests that the newly listed group is likely to include growth market firms. These firms tend to differ from main market firms, as they are often smaller, face fewer listing and disclosure requirements, and fall outside the scope of the NFRD, which together may contribute to higher information asymmetry between the firm and external investors (Economidou et al., 2023, p. 170). This means that the finding that the combined ESG score, environmental score, and social score are more positively associated with ROA and ROCE for newly listed firms may partly reflect a listing venue effect rather than recent listing status alone, as the two were difficult to separate in the sample. Future research could therefore obtain listing segment data and separate main market firms from growth market firms, to examine whether the more positive relationship between sustainability and firm performance for newly listed firms is driven by recent listing status, listing venue, or the higher information asymmetry that may be common to both. This would help clarify whether sustainability is particularly beneficial for newly listed firms specifically, or for smaller growth market firms more broadly.

The study is also limited to listed firms in the Nordic region during the period 2010-2025. The Nordic region is characterized by developed capital markets, high transparency, and strong sustainability expectations (Saha & Khan, 2024, p. 795) which means that the findings may not generalize to other institutional contexts. The finding that the composite ESG score does not show a significant relationship with firm performance in the Nordic region raises the question of whether this reflects a sustainability maturity effect that may also be present in other developed European markets. Future research could compare the Nordic region with other European markets to examine whether the pillar-specific and context-dependent nature of the relationship found in this study holds more broadly, or whether it is specific to the sustainability-mature Nordic context.

This study uses ROA, ROCE, and Tobin's Q as measures of firm performance, capturing operating, financial, and market performance. These measures do not cover all possible dimensions of firm performance, as previous research has also used measures such as return on equity, net profit margin, and firm survival (Economidou et al., 2023; Eliwa et al., 2021; Fu et al., 2023; Saha & Khan, 2024). Since the findings show that the relationship between sustainability and firm performance differs across performance

measures, future research could include additional performance measures to provide a more comprehensive picture of how sustainability is associated to different aspects of firm performance.

Finally, although this study includes robustness checks using lagged ESG scores, a country separation comparison, and an alternative IPO cutoff year, the study is still limited by its quantitative research design based on panel OLS regressions. Future research could use more advanced or complementary research designs and dissect other specific firm groups, such as smaller newly listed firms or high growth firms. Future research could also examine the specific mechanisms through which ESG signals may reduce information asymmetry, investor uncertainty, and agency-concerns for newly listed firms, such as changes in analyst coverage, institutional investor attention following ESG disclosure. Also, future studies could examine indirect channels through which ESG may affect firm performance more broadly, such as financing conditions (Eliwa et al., 2021, pp. 16-17), financial constraints (Qian et al., 2024, p. 12), and investor perception as a mediating mechanism (Tang et al., 2024, p. 7).

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Appendixes

Appendix 1. List of firms

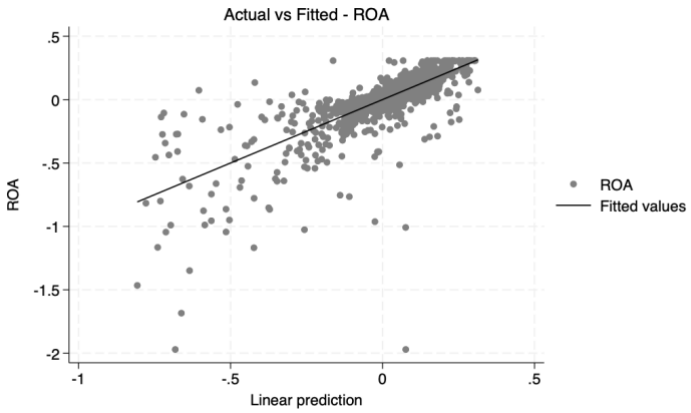
Firm name	Firm name	Firm name
1. AAC Clyde Space AB	2. AAK AB (publ)	3. AF Gruppen ASA
4. ALK-Abello A/S	5. ALM Equity AB	6. AP Moeller - Maersk A/S
7. AQ Group AB	8. Absolent Air Care Group AB	9. AcadeMedia AB
10. Acarix AB	11. Acast AB (publ)	12. Acconeer AB
13. Acroud AB	14. AddLife AB	15. Addnode Group AB (publ)
16. Addtech AB	17. Afry AB	18. Aino Health AB (publ)
19. Akastor ASA	20. Aker ASA	21. Aker BP ASA
22. Aker Biomarine ASA	23. Aker Solutions ASA	24. Akva Group ASA
25. Alcadon Group AB	26. Alfa Laval AB	27. Alimak Group AB (publ)
28. Alligator Bioscience AB	29. Alligo AB	30. Alma Media Oyj
31. Ambea AB (publ)	32. Ambu A/S	33. Anora Group Oyj
34. Arendals Fossekompani ASA	35. Arjo AB (publ)	36. Aspo Oyj
37. Assa Abloy AB	38. Atea ASA	39. Atlas Copco AB
40. Atria Oyj	41. Atrium Ljungberg AB	42. Attendo AB (publ)
43. Austevoll Seafood ASA	44. AutoStore Holdings Ltd	45. Axfood AB
46. BE Group AB (publ)	47. BHG Group AB	48. BICO Group AB
49. BTS Group AB	50. Bactiguard Holding AB	51. Balco Group AB
52. Bang & Olufsen A/S	53. Bavarian Nordic A/S	54. Beijer Alma AB
55. Beijer Ref AB (publ)	56. Bergman & Beving AB	57. Berner Industrier AB
58. Betsson AB	59. Better Collective A/S	60. Bewi ASA
61. Bilia AB	62. Billerud AB (publ)	63. Bimobject AB
64. BioArctic AB	65. BioInvent International AB	66. Biogaia AB
67. Biovica International AB	68. Bittium Oyj	69. Bjorn Borg AB
70. Boliden AB	71. Bonava AB (publ)	72. Bonesupport Holding AB
73. Bong AB	74. Bonheur ASA	75. Boozt AB
76. Borregaard ASA	77. Boule Diagnostics AB	78. Bouvet ASA
79. Bravida Holding AB	80. Brim hf	81. Broedrene A & O Johansen A/S
82. Bufab AB (publ)	83. Bulten AB	84. Byggma ASA
85. Byggmax Group AB	86. C Rad AB	87. CTT Systems AB
88. Cadeler A/S	89. Camurus AB	90. Carlsberg A/S
91. Castellum AB	92. Catena AB	93. Cdon AB
94. CellaVision AB	95. Chemometec A/S	96. Cibus Nordic Real Estate AB
97. Citycon Oyj	98. Clas Ohlson AB	99. Clavister AB
100. Climeon AB (publ)	101. Cloetta AB	102. Coloplast A/S
103. Columbus A/S	104. Consti Oyj	105. ContextVision AB
106. Coor Service Management Holding AB	107. Copenhagen Airports A/S	108. Corem Property Group AB
109. Crown Energy AB	110. DFDS A/S	111. DSV A/S
112. Dampskibsselskabet Norden A/S	113. Dedicare AB (publ)	114. Demant A/S
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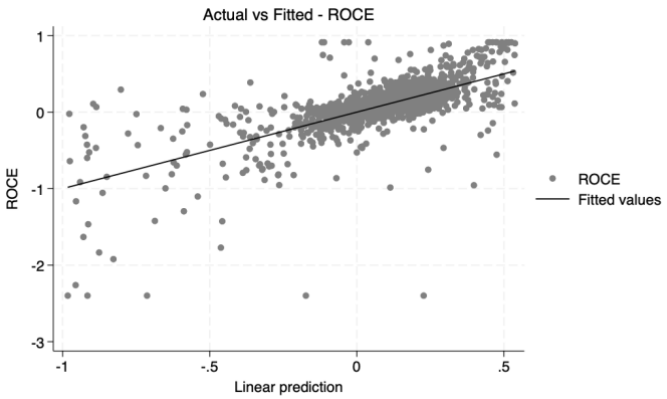
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130. Elanders AB	131. Electrolux AB	132. Electrolux Professional AB (publ)
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136. Elmera Group ASA	137. Eltel AB	138. Embla Medical hf
139. Embracer Group AB	140. Enad Global 7 AB (publ)	141. Endur ASA
142. Enea AB	143. Enento Group Oyj	144. Eniro Group AB
145. Entra ASA	146. Enzymatica AB (publ)	147. Ependion AB
148. Epiroc AB	149. Episurf Medical AB	150. Equinor ASA
151. Essity AB (publ)	152. Etteplan Oyj	153. Europris ASA
154. Evolution AB (publ)	155. Ework Group AB	156. Exel Composites Oyj
157. FLSmidth & Co A/S	158. Fabege AB	159. Fagerhult Group AB
160. Fasadgruppen Group AB (publ)	161. FastPartner AB	162. Fastighets AB Balder
163. Ferronordic AB	164. Festi hf	165. Fingerprint Cards AB
166. Finnair Oyj	167. Fiskars Oyj Abp	168. Flerie AB
169. Flugger group A/S	170. FormPipe Software AB	171. Fortum Oyj
172. G5 Entertainment AB (publ)	173. GN Store Nord A/S	174. Gabriel Holding A/S
175. Garo AB	176. Genmab A/S	177. Genova Property Group AB
178. Genovis AB	179. Getinge publ AB	180. Glaston Oyj Abp
181. Gofore Oyj	182. GomSpace Group AB	183. Granges AB
184. Green Landscaping Group AB (publ)	185. GreenMobility A/S	186. Grieg Seafood ASA
187. H & M Hennes & Mauritz AB	188. H Lundbeck A/S	189. H+H International A/S
190. HKFoods Oyj	191. HMS Networks AB	192. Hagar hf
193. Hampidjan hf	194. Hansa Biopharma AB	195. Hanza AB
196. Harvia Oyj	197. Heimstaden AB	198. Heliospectra AB (publ)
199. Hemnet Group AB (publ)	200. Hexagon AB	201. Hexagon Composites ASA
202. Hexatronic Group AB	203. Hexpol AB	204. Hiab Oyj
205. Holmen AB	206. Hufvudstaden AB	207. Huhtamaki Oyj
208. Humana AB	209. HusCompagniet A/S	210. Husqvarna AB
211. Hydrogenpro ASA	212. I-Tech AB	213. ISS A/S
214. ITAB Shop Concept AB	215. Iceland Seafood Internat hf	216. Icelandair Group hf
217. Immunovia AB (publ)	218. Incap Oyj	219. Indutrade AB
220. Infant Bacterial Therapeutics AB	221. Inission AB	222. Instalco AB
223. Intea Fastigheter AB (publ)	224. Intervacc AB	225. Invisio AB
226. Inwido AB (publ)	227. Irisity AB (publ)	228. Itera ASA
229. JLT Mobile Computers AB (publ)	230. JM AB	231. John Mattson Fastighetsforetagen AB
232. K-Fast Holding AB	233. K2A Knaust & Andersson Fastigheter AB	234. Kamux Oyj
235. Karnov Group AB (publ)	236. Kdventures AB	237. Kemira Oyj
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241. Knowit AB (publ)	242. Kone Corp	243. Konecranes Oyj

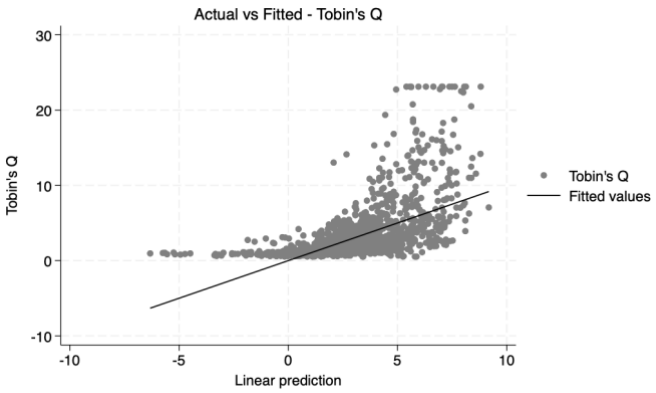
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250. Lime Technologies AB (publ)	251. Lindab International AB	252. Lindex Group Oyj
253. Link Mobility Group Holding ASA	254. Logistea AB	255. Loomis AB
256. Lumo Kodit Oyj	257. Luotea Oyj	258. MAG Interactive AB (publ)
259. MT Hoejgaard Holding A/S	260. Maha Capital AB	261. Marimekko Oyj
262. Matas A/S	263. Medicover AB	264. Medivir AB
265. Meko AB	266. Mendus AB (publ)	267. Mentice AB
268. Metacon AB (publ)	269. Metsa Board Oyj	270. Metso Oyj
271. Micro Systemation AB (publ)	272. Midsona AB	273. Midsummer AB
274. MilDef Group AB	275. Minesto AB	276. Mips AB
277. Modern Times Group MTG AB	278. Moment Group AB	279. Mowi ASA
280. Multiconsult ASA	281. Munters Group AB	282. Musti Group Oyj
283. Mycronic AB (publ)	284. NCAB Group AB (publ)	285. NNIT A/S
286. NP3 Fastigheter AB	287. NRC Group ASA	288. NTG Nordic Transport Group AS
289. Ncc AB	290. Nederman Holding AB	291. Nel ASA
292. Nelly Group AB (publ)	293. Neobo Fastigheter AB (publ)	294. Neste Oyj
295. Net Insight AB	296. Netcompany Group A/S	297. New Wave Group AB
298. Nibe Industrier AB	299. Nilfisk Holding A/S	300. Nilorngruppen AB
301. Nitro Games Oyj	302. Nkt A/S	303. NoHo Partners Oyj
304. Nobia AB	305. Nokia Oyj	306. Nokian Renkaat Oyj
307. Nolato AB	308. Nordic Semiconductor ASA	309. Norsk Hydro ASA
310. Norske Skog ASA	311. North Media A/S	312. Norwegian Air Shuttle ASA
313. Note AB (publ)	314. Novo Nordisk A/S	315. Novozymes A/S
316. Nyfosa AB	317. OEM International AB	318. Odfjell SE
319. Oersted A/S	320. Olvi Oyj	321. Oncopeptides AB
322. Orexo AB	323. OrganoClick AB	324. Oriola Oyj
325. Orion Oyj	326. Orkla ASA	327. Orron Energy AB
328. Otello Corporation ASA	329. Outokumpu Oyj	330. Ovzon AB (publ)
331. Oxe Marine AB	332. Pandora A/S	333. Pandox AB
334. Paradox Interactive AB (publ)	335. Paxman AB (publ)	336. Peab AB
337. Per Aarsleff Holding A/S	338. Pihlajalinna Oyj	339. Platzer Fastigheter Holding AB (publ)
340. Polygiene Group AB	341. Ponsse Oyj	342. Powercell Sweden AB (publ)
343. Precise Biometrics AB	344. Prevas AB	345. Pricer AB
346. Proact IT Group AB	347. Profilgruppen AB	348. Prosafe SE
349. Public Property Invest ASA	350. Q linea AB	351. Qt Group Oyj
352. REC Silicon ASA	353. Railcare Group AB	354. Rapala VMC Oyj
355. Raute Oyj	356. RaySearch Laboratories AB (publ)	357. Reitir fasteignafelag hf
358. Rejlers AB (publ)	359. Rockwool A/S	360. Rottneros AB
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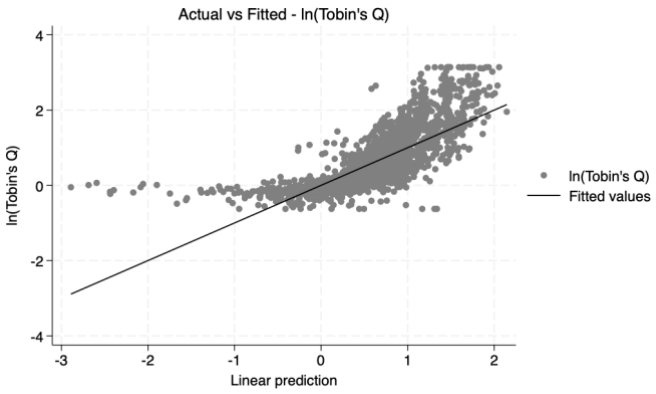
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373. Sats ASA	374. Scandi Standard AB (publ)	375. Scandic Hotels Group AB
376. Scandinavian Tobacco Group A/S	377. Scanfil Oyj	378. Scatec ASA
379. Schouw & Co A/S	380. SciBase Holding AB (publ)	381. Sectra AB
382. Securitas AB	383. Sedana Medical AB (publ)	384. Selvaag Bolig ASA
385. Sensys Gatso Group AB	386. Senzime AB (publ)	387. Siili Solutions Oyj
388. Sinch AB (publ)	389. SinterCast AB	390. Sivers Semiconductors AB
391. Skanska AB	392. SkiStar AB	393. Smart Eye AB (publ)
394. Softronic AB	395. Solar A/S	396. Solteq Oyj
397. Starbreeze AB	398. Stillfront Group AB (publ)	399. Stora Enso Oyj
400. Storskogen Group AB (publ)	401. Storytel AB (publ)	402. Strongpoint ASA
403. Studsvik AB	404. Summa Defence Oyj	405. Suominen Oyj
406. Svedbergs Group AB	407. Svenska Cellulosa Aktiebolaget SCA	408. Sweco AB (publ)
409. Swedencare AB (publ)	410. Swedish Orphan Biovitrum AB (publ)	411. Syn hf
412. SyntheticMR AB (publ)	413. Systemair AB	414. TCM Group A/S
415. TGS ASA	416. TagMaster AB	417. Talenom Oyj
418. Tele2 AB	419. Telefonaktiebolaget LM Ericsson	420. Telenor ASA
421. Teleste Oyj	422. Telia Company AB	423. Teneo AI AB
424. Terveystalo Oyj	425. Thor Medical ASA	426. Thule Group AB
427. Tieto Oyj	428. Tivoli A/S	429. Tobii AB
430. Tokmanni Group Oyj	431. Tomra Systems ASA	432. Trelleborg AB
433. Troax Group AB (publ)	434. UPM-Kymmene Oyj	435. VBG Group AB (publ)
436. Vaisala Oyj	437. Valmet Corp	438. Var Energi ASA
439. Veidekke ASA	440. Vend Marketplaces ASA	441. Verkkokauppa.com Oyj
442. Verve Group SE	443. Vestas Wind Systems A/S	444. Viaplay Group AB (publ)
445. Vicore Pharma Holding AB	446. Viking Line Abp	447. Vitec Software Group AB (publ)
448. Vitrolife AB	449. Vivesto AB	450. Volati AB
451. Volvo AB	452. Volvo Car AB	453. Wallenius Wilhelmsen ASA
454. Wallenstam AB	455. Wartsila Oyj Abp	456. Wastbygg Gruppen AB (publ)
457. Webstep ASA	458. Wihlborgs Fastigheter AB	459. Wilh Wilhelmsen Holding ASA
460. XANO Industri AB	461. XMReality AB (publ)	462. Xbrane Biopharma AB
463. Xspray Pharma publ AB	464. Xvivo Perfusion AB	465. YIT Oyj
466. Yara International ASA	467. Zalaris ASA	468. Zealand Pharma A/S

Appendix 2. Scatterplots of dependent variables

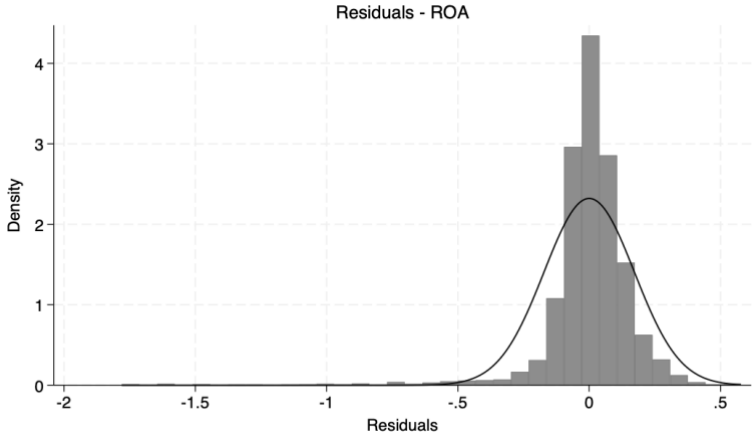


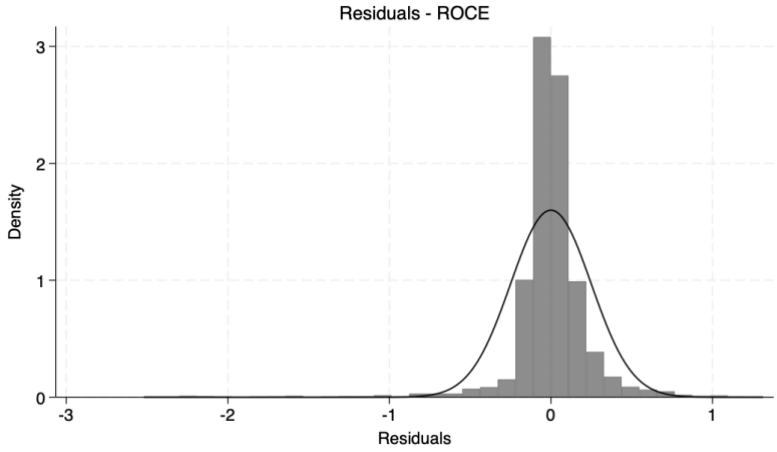


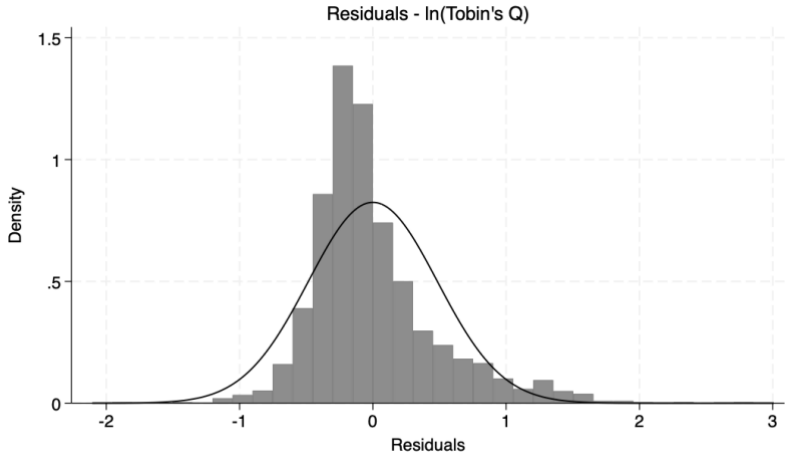




Appendix 3. Histogram of dependent variables









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